Elliptic Curves

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June 2026

Contents

Preface

These are lecture notes for a course on the arithmetic of elliptic curves aimed at peer mentors and counselors that I taught at Ross/Ohio 2026. The course covers the basic theory of elliptic curves at the level of [1] or [2]. The course assumes only a tiny bit more than familiarity with the material on the Ross first-year sets, and a tiny bit more.

Chapter 1

Lecture Notes

1.1 26/06/15 - Introduction I: Diophantine Equations, Plane Curves, and Bachet's Example

One of the oldest and most natural questions in mathematics goes back at least to the 3rd century CE, and probably much more. This question is the study of solutions to *Diophantine equations*, named after Diophantus of Alexandria, who "formulated and solved many such problems" ([2, p. xv]).

A *Diophantine equation* is an equation of the form

$$f(x_1, \ldots, x_n) = 0$$

where $n \in \mathbb{Z}_{\geq 1}$, and $f \in \mathbb{Q}[x_1, \ldots, x_n]$ is a polynomial in n variables with rational coefficients. To *solve it* means to find all rational solutions $(x_1, \ldots, x_n)$ to the equation, although what it means to "find" all solutions can indeed be quite subjective. What is usually meant and hoped for is some sort of complete description or parametrization of such solutions. Variations of this problem ask for solutions over a ring R to similar equations defined over R instead of $\mathbb{Q}$ for various rings R such as $\mathbb{Z}$, a number field K, or its ring of integers $\mathscr{O}_K$. Of course, there is a cool interaction between solutions over $\mathbb{Z}$ and $\mathbb{Q}$; see 1.1.3 for one example.

There are (at least) two measures of the complexity of a given Diophantine equation: the number n of variables, and the degree $d \in \mathbb{Z}_{\geq 1}$ of the polynomial f. Let's study some examples for small n and d.

Example 1.1.1 (Single-Variable Equations). The case of $n = 1$ corresponds to equations of a single variable $x = x_1$.

(a) When $d = 1$, we are looking at linear equations in one variable, which are of the form $ax + b = 0$ for some $a, b \in \mathbb{Q}$ with $a \neq 0$. I trust the reader knows how to solve such equations.

(b) When $d = 2$, we are looking at solving quadratic equations over $\mathbb{Q}$, which look like $ax^2 + bx + c = 0$ for $a, b, c \in \mathbb{Q}$ with $a \neq 0$. Our insistence on asking for rational solutions means that there are essentially four things that can happen, depending on the discriminant $\Delta = b^2 - 4ac$; these are:

  (i) There is a unique root of "multiplicity two"; this happens iff $\Delta = 0$ (e.g., $x^2 - 4x + 4 = 0$).

  (ii) The two distinct roots are non-real complex conjugates; this happens iff $\Delta < 0$ (e.g., $x^2 - 5x + 5 = 0$).

  (iii) The two distinct roots are irrational but real Galois conjugates; this happens iff $\Delta > 0$ but Δ is not a perfect square (e.g., $x^2 - 5x + 7 = 0$).

  (iv) The two distinct roots are both rational; this happens iff Δ is a perfect square (e.g., $x^2 - 5x + 6 = 0$).

(c) When $d = 3, 4$, more complicated formulae exist for the solutions, but these are not always helpful. Sometimes we can use elementary calculus to estimate the number of real roots (see, e.g., 2.1.1). When $d \geq 5$, a famous theorem says that no such formula in terms of radicals even exists in general. However, as long as we are only interested in rational solutions, there exists a complete algorithmic procedure to find them all; this is contained in 1.1.2.

Theorem 1.1.2 (Rational Root). Let $d \in \mathbb{Z}_{\geq 1}$, let $a_0, \ldots, a_d \in \mathbb{Z}$, and consider the equation

$$a_0 x^d + a_1 x^{d-1} + \cdots + a_d = 0.$$

Suppose that $x = p/q$ is a rational solution to this equation, where $p, q \in \mathbb{Z}$ are coprime and $q \neq 0$. Then $p \mid a_d$ and $q \mid a_0$. In particular, if $a_0 = 1$ (i.e., the equation is monic), then every rational root is an integer.

Remark 1.1.3. Do you see why this gives a complete (algorithmic) answer (in theory, if we can find all factors of a given number) of finding all rational (and hence integral) roots to a single-variable rational polynomial?

Proof of 1.1.2. Since p/q is a solution to the equation, we have

$$a_0 p^d + a_1 p^{d-1} q + \cdots + a_d q^d = 0.$$

Therefore, $q \mid a_0 p^d$; since $(p, q) = 1$, this forces $q \mid a_0$. The part with p is similar and left to the reader. ■

Exercise 1.1.4. Finish the proof by showing that $p \mid a_d$. You might have to be slightly more careful than for q because p could be zero; what would that mean?

We didn't really use any special properties of the integers here–other than the fact that they admit unique factorization. Therefore, the same result and proof works also with $\mathbb{Z}$ replaced by any unique factorization domain (UFD) R; see 2.1.2.

Example 1.1.5 (Plane Curves). The next simplest case of $n = 2$ corresponds to *plane curves*. These correspond to a single equation relating two variables $x = x_1$ and $y = x_2$; the corresponding locus can be plotted as a subset of the Euclidean plane $\mathbb{R}^2$.

(a) When $d = 1$, we are talking about linear equations of the form $ax + by + c = 0$, with $a, b, c \in \mathbb{Q}$ and not both a and b zero. The theory of rational solutions here is familiar from high-school coordinate geometry, and the theory of integer solutions is also well-understood, although it might need a little thought (2.1.3).

(b) When $d = 2$, we are looking at *conic sections*, so named because these curves can be obtained by slicing a (doubly infinite) cone by planes at various angles (see this excellent video by 3Blue1Brown illustrating this). Examples include circles (e.g., $x^2 + y^2 = 1$), parabolae (e.g., $y = x^2$), ellipses (e.g., $x^2+2y^2 = 1$), hyperbolae (e.g., $xy = 1$ or $3x^2-2y^2 = 1$), pairs of lines lines (e.g., $y^2-x^2 = 0$), and even "doubled lines" (e.g., $x^2 = 0$) (whatever that means). Plot these on Desmos! Here, the theory of rational solutions is very well-understood. These curves may or may not have any rational points (e.g., $x^2 + y^2 = 1$ and $x^2 + 3y^2 = 2$ respectively; see 2.1.4(a)), but if there is at least one, then we understand all of them very well. We will discuss this more next time.

(c) When $d = 3$, we are looking at *cubic curves*. Again these may or may not have any rational points at all (e.g., $y^2 = x^3 + 1$ and $x^3 + 2y^3 = 4$ respectively; see 2.1.4(b)), but the theory of cubic curves, even those with rational points (these are called *elliptic curves*), is quite subtle. The object of this summer will be to study precisely these objects.

We mention in closing that the cases $d \geq 4$ or $n \geq 3$ are even more complicated; the study of such diophantine equations properly belongs to the field of math known as *arithmetic geometry*. We will not say much about such objects in general, except perhaps for some specific examples.

In 1.1.5, more or less the same discussion can be had with $\mathbb{Q}$ replaced with essentially any field K. (We will systematically make this replacement and work with general fields starting very soon.)

For another Diophantine equation, consider

Example 1.1.6 (Bachet's Example). When does a cube differ from a square by 1? Phrased algebraically, this amounts to solving the equation

$$y^2 = x^3 + 1$$

in pairs of integers (x, y). Five such solutions are $(-1, 0), (0, \pm 1)$, and $(2, \pm 3)$. Are there more?

More generally, given a $c \in \mathbb{Z}$, we can ask for integer solutions to the equation

$$y^2 = x^3 + c. \tag{1.1.7}$$

The case $c = -2$ was studied extensively by French mathematicians Fermat and Bachet in the early 17th century; see the discussion of the history of this problem in [3, Ch. 17, §10]. Bachet observed in 1621 that if (x_0, y_0) is a solution to (1.1.7) with $y_0 \neq 0$, then so is

$$(x_1, y_1) := \left(\frac{x_0^4 - 8cx_0}{4y_0^2}, \frac{-x_0^6 - 20cx_0^3 + 8c^2}{8y_0^3}\right). \tag{1.1.8}$$

This is quite crazy! It's very easy algebra to verify that it works, but how does one come up with such a thing?

Let's assume this for a moment. Given a "seed solution" (x_0, y_0), we can iterate the map $(x_0, y_0) \mapsto (x_1, y_1)$ to produce an infinite sequence (x_n, y_n) of solutions to 1.1.7. For instance, if $c = 1$, then starting from the seed $(x_0, y_0) = (0, 1)$ returns the sequence

$$(0, 1), \quad (0, 1), \quad (0, 1), \ldots.$$

Boring! If we start from the seed $(2,3)$, then we get the sequence

$$(2,3), \quad (0,-1), \quad (0,-1), \ldots.$$

That's slightly more interesting; it's cool that it stabilizes.

Let's take a different c. If we take $c = 3$ and the seed $(x_0, y_0) = (1,2)$ gets us the sequence

$$(1,2), \quad \left(\frac{-23}{4^2}, \frac{11}{4^3}\right), \quad \left(\frac{2540833}{88^2}, \frac{4050085583}{88^3}\right), \ldots.$$

That's very interesting! These numbers seem to get pretty huge pretty quickly. (We will prove later that the number of decimal digits in the numerator and denominator in lowest terms roughly doubles with each iteration.)

When do we get infinitely many distinct solutions in the sequence (x_n, y_n)? It is a theorem of Fueter from 1930 (reproven by Mordell in 1966), which we will prove later, that if $c \notin \{1, -432\}$, then for any seed (x_0, y_0) with $x_0 y_0 \neq 0$, the sequence (x_n, y_n) gives us infinitely many rational solutions.

What about integer solutions? It is a theorem of Thue from 1908 that for any nonzero c, there are only finitely many integer solutions (x, y) to 1.1.7. Therefore, even if there are infinitely many solutions produced by Bachet's formula (as for $c \notin \{1, -432\}$), only finitely many can be integral. We may or may not get to this result by the end of the course; if you're particularly interested in seeing a proof, I can try to incorporate it.

All of this still begs the question: where does Bachet's formula (1.1.8) really come from? The beautiful answer is that it really comes from *geometry*.[1]

The equation $y^2 = x^3 + c$ defines a *cubic curve*, say C, in the plane. This cubic curve has the property that it is *smooth*, i.e., has a well-defined tangent line at every point; equivalently, at every point of the curve, at least one of the partial derivatives $2y$ or $3x^2$ is nonzero.[2] Given a point $P_0 = (x_0, y_0)$ on this cubic curve C, we can draw the tangent line L to C at P_0. Then an elementary case of *Bézout's Theorem* tells us that L will intersect C in a third point P_1 on C; this is quite clear if we parametrize L and restrict the equation of C via this parametrization to get a cubic equation in one variable whose roots correspond to the intersection points. "Two" of these intersection points are at P_0 (i.e., P_0 corresponds to a double root of this cubic equation), and since a cubic has three roots, there must be a third point–this is P_1. The coordinates of P_1 are exactly the solution (x_1, y_1) expressed in Bachet's formula. In older texts, this procedure $P_0 \mapsto P_1$ is called the "chord and tangent process".

Let's work this out explicitly. Suppose we are given a point (x_0, y_0) on the curve (1.1.7). An easy exercise in elementary geometry or calculus tells us that the slope of the curve 1.1.7 at (x_0, y_0) when $y_0 \neq 0$ is $(3x_0^2)/(2y_0)$, so the equation to the tangent line is

$$y = \left(\frac{3x_0^2}{2y_0}\right) x + \left(y_0 - \frac{3x_0^3}{2y_0}\right). \tag{1.1.10}$$

Substitute this expression for y into 1.1.7 to get the cubic equation

$$x^3 - \left[\left(\frac{3x_0^2}{2y_0}\right) x + \left(y_0 - \frac{3x_0^3}{2y_0}\right)\right]^2 + c = 0.$$

This cubic equation has a double root at $x = x_0$. Let's call its third root x_1. Then Vieta's formula allow us to compute the the product of the roots as the negative of the constant term, which gives us

$$x_0^2 x_1 = \left(y_0 - \frac{3x_0^3}{2y_0}\right)^2 - c,$$

giving an expression for x_1 in terms of x_0, y_0, and c. Plugging this into 1.1.10 gives us our y_1. A little bit of simplification using that (x_0, y_0) lies on 1.1.7–left to the reader (2.1.5)–then expresses (x_1, y_1) in the promised form 1.1.8.

[1] I say *really*, because this is probably not how Bachet found it.

[2] This requires us to work over a field of characteristic other than 2 or 3, so $\mathbb{Q}$ is certainly OK.

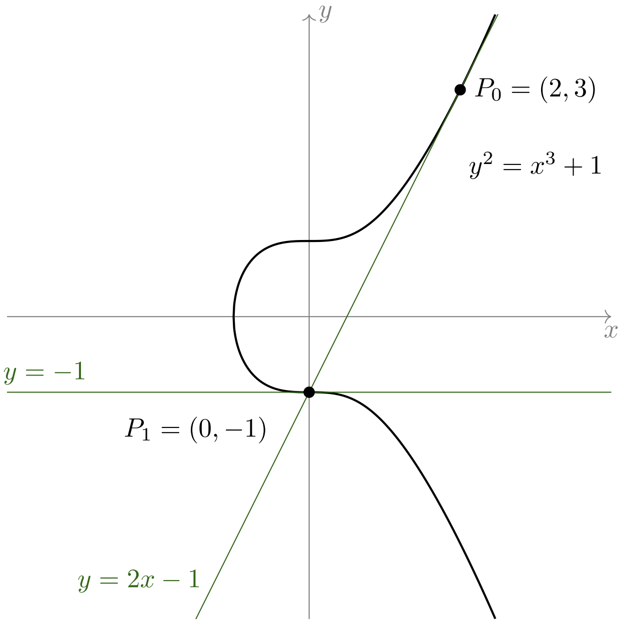


Figure 1.1.9: The "chord and tangent process" which yields Bachet's formula, illustrated for $c = 1$ and seed $P_0 = (x_0, y_0) = (2, 3)$.

The above procedure also has an analog when you are given two points, say P, Q on C: you take the line L joining P and Q, and consider the third intersection points R of P and Q with C (this is the "chord" part of the "chord and tangent process"). It is a fascinating conjecture of Poincaré from around 1901, proven by Mordell 1922 (and then generalized by Weil to number fields in 1928), which says that given any such smooth cubic C over $\mathbb{Q}$, there are finitely many "seed points" $P_i \in C(\mathbb{Q})$ such that *every* other rational point on C can be obtained by iteratively applying the chord and tangent processes to the points P_i. This is now known as the Mordell Theorem or the Mordell-Weil Theorem, and one of our goals for this summer will be to prove this. This cool interplay between algebra, number theory, and geometry is the tip of the iceberg which is the theory of elliptic curves, and what we shall study this summer.

1.2 26/06/16 - Introduction II: Pythagorean Triples and Fermat for Exponent 3

Let us now study some special cases of the *Fermat problem*, the statement of which is colloquially know as *Fermat's Last Theorem*. Recall that this says that if $n \in \mathbb{Z}_{\geq 3}$ and $X, Y, Z \in \mathbb{Z}$ are such that $X^n + Y^n = Z^n$, then the solution (X, Y, Z) is trivial, i.e., $XYZ = 0$. We will discuss and prove the cases $n = 3$ and $n = 4$.

1.2.A Pythagorean Triples

Before we do that however, it is helpful to review the case $n = 2$ of Pythagorean triples. Here we are trying to solve $X^2 + Y^2 = Z^2$ in the integers. The first thing we note is that this is a *homogeneous equation*; this means that if (X, Y, Z) is a solution, then so is $(\lambda X, \lambda Y, \lambda Z)$ for any λ. This is very helpful, and allows some geometric and arithmetic simplifications. We will develop the language of projective curves in the next lecture to deal with the geometric situation more generally, and we will deal with the arithmetic implications below, but for now it suffices to note that if $Z \neq 0$ (the solutions with $Z = 0$ are easy to understand!), then we can divide throughout by Z to get that $(x, y) := (X/Z, Y/Z)$ satisfies

$$x^2 + y^2 = 1.$$

This curve is familiar to us as the circle C. Thus, Pythagorean triples give rise to points on the circle. (This is a fundamental insight going back even to Descartes!) What is more, if X, Y, Z are integers, then x and y are rational numbers; therefore, if we have a good understanding of rational points on the circle $x^2 + y^2 = 1$, then we will have a way to find all Pythagorean triples. For instance, if we could parametrize all points on the circle, then we can hope to parametrize all Pythagorean triples.

Now the circle is a curve of degree two, i.e., a conic. This has the important consequence that a general line meets it in *two points* (some are tangent, and some might need you to allow complex numbers). This leads us to the following simple strategy: if we pick any rational point P on C and project from it to a general line L, then we will get (essentially) a bijective correspondence between points on this line and on the circle. (Make sure you believe this!) Since it is very easy to parametrize points on a line, this gives us a way of parametrizing points on the conic curve. Let's see this work out in practice.

Take the point P to be $(-1, 0)$ and the line L to be the x-axis. Given a point (x_0, y_0) on the circle C, the line joining P and (x_0, y_0) meets the line L in the point $(0, t)$ where $t = y_0/(x_0 + 1)$. Note that this does not work when $x_0 = -1$ corresponding to P itself, when the the limiting line is the tangent line $x = -1$, which does not meet L–in the affine plane. We'll figure out how to fix this next time.

Going the other way, we can parametrize points on L as $(0, t)$ for varying t. The line joining $(-1, 0)$ and $(0, t)$ has equation

$$y = tx + t,$$

which we would like to intersect with the circle. Plugging this in to the equation for the cirlce gives us the quadratic equation

$$x^2 + (tx + t)^2 = 1,$$

which factors as

$$(x + 1)((t^2 + 1)x + (t^2 - 1)) = 0.$$

One root is the expected $x = -1$; it's the other one which we are after, which has $x = (1 - t^2)/(1 + t^2)$. Using $y = tx + t$, this gives us the other point of intersection

$$(x, y) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right).$$

(Those who have seen a little trigonometry might recognize this is as the formula for $(\cos\theta, \sin\theta)$ in terms of $\tan(\theta/2)$; why is that?). These operations are visibly inverses to each other and the resulting maps

$$(x, y) \mapsto y/(x + 1) \text{ and } t \mapsto \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right)$$

are almost inverses to each other (check!). The only point that gives us some trouble is the point $P = (-1, 0)$ itself, which should somehow correspond to "$t = \infty$". Now, the above discussion works over any field (but see 1.2.1). In particular, if t is a rational number, then so is (x, y), and conversely. In this way, we have completely figured out all rational points on C.

Remark 1.2.1. So far we haven't really used many properties of the rational numbers. The same algebra can be carried out over more or less any field K (even if it's not easy visualize what that might mean), with some important caveats. Here are two things to think about:

(a) What (if anything) goes wrong when you work with a field of characteristic two such as $K = \mathbb{F}_2$ instead?
(b) If we work over the field $K = \mathbb{C}$, what happens to the values $t = \pm i$ of the parameter t?

Remark 1.2.2. Note that we could have "dehomogenized with respect to other variables" too. For instance, if $Y \neq 0$, then we may set $(s, t) := (X/Y, Z/Y)$ instead to get instead the equation $s^2 + 1 = t^2$, or equivalently $t^2 - s^2 = 1$. This would have led to a parametrization similar to the one above. This additional symmetry–while not useful here–can be extremely useful in other contexts; we will study this systematically next time.

Let us see what consequence this has for the arithmetic. Suppose we are interested in finding integer solutions X, Y, Z to $X^2 + Y^2 = Z^2$ with $X, Y, Z \neq 0$. We may assume by flipping signs if needed that $X, Y, Z > 0$. We may also divide throughout by the greatest common divisor of all X, Y, Z to assume that X, Y, Z have no common factors as a *triple*. This has the consequence that X, Y, Z have no common factors *pairwise*: if a prime p divides two of X, Y, Z, then the equation $X^2 + Y^2 = Z^2$ forces it to divide the third. (Therefore, things like $(6, 10, 15)$ will never work.) With these simplifications, we can show

Theorem 1.2.3. Let X, Y, Z be pairwise coprime positive integers such that $X^2 + Y^2 = Z^2$. Then there are unique coprime $m, n \in \mathbb{Z}$ of different parity with $m > n > 0$ such that exactly one of (X, Y, Z) or (Y, X, Z) is
$$(m^2 - n^2, 2mn, m^2 + n^2).$$

Proof. The point $(x, y) = (X/Z, Y/Z)$ lies on the circle $C : x^2 + y^2 = 1$ and is different from $(-1, 0)$, so the above discussion tells us that there is a unique $t \in \mathbb{Q}$ such that
$$\left(\frac{X}{Z}, \frac{Y}{Z}\right) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right).$$
Since x and y are positive, we conclude that $0 < t < 1$ (check!). Write $t = n/m$ for coprime positive $m, n \in \mathbb{Z}$ such that $m > n > 0$, so that
$$\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right) = \left(\frac{m^2-n^2}{m^2+n^2}, \frac{2mn}{m^2+n^2}\right).$$
We claim that the fractions on the right are in their lowest terms (up to small factors of two; see below). Why is this?

Let's analyze the parities of m and n. Since they are coprime, they can't both be even. If they have different parity, then $m^2 \pm n^2$ are odd, and this forces
$$\gcd(m^2 + n^2, m^2 - n^2) = \gcd(2mn, m^2 + n^2) = 1.$$
i.e., these fractions are in lowest form (check!). This proves that (X, Y, Z) is the of the form desired.

Finally, can they both be odd? If this is the case, then the integers $m_1 = (m + n)/2$ and $n_1 := (m - n)/2$ are coprime, satisfy $m_1 > n_1 > 0$, have different parity (check!), and further we have that
$$\left(\frac{m^2-n^2}{m^2+n^2}, \frac{2mn}{m^2+n^2}\right) = \left(\frac{2m_1n_1}{m_1^2+n_1^2}, \frac{m_1^2-n_1^2}{m_1^2+n_1^2}\right),$$
and so the argument from the previous paragraph implies $(Y, X, Z) = (m_1^2 - n_1^2, 2m_1n_1, m_1^2 + n_1^2)$, finishing the proof. ■

Exercise 1.2.4. Fill in the details in the above proof: show that if $m, n \in \mathbb{Z}$ are coprime and of different parity, then $\gcd(m^2+n^2, m^2-n^2) = \gcd(2mn, m^2+n^2) = 1$, and show that if $m > n > 0$ are both odd, then $m_1 = (m+n)/2$ and $n_1 := (m-n)/2$ are coprime and have different parity.

In the above study, we didn't really use any properties of the curve C except that it is of degree two–and that it had a rational point P to get our method started. Therefore, the same argument applies to any conic C with a rational point $P \in C(\mathbb{Q})$. This allows us to give complete parametrizations of all rational points of such objects; see, for instance, 2.1.6(a). (Get a hold on the integer points this way might be a little harder; see 2.1.6(b).)

Clearly, then we would have a complete understanding of rational solutions to all conic sections C if we could just prove that they all contained some rational point P. Sadly, this is not the case (2.1.4(a)). How can we figure out when a given conic has a rational point, and when it doesn't?

Example 1.2.5. Let's look at a few examples.

(a) Does the curve
$$4x^2 + 12xy + 9y^2 + 1 = 0$$
have any rational points? No. This is because it can be written as $(2x+3y)^2+1 = 0$, which doesn't have any real solutions (x, y), let alone rational ones. We say such curves are *locally obstructed at* ∞ or *locally obstructed over* $\mathbb{R}$.

(b) Does the curve
$$x^2 + xy + y^2 = 2$$
have any rational points? No, but this is not because it doesn't have any real points (e.g., $(\sqrt{2}, 0)$ works). This is more subtle. Suppose there is a rational solution (x, y). Write $x := X/Z$ and $y := Y/Z'$ for integers $X, Y, Z, Z' \in \mathbb{Z}$ with $Z, Z' > 0$ and $\gcd(X, Z) = \gcd(Y, Z') = 1$. Plugging this into the equation and multiplying through by $Z^2(Z')^2$ gives us
$$X^2(Z')^2 + XYZZ' + Y^2Z^2 = 2Z^2(Z')^2.$$
This equation implies that $Z \mid (Z')^2$ and so by symmetry that $Z' \mid Z^2$. This is enough to conclude that Z and Z' have the same prime factors, but this is not enough to conclude that $Z = Z'$ (consider $2^2 3^3$ vs. $2^3 3^2$). We need to be a little more careful. For this, let p be a prime dividing both Z and Z', and let $\alpha := v_p(Z)$ (resp. $\alpha' := v_p(Z')$), so $\alpha, \alpha' \geq 1$. We have that $v_p(X) = v_p(Y) = 0$, that $\alpha \leq 2\alpha'$ and $\alpha' \leq 2\alpha$. We have to show that $\alpha = \alpha'$. Suppose to the contrary that $\alpha \neq \alpha'$; we may then assume without loss of generality that $\alpha' > \alpha$ (the other case being symmetrical). In this case, the p-adic valuation of the terms on the left is $2\alpha', \alpha+\alpha'$, and 2α, which forces the sum to have p-adic valuation 2α. The right hand side has v_p equal to either $2\alpha + 2\alpha$ if $p \neq 2$, and $2\alpha + 2\alpha' + 1$ if $p = 2$, and we easily see that none of these options work. Thus, we must have $\alpha = \alpha'$. Since this is true for all p and $Z, Z' > 0$, we conclude that $Z' = Z$, and so we can remove it to get the homogeneous equation
$$X^2 + XY + Y^2 = 2Z^2 \tag{1.2.6}$$
we were hoping for. Now we are almost done. Let's consider the parities of X and Y. Well, it cannot be that both of them are odd or that precisely one is odd, because then the left side of 1.2.6 is odd. Therefore, both of them must be even, say $X = 2X_1$ and $Y = 2Y_1$. This then gives us the equation $2(X_1^2 + X_1Y_1 + Y_1^2) = Z^2$, which forces Z to be even as well–but $\gcd(X, Z) = 1$, and this is a contradiction. This proves that there couldn't have been any solutions to begin with.
We say that the curve $x^2 + xy + y^2 = 2$ is *locally obstructed at 2*. Note that this does not mean it has no solutions mod 2 (i.e., that $C(\mathbb{F}_2) = \emptyset$, which is false); rather, it means that it has no *2-adic solutions*, i.e., $C(\mathbb{Q}_2) = \emptyset$. (Don't worry if you don't know yet what that means.)

Similarly, curves can be locally obstructed at other primes; see 2.1.7.

For right now, you can take local obstruction at a prime $p > 0$ to mean that "the arithmetic doesn't work out at p". Clearly, if there is a local obstruction at any prime (including the "prime" $p = \infty$ corresponding to $\mathbb{Q}_\infty = \mathbb{R}$), then there is no rational solution. But the converse is not that clear. Can we have a curve that has solutions *everywhere locally*, but none globally?

It turns out that this "failure of the local-to-global principle" cannot happen for conics, but it can happen for cubics. The fact that it cannot happen to conics is known as the Hasse-Minkowski

Theorem. This holds more generally not just for conic curves, but for quadratic forms in any number of variables. Proving this would take us a little far afield, so I will leave the simplest case of quadratic equations as an exercise (2.1.10), and refer the reader to [4, Ch. IV, Thm. 8] for the general proof.

The really interesting thing I want to draw your attention to, however, is that this result fails in general for curves of degree ≥ 3: there are cubic curves which have solutions in $\mathbb{Q}_p$ for all $p \leq \infty$, but no solutions in $\mathbb{Q}$. One of the simplest examples of this sort is the curve due to Selmer with homogeneous equation

$$3X^3 + 4Y^3 + 5Z^3 = 0. \tag{1.2.7}$$

(This is the homogeneous equation; the inhomogeneous equation is $3x^3 + 4y^3 = -5$.) We will revisit this curve later in the course, and develop tools to study such "obstructions to the Hasse-Minkowski principle". For now, let us turn to

1.2.B Fermat for Exponent 3

The next case of Fermat's problem is $n = 3$, where we are solving $X^3 + Y^3 = Z^3$. Again, this is a homogeneous equation. It is now believed that Fermat did not have a proof for this case, and that the first proof was given by Euler in 1770 (although see the discussion on the Wikipedia page). Euler's proof used the arithmetic of $\mathbb{Z}[\omega]$, where $\omega = (-1 + \sqrt{-3})/2$ is a primitive cube root of unity.

We'll pursue a totally different idea. The idea is that the curve $X^3 + Y^3 = Z^3$ is a cubic plane curve (again, given the understanding of (de)homogenization), and has a rational inflection point; this allows us to put it in *Weierstrass form.* More generally, we can consider for any nonzero α the equation

$$X^3 + Y^3 = \alpha Z^3.$$

This equation has rational point $[1 : -1 : 0]$. Let's move this a little bit, specifically to $[0 : 1 : 0]$: consider the "change of coordinates" given by

$$(X_1, Y_1, Z_1) = (12\alpha Z, 36\alpha(X - Y), X + Y). \tag{1.2.8}$$

Exercise 1.2.9. Figure out how to invert this change of coordinates: express (X, Y, Z) in terms of (X_1, Y_1, Z_1).

We will see below why this change of coordinates is a good one; we will explain later where it comes from. Given this "inverse change of coordinates", we note that the curve above transforms to the one of the form

$$Y_1^2 Z_1 = X_1^3 - 432\alpha^2 Z_1^3,$$

which after dehomogenezing (dividing throughout by Z_1^3) is the curve

$$y^2 = x^3 - 432\alpha^2.$$

This is nothing but our example from 1.1.6, with $c = -432\alpha^2$, and we are back to finding rational points on this cubic curve. The Fermat case corresponds to $\alpha = 1$, and hence $c = -432$. (This might explain whey $c = -432$ is special in 1.1.6.) This is again what we call an *elliptic curve.* We will show later using techniques for elliptic curves that the only points on here are the points $(12, \pm 36)$.

Exercise 1.2.10. Which points on $X^3 + Y^3 = Z^3$ do the points $(12, \pm 36)$ on $y^2 = x^3 - 432$ correspond to?

This finishes the proof of Fermat's last theorem modulo the claim made about the curve $y^2 = x^3 - 432$, which we will prove later. I also still need to explain where this change of coordinates comes from; that will be done sooner. (Hopefully this motivates you to keep coming to this course!) We'll deal with the case $n = 4$ next time.

1.3 26/06/17 - Introduction III: Fermat for Exponent 4 and Infinite Descent; Review of Projective Geometry I: Basics

Let's first finish up our discussion of the Fermat problem for the exponent $n = 4$.

1.3.A Fermat for Exponent 4, Proof 1

Let's try to solve the equation

$$X^4 + Y^4 = Z^4.$$

This is the only case for which Fermat is believed to have a proof, and Fermat's proof uses a very important strategy, known now as "Fermat's method of infinite descent". Let us now review (in modern terminology) Fermat's proof. In fact, we will show something stronger.

Theorem 1.3.1 (Fermat). Let $X, Y, W \in \mathbb{Z}_{\geq 0}$ be such that $X^4 + Y^4 = W^2$. Then $XYW = 0$.

Proof. Suppose to the contrary there is a solution (X, Y, W) with $X, Y, W > 0$, and pick such a solution with the smallest possible W. This implies in particular that X^2, Y^2, W are pairwise coprime (why?) and hence using 1.2.3 that (possibly up to permuting X and Y, which we do), there are unique coprime $m, n \in \mathbb{Z}$ of different parity with $m > n > 0$ such that

$$(X^2, Y^2, W) = (m^2 - n^2, 2mn, m^2 + n^2);$$

in particular, X is odd. Now $X^2 = m^2 - n^2$, so that $X^2 + n^2 = m^2$. Again X, n, m are pairwise coprime positive integers with X odd, so 1.2.3 applied once again gives us unique coprime $p, q \in \mathbb{Z}$ of different parity with $p > q > 0$ and such that

$$(X, n, m) = (p^2 - q^2, 2pq, p^2 + q^2).$$

In particular, n is even and hence m is odd. Finally, $m = p^2 + q^2$ implies that m, p, and q are pairwise coprime. Therefore

$$Y^2 = 2mn = 4mpq$$

implies that p, q, m are all squares. Write $(p, q, m) = (X_1^2, Y_1^2, W_1^2)$ for some integers $X_1, Y_1, W_1 > 0$. Then

$$X_1^4 + Y_1^4 = p^2 + q^2 = m = W_1^2,$$

so that (X_1, Y_1, W_1) is also a solution. Finally,

$$W_1 \leq W_1^2 = m \leq m^2 < m^2 + n^2 = W$$

tells us that this is a "smaller" solution, giving us the required contradiction. ∎

Can you see why this is called "infinite descent"?

The machinery of elliptic curves (specifically descent on them) is a vast modern generalization of this technique due to Fermat; this is where the name "descent" in the theory of elliptic curves comes from. Now Fermat's equation for $n = 4$ does not lend itself directly to the technology of elliptic cuvres, because it not one: $X^4 + Y^4 = Z^4$ defines a *quartic* curve in the plane which is a *canonical curve of genus two* (whatever that means). However, there is a clever way to massage it to give points on an elliptic curve; it is this curve that we do "descent" on. Since it's fun to figure out how to do this massaging, I will leave this as an exercise (2.1.12). We will figure out how to do descent on that curve–and hence prove Fermat's result in a second way–in a more general context much later.

In closing, let me remark that the case $n \geq 5$ of Fermat's theorem is genuinely very difficult, and it is beyond my abilities to go into the details here (see the history on the Wikipedia page, for instance). However, Andrew Wiles's proof from 1994 of it still nontrivially uses the theory of elliptic curves, following an idea going back to Frey from 1984. If we have time and interest, I am happy to explain the outline of the proof towards the end of the course; let me know if this is something you would like to see.

To fruitfully study plane curves, we need some basic understanding of the projective geometry of the plane. Let's take a detour to rapidly review what we need; a much fuller account can be found at [5].

1.3.B Basics

Example 1.3.2. Fix a field K, and for $s \in K$ consider the intersection of the two lines

$$sy = x \text{ and } x = 1.$$

When $s \neq 0$, there is a unique intersection point, namely $(1, 1/s)$, but when $s = 0$, there is no intersection at all, and the lines are parallel. What has happened is that as $s \to 0$, the intersection point has "escaped to infinity".

Such behavior can be extremely troublesome while doing geometry, and it is helpful to develop basic vocabulary to deal with "points at infinity". The modern approach to doing this involves working with *homogeneous coordinates on projective space.*

Definition 1.3.3. Let K be a field and $n \in \mathbb{Z}_{\geq 0}$. We define the *set of K-points of projective n-space over K* to be

$$\mathbb{P}^n(K) := \{[X_0 : X_1 : \cdots : X_n] : X_0, \ldots, X_n \in K \text{ not all zero}\} / \sim$$

where $\sim$ means that we identify points with the same coordinates up to scaling by a nonzero number, i.e., for all $X_0, \ldots, X_n \in K$, not all zero, and $\lambda \in K^\times$ we have

$$[X_0 : \cdots : X_n] \sim [\lambda X_0 : \cdots : \lambda X_n].$$

The coordinates $X_0, \ldots, X_n$ are called *homogeneous coordinates*, and are well-defined only up to simultaneous scaling.

Unimportant Remark 1.3.4. For those who have seen linear algebra, the set $\mathbb{P}^n(K)$ can be identified with the set of one-dimensional K-linear subspaces of K^{n+1} (how?). For those who have seen a little topology, when $K = \mathbb{R}$ or $K = \mathbb{C}$ (or more generally any topological field such as $K = \mathbb{Q}_p$), then $\mathbb{P}^n(K)$ can be given a natural topology, which for $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}_p$ even makes it a(n) (analytic) manifold.

Example 1.3.5.

(a) When $n = 0$, the space $\mathbb{P}^0(K)$ is just one point.

(b) When $n = 1$, the points of $\mathbb{P}^1(K)$ look like $[S : T]$, where $S, T \in K$ are not both zero, and we can simultaneously scale S and T by the same nonzero element λ of K. In particular, when $S \neq 0$, then scaling by $\lambda = S^{-1}$ gives the unique representative $[1 : t]$ where $t := T/S \in K$. In this way, $\mathbb{A}^1(K)$ sits naturally in $\mathbb{P}^1(K)$. The only point not contained in it is the point $[0 : 1]$ corresponding to $S = 0$; this can be thought of as corresponding to "$t = 1/0 = \infty$" and will often be called the *point at infinity* and denoted ∞. Therefore, $\mathbb{P}^1(K) = \mathbb{A}^1(K) \amalg \{\infty\}$ is the "one-point compactification" of $\mathbb{A}^1(K)$. (Warning: This is not true for higher n.) This allows us to handle infinity easily–no actual division by zero needed.

(c) When $n = 2$, we are looking at triples $[X : Y : Z]$ of $X, Y, Z \in K$, not all zero, up to simultaneous scaling. Again, if $Z \neq 0$, then scaling by $\lambda = Z^{-1}$ gives us the unique representative $[x : y : 1]$ where $(x, y) = (X/Z, Y/Z)$, and again we get an $\mathbb{A}^2(K) \subset \mathbb{P}^2(K)$. The complement corresponds to $Z = 0$, which is the set of points $[X : Y : 0]$, in natural bijection with $\mathbb{P}^1(K)$ itself. This locus is called the *line at infinity*; the point $[X : Y : 0]$ should be thought of as the point infinitely far away in the direction of the vector $[X : Y]$. In this way, $\mathbb{P}^2(K) = \mathbb{A}^2(K) \amalg \mathbb{P}^1(K)$. Said yet another way, we have

$$\big\{[X : Y : Z] \in \mathbb{P}^2(K) : Z \neq 0\big\} = \big\{[X/Z : Y/Z : 1] \in \mathbb{P}^2(K)\big\} = \{(x, y) \in \mathbb{A}^2(K)\},$$

where again $(x, y) = (X/Z, Y/Z)$.

Unimportant Remark 1.3.6.

(a) Continuing in the above fashion, we can describe a decomposition of $\mathbb{P}^n(K)$ of the form

$$\mathbb{P}^n(K) = \mathbb{A}^n(K) \amalg \mathbb{P}^{n-1}(K) = \mathbb{A}^n(K) \amalg \mathbb{A}^{n-1}(K) \amalg \cdots \amalg \mathbb{A}^1(K) \amalg \mathbb{A}^0(K).$$

(b) We will probably not seriously need $\mathbb{P}^n(K)$ for $n \geq 3$ in this course, but it may be helpful to note that much of what follows will apply with minor modifications to the general case.

(c) If you have seen barycentric coordinates before, then homogeneous coordinates on projective space $\mathbb{P}^2$ can be thought of as barycentric coordinates with the coordinate (or pure weight) points at $[1:0:0], [0:1:0]$, and $[0:0:1]$.

All faults of affine geometry are rectified in projective geometry, and this boils down to basic linear algebra/matrix theory; as an example, let's see how to fix 1.3.2.

Example 1.3.7. The "projectivization" of the lines of 1.3.2 are the lines

$$sY = X \text{ and } X = Z$$

in $\mathbb{P}^2(K)$. These two *always* have a unique intersection point, namely $[s:1:s]$, in $\mathbb{P}^2(K)$. Note that for $s = 0$, this is the point $[0:1:0]$ at infinity along the vertical direction, as expected.

Example 1.3.8. The "projectivization" of the parametrization in §1.2.A is the map $\mathbb{P}^1 \to \mathbb{P}^2$ given by

$$[S:T] \mapsto [S^2 - T^2 : 2ST : S^2 + T^2],$$

with image $X^2 + Y^2 = Z^2$. (Why is this map well-defined? For this to be true, we have to check that $S^2 - T^2, 2ST, S^2 + T^2$ are never simultaneously zero. This requires our base field to have characteristic other than 2.) When $\operatorname{ch} K \neq 2$, this gives us a *genuine* bijection (even isomorphism) between $\mathbb{P}^1(K)$ and $C(K)$, where $C = \mathbb{V}(X^2 + Y^2 - Z^2) \subset \mathbb{P}^2_K$. This way of looking at it solves various mysteries: for instance, the unique point mapping to $P = (-1, 0) = [-1:0:1]$ is precisely $[S:T] = [0:1]$, also known as "$t = \infty$". Similarly, over $K = \mathbb{C}$ the points $[1:\pm i]$ map to the points $[1:\pm i:0]$ in $\mathbb{P}^2(\mathbb{C})$, known as the "circular points at infinity"; these are not visible over $\mathbb{Q}$ or $\mathbb{R}$.

Let us study solutions of polynomial equations on projective spaces; for simplicity, we will stick to $n = 3$ and use homogeneous coordinates X, Y, Z. On projective space, the vanishing locus of an arbitrary $F \in k[X, Y, Z]$ does not make sense in general: if $F = X^2 - Y$, for instance, then $(1, 1, 0)$ lies on F but $(2, 2, 0)$ does not. However, the one case in which this problem does *not* arise is when F is *homogeneous*, i.e., when all monomials appearing in F have the same degree, say $d \in \mathbb{Z}_{\geq 0}$ (c.f. 2.1.11). In this case we have

$$F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z)$$

for $\lambda \in K^\times$, so it makes sense to speak of the *curve* $C = \mathbb{V}(F)$ cut out by F in projective space. In other words, C consists of K-points

$$C(K) := \{[X:Y:Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0\}.$$

Example 1.3.9. If $F = Y^2Z - X^3 - Z^3$ and $C = \mathbb{V}(F) \subset \mathbb{P}^2_K$, then $\mathscr{O} := [0:1:0] \in C(K)$ but $[1:0:0] \notin C(K)$.

Clearly, replacing F by a nonzero scalar multiple μF for $\mu \in K^\times$ is not going to change the vanishing locus, and in this way we get an identification between curves and nonzero homogeneous polynomials in $K[X, Y, Z]$ up to scaling. We take this to be our definition.

Definition 1.3.10. A *projective plane curve* C over a field K is an equivalence class of nonzero homogeneous polynomials $F \in K[X, Y, Z]$ under the equivalence relation of scaling: F and μF for $\mu \in K^\times$ define the same curve $C = \mathbb{V}(F) = \mathbb{V}(\mu F)$. A representative F of this equivalence class is said to be a *defining equation* of the curve. The degree of any defining equation is then called the *degree* of the curve.

We emphasize that this is an *algebraic* definition; the curve is determined by its equation and not its set of K-points, of which there may not be any (e.g., $C = \mathbb{V}(X^2 + Y^2 + Z^2)$ over $K = \mathbb{R}$). A curve of degree one (resp. two, three, four, etc.) is called a *line* (resp. *conic*, *cubic*, *quartic*, etc.). However, it is still natural to think of the curve C–to a first approximation–in terms of its K-points $C(K)$.

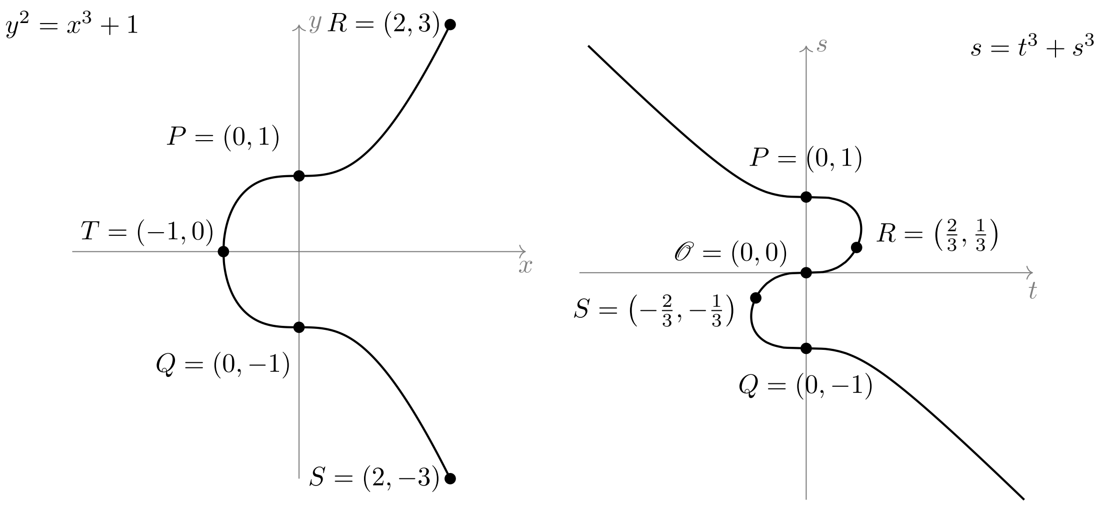
How do we visualize a projective curve C? Let us now briefly discuss something we have seen in practice already. Given a projective curve $C = \mathbb{V}(F) \subset \mathbb{P}^2_K$ over some field K and of degree $d \in \mathbb{Z}_{\geq 1}$, we can dehomogenize it with respect to any of the variables X, Y, or Z to get an affine curve, where dehomogenization, say, with respect to the variable Z amounts to either dividing throughout by Z^d and using coordinates $x := X/Z$ and $y := Y/Z$, or equivalently looking at points in the affine patch $Z \neq 0$, where we can normalize to $Z = 1$. Algebraically, this amounts to replacing $F \in K[X, Y, Z]$ by $f(x, y) := F(x, y, 1) \in K[x, y]$. In other words,

$$\begin{aligned}
C(K) \cap \{Z \neq 0\} &= \{[X : Y : Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0 \text{ and } Z \neq 0\} \\
&= \{(x, y) \in \mathbb{A}^2(K) : F(x, y, 1) = 0\} \\
&= \{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}.
\end{aligned}$$

There is nothing special about Z here, and we can equally well dehomogenize with respect to say Y, by taking coordinates $t := X/Y$ and $s := Z/Y$ instead. These different "local patches" of the curve can then be glued together to get a fuller picture of the projective curve.

Example 1.3.11. Consider the projective curve $C := \mathbb{V}(Y^2Z - X^3 - Z^3) \subset \mathbb{P}^2_K$ (for any field K). The affine patch $Z \neq 0$ looks like the familiar curve $y^2 = x^3 + 1$, and the affine patch $Y \neq 0$ looks like $s = t^3 + s^3$. The first patch has all points on C except for $\mathscr{O} = [0 : 1 : 0]$; the second patch has all points except for $[-\omega^j : 0 : 1]$ for $j = 0, 1, 2$, where $\omega = \zeta_3$ is a primitive cube root of unity. (Some of these point may not exist in $C(K)$. If K has characteristic three, there is only one point $[-1 : 0 : 1]$.) These two patches are related by the coordinate change

$$(t, s) = \left(\frac{x}{y}, \frac{1}{y}\right) \text{ and } (x, y) = \left(\frac{t}{s}, \frac{1}{s}\right).$$



(a) The affine chart $Z \neq 0$.

(b) The affine chart $Y \neq 0$.

Figure 1.3.12: Two affine patches of the curve $Y^2Z = X^3 + Z^3$ with a few marked points. Note that the point $\mathscr{O}$ is not visible in the first patch, and the point T is not visible in the second patch, but together they cover everything. Note also that the line at infinity in the first patch becomes the t-axis $s = 0$ in the second patch, and that this line has a "triple" intersection with the curve at $\mathscr{O}$, i.e., is an inflectional tangent to C at $\mathscr{O}$.

I would recommend playing with Desmos3D and plotting $Y^2Z = X^3 + Z^3$ and seeing how it intersects the loci $Z = 1$, $Y = 1$, and $X^2 + Y^2 + Z^2 = 1$ to see how these pictures are related to each other.

Going in the other direction, given an affine curve $C \subset \mathbb{A}^2_K$ given by the vanishing of say $f(x, y) \in K[x, y]$, we can look at its *projective closure*. This corresponds to replacing f by its *homogenization* $F \in K[X, Y, Z]$ obtained by replacing x by X/Z and y by Y/Z and clearing denominators, or (equivalently) by replacing x by X and y by Y, and inserting suitable powers of Z in the monomials to get a homogeneous polynomial of lowest possible degree. This projective closure is usually denoted by $\overline{C} \subset \mathbb{P}^2_K$. We can recover C from $\overline{C}$ by dehomogenizing with respect to Z once again.

Example 1.3.13. The line $2x - 5y + 3 = 0$ in $\mathbb{A}^2_K$ has projective closure $2X - 5Y + 3Z = 0$ in $\mathbb{P}^2_K$. To obtain the points at infinity on this projective closure, we set $Z = 0$ to get the unique point $[5 : 2 : 0]$ in the direction of this line. Make sure this makes sense to you!

Example 1.3.14. The rectangular hyperbola $x^2 - y^2 = 1$ in $\mathbb{A}^2_K$ has projective closure $X^2 - Y^2 = Z^2$ in $\mathbb{P}^2_K$. To obtain the points at infinity of this projective closure, we set $Z = 0$ to get the two points $[1 : \pm 1 : 0]$. Of course, this is as expected: these correspond to the asymptotes of the hyperbola.

Exercise 1.3.15. Are the operations of homogenization and dehomogenization inverses?

1.4 25/06/18: Review of Projective Geometry II: Projective Changes of Coordinates

Perhaps I didn't do a great job of explaining projective space last time, so let me try again.

1.4.A More on Curves and Homogenization

Let us review some things from last time. Given a field K, we met the *affine plane* $\mathbb{A}^2(K)$ over K, which is just the good old plane: it consists of all points (x, y) where $x, y \in K$:

$$\mathbb{A}^2(K) = \{(x, y) : x, y \in K\}.$$

Note that for any field extension L/K, we have a natural inclusion $\mathbb{A}^2(K) \subset \mathbb{A}^2(L)$.

Given a nonzero $f \in K[x, y]$, we can look at its *vanishing locus* in $\mathbb{A}^2(K)$, which is just

$$\{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}.$$

This is what we want to call a curve. Note that for any $\lambda \in K^\times$, the polynomials f and λf have the same vanishing loci; they define the same *curve*. This motivates the algebraic

Definition 1.4.1.

- An *affine curve* is an equivalence class of nonzero $f \in K[x, y]$ under the equivalence relation $f \sim \lambda f$ for nonzero $f \in K[x, y]$ and $\lambda \in K^\times$.

The curve defined by a nonzero f will be denoted $C := \mathbb{V}(f)$, so that $\mathbb{V}(f) = \mathbb{V}(\lambda f)$ for all $\lambda \in K^\times$.

- Such an f as above is called *a defining equation* for the curve C, and the *degree of the curve* C is defined to be the degree of any defining equation for C.

A(n) (affine) curve of degree one (resp. two, three, four, etc.) is called a(n) (affine) *line* (resp. *conic*, *cubic*, *quartic*, etc.).

Given such an f and the associated curve $C = \mathbb{V}(f) \subset \mathbb{A}^2_K$, we can then look at its K-points

$$C(K) := \{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}.$$

The reason for keeping this notation separate is because given a polynomial $f \in K[x, y]$, we would like to study its solutions over any field extension L/K. For instance, K could be $\mathbb{Q}$, and L could be $\mathbb{R}$ or $\mathbb{C}$ or $\mathbb{Q}_p$ or some number field like $\mathbb{Q}[i]$ or $\mathbb{Q}[\sqrt{5}]$. The above notation easily lets us do that, so we get a natural inclusion $\mathbb{A}^2(K) \subset \mathbb{A}^2(L)$ and $C(K) \subset C(L)$.

Example 1.4.2. Let $K = \mathbb{Q}$, let $f(x, y) = x^2 + y^2 - 1 \in \mathbb{Q}[x, y]$, and let $C := \mathbb{V}(f)$. Then $C(\mathbb{Q}) \subset C(\mathbb{R}) \subset C(\mathbb{C})$. We have that $(1, 0) \in C(\mathbb{Q})$, that $(1/\sqrt{2}, 1/\sqrt{2}) \in C(\mathbb{R}) \smallsetminus C(\mathbb{Q})$, and $(i, \sqrt{2}) \in C(\mathbb{C}) \smallsetminus C(\mathbb{R})$.

This algebraic definition also allows us to work with curves that have no points over the base field.

Example 1.4.3. Let $K = \mathbb{Q}$, let $f(x, y) := x^2 + y^2 + 1 \in \mathbb{Q}[x, y]$, and let $C := \mathbb{V}(f)$. Then $C(\mathbb{Q}) = C(\mathbb{R}) = \emptyset$ but $C(\mathbb{C}) \neq \emptyset$, because for instance $(0, i) \in C(\mathbb{C})$. In particular, C is not "empty"; it's just that $\mathbb{R}$ is not a good enough field to see any points on it.

Exercise 1.4.4. Let K be an algebraically closed field. Then for any nonzero $f \in K[x, y]$ of positive degree, if we set $C := \mathbb{V}(f)$, then $C(K) \subset \mathbb{A}^2(K)$ is nonempty. (Note that of course this is not true if K is not algebraically closed, as we have seen above in 1.4.3.)

The abstract curve $C = \mathbb{V}(f) \subset \mathbb{A}^2_K$ arising from this algebraic definition encodes this data of points in every field extension L/K.

Unimportant Remark 1.4.5. The objects $\mathbb{A}^2_K$ and $\mathbb{A}^2(K)$ are slightly different; the latter is the set of K-points of the former (and is probably more properly written $\mathbb{A}^2_K(K)$). The first is a "formal object" that records the data of all $\mathbb{A}^2(L)$ for all extensions L/K. The same goes for the curve C and its L-points $C(L)$ for any L/K. The subscript K in the formal object reminds us what field we are doing the algebra; for instance, if $K = \mathbb{F}_3$ and $C = \mathbb{V}((1 \mod 3)y + (2 \mod 3)x + (1 \mod 3)) \subset \mathbb{A}^2_{\mathbb{F}_3}$, then the subscript reminds us that it makes no sense to talk about $C(\mathbb{Q})$, although it makes perfect sense to talk about $C(\mathbb{F}_3)$ or more generally $C(L)$ for any $L/\mathbb{F}_3$. (For the cognoscenti: $\mathbb{A}^2_K := \operatorname{Spec} K[x, y]$ and $C := \operatorname{Spec} K[x, y]/(f)$.)

Remark 1.4.6. One strange consequence of the above definition is that sometimes you can have two curves $C, C' \subset \mathbb{A}^2_K$ such that for all fields L/K we have $C(L) = C'(L)$ even though $C \neq C'$. For instance, C and C' might even have different degrees, e.g., if $C = \mathbb{V}(x)$ and $C' = \mathbb{V}(x^2)$. Why do you think this is a good thing? Is it?

More or less the same things as above can be done in the projective plane. As above,

$$\mathbb{P}^2(K) = \{[X : Y : Z] : X, Y, Z \in K \text{ not all zero}\}/\sim$$

where the $\sim$ means that we identify points $[X : Y : Z]$ and $[\lambda X : \lambda Y : \lambda Z]$ for all $\lambda \in K^\times$. As before, for each L/K, we have $\mathbb{P}^2(K) \subset \mathbb{P}^2(L)$. Finally, for any K, the affine space $\mathbb{A}^2(K)$ sits inside $\mathbb{P}^2(K)$ by sending (x, y) to $[x : y : 1]$, with the complement being the "line at infinity" given by $Z = 0$.

Given a nonzero *homogeneous* $F \in K[X, Y, Z]$, we can look at its *vanishing locus* in $\mathbb{P}^2(K)$, which is just

$$\{[X : Y : Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0\}.$$

As before, we have

Definition 1.4.7.

- A *projective curve* is an equivalence class of nonzero homogeneous $F \in K[X, Y, Z]$ under the equivalence relation $F \sim \lambda F$ for nonzero homogeneous $F \in K[X, Y, Z]$ and $\lambda \in K^\times$.

The curve defined by a nonzero F will be denoted $D := \mathbb{V}(F)$, so that $\mathbb{V}(F) = \mathbb{V}(\lambda F)$ for all $\lambda \in K^\times$.

- Such an F as above is called *a defining equation* for the curve D, and the *degree of the curve* D is defined to be the degree of any defining equation for D.[a]

[a] For the cognoscenti: D stands for "divisor".

As before, we have projective lines, conics, cubics, quartics, etc., and as before for any $D = \mathbb{V}(F) \subset \mathbb{P}^2_K$ it makes sense to talk about $D(L)$ for any L/K, with $D(K) \subset D(L)$.

Example 1.4.9. A projective line has the form $D = \mathbb{V}(aX + bY + cZ)$ for $a, b, c \in K$ not all zero, and where we may simultaneously scale a, b, and c. In particular, the set of all lines in $\mathbb{P}^2_K$ it itself a $\mathbb{P}^2(K)$, called the *dual projective plane*. (This duality has some lots of interesting consequences.) In general, this simultaneous scaling business feels awfully similar to what we have done above in the definition of projective spaces itself. This is no coincidence. The "space" of all curves of a given degree $d \in \mathbb{Z}_{\geq 0}$ is itself a projective space $\mathbb{P}^N(K)$ for some N (which?).

Remark 1.4.10. Doing the same procedure as above for arbitrary n gives us a *hypersurface* $D = \mathbb{V}(F) \subset \mathbb{P}^n_K$ associated to any nonzero homogenous $F \in K[X_0, \ldots, X_n]$. When $n = 1$, for any field extension L/K, the set $D(L)$ is then finite and counts the number of roots–including those at infinity–of a (homogenized) single-variable equation in L (do you see why?). For example, if $K = \mathbb{Q}$ and $F = S^2 - 5ST + 5T^2$, then $D(\mathbb{Q}) = D(\mathbb{R}) = \emptyset$ but $D(\mathbb{C})$ consists of two points (c.f. 1.1.1(b)(i)).

Finally, as before we have the notions of homogenization and dehomogenization. We have that the locus $Z \neq 0$ (i.e., the complement of the line $\mathbb{V}(Z)$ at infinity) in $\mathbb{P}^2_K$ is naturally an $\mathbb{A}^2_K$:

$$\mathbb{A}^2_K = \{Z \neq 0\} \subset \mathbb{P}^2_K.$$

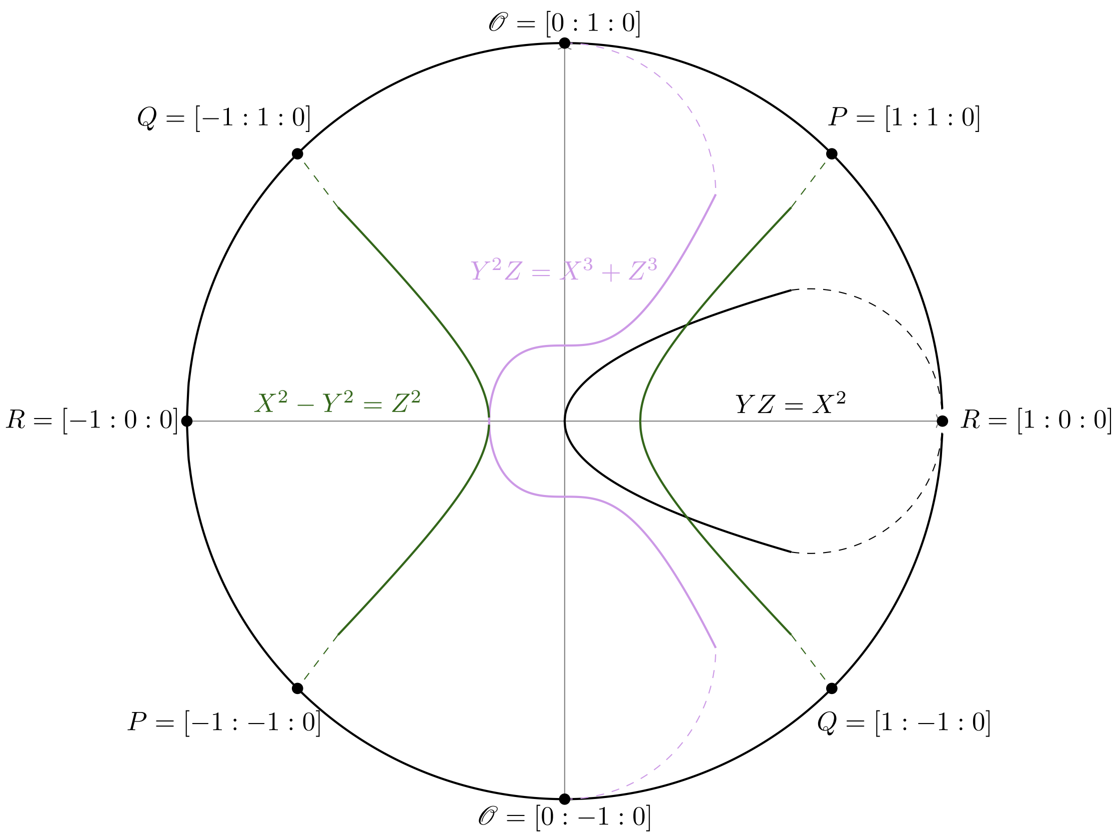


Figure 1.4.8: Some curves in the projective plane and their points at infinity.

For a projective curve $D = \mathbb{V}(F) \subset \mathbb{P}^2_K$, the intersection

$$C := D^\circ := D \cap \mathbb{A}^2_K = \mathbb{V}(f) \subset \mathbb{A}^2_K,$$

where $f(x,y) := F(x,y,1) \in K[x,y]$. This is the dehomogenization with respect to Z. Conversely, given an $f(x,y) \in K[x,y]$ of degree $d \in \mathbb{Z}_{\geq 1}$ and corresponding affine curve $C = \mathbb{V}(f)$, its *projective completion* is the projective curve

$$D := \overline{C} = \mathbb{V}(F) \subset \mathbb{P}^2_K,$$

where $F := Z^d f(X/Z, Y/Z) \in K[X,Y,Z]$. These two operations are basically inverses, but this is not literally true. For instance, we certainly have $(\overline{C})^\circ = C$ and $D \subset \overline{D^\circ}$, but equality need not hold in the latter–we may lose components corresponding to the line at infinity.

Exercise 1.4.11. What on earth does this last line mean? Are the operations of homogenization and dehomogenization inverses to each other?

However, in projective space the coordinates X, Y, Z are completely symmetrical. This flexibility of homogenization with respect to one variable, applying a linear transformation of coordinates, and then dehomogenization with respect to another yields lots of birational transformations of curves.

Example 1.4.12. Consider the affine curve $x^3 + y^3 = \alpha$ from 1.2.B. The transformation of coordinates given by

$$(x_1, y_1) := \left(\frac{12\alpha}{x+y}, \frac{36\alpha(x-y)}{x+y} \right)$$

is a birational transformation onto the curve $y_1^2 = x_1^3 - 432\alpha^2$. Of course, this arises from homogenization first to $X^3 + Y^3 = \alpha Z^3$, applying the transformation 1.2.8, and then dehomogenization with respect to Z_1. (Make sure to convince yourself that this is a birational transformation!)

Here we used a linear change of coordinates in the middle. Let's study these a little more now.

Unimportant Remark 1.4.13. Sometimes there are equations "over $\mathbb{Z}$", which can then be reduced mod p for various primes p. For instance, we could look at the curve C given by $Y^2Z = X^3 + Z^3$, or equivalently $y^2 = x^3 + 1$, and count points over $\mathbb{F}_p$ for various p. One would like to say that the original curve is defined over something like $\mathbb{Z}$ (i.e., $C \subset \mathbb{P}^2_{\mathbb{Z}}$) and these $\mathbb{F}_p$-counts are the usual counts $C(\mathbb{F}_p)$ for the "extensions $\mathbb{F}_p/\mathbb{Z}$". This is subtle, but can be made to work using the modern language of schemes. Indeed, figuring out the right definition of $\mathbb{P}^2_{\mathbb{Z}}$ (it may not be what you think!) was an impetus for the development of modern scheme theory.

$$
\begin{tikzcd}
C(\mathbb{Z}) \arrow[r, hook] \arrow[d, dashed, leftrightarrow] & C(\mathbb{Z}_p) \arrow[r] \arrow[d, dashed, leftrightarrow] & \overline{C}(\mathbb{F}_p) \arrow[r, hook] & \overline{C}(\overline{\mathbb{F}}_p) \\
C(\mathbb{Q}) \arrow[r, hook] \arrow[dr] \arrow[ddr] & C(\mathbb{Q}_p) \arrow[r] & C(\overline{\mathbb{Q}}_p) \arrow[r, hook] & C(\mathbb{C}_p) \arrow[d, leftrightarrow, "\simeq"] \\
 & C(\mathbb{R}) \arrow[rr, hook] & & C(\mathbb{C}) \\
 & C(\mathbb{Q}[\sqrt[3]{2}]) \arrow[u, hook] \arrow[r, hook] & C(\mathbb{Q}[\sqrt[3]{2}, \zeta_3]) \arrow[r, hook] & C(\overline{\mathbb{Q}}) \arrow[u, hook]
\end{tikzcd}
$$

Figure 1.4.14: A figure explaining the relationship between the K-points of a projective curve C defined over $\mathbb{Q}$ (and lifted to $\mathbb{Z}$) for various K. Some of the terminology and relationships will be explained later.

1.4.B Projective Changes of Coordinates

One advantage of working with projective spaces is the presence of lots of extra symmetries not visible in the affine plane. These come from linear changes of coordinates and look like

$$(X_1, Y_1, Z_1) = (aX + bY + cZ, dX + eY + fZ, gX + hY + iZ) \tag{1.4.15}$$

for $a, b, c, d, e, f, g, h, i \in K$. Clearly, not all choices of these "transformation coordinates" will work: you certainly can't take all of them to be zero. But, in fact the situation is a little stricter: you cannot take $(a, b, c, d, e, f, g, h, i) = (1, 0, 0, 1, 0, 0, 1, 0, 0)$ either. Similarly, but perhaps more subtly, you can't take $(a, b, c, d, e, f, g, h, i) = (2, 3, 5, -1, 2, -2, 0, 7, -1)$ either (why?). To see exactly what this constraint is, it is helpful to write this change of coordinates in the form

$$\begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}. \tag{1.4.16}$$

The idea is that if this transformation is to work, then there better not be a triple of numbers X, Y, Z, not all zero, such that $X_1 = Y_1 = Z_1 = 0$. This amounts to saying that the matrix A as above "representing this transformation" has no kernel, or equivalently (by a little linear algebra) has nonzero determinant. Conversely, if this matrix has nonzero determinant, then we get a way to invert this transformation by applying A^{-1} on the left. This is therefore the correct condition for invertibility: $\det A \neq 0$.

This matrix representation is also very helpful when we compose changes of coordinates: this just amounts to multiplying the corresponding change-of-coordinate matrices. This would lead us to believe that the group (not abelian!) of changes of coordinates in the group of invertible matrices, $\mathrm{GL}_3(K)$. But it's a little more subtle than that. For instance, the change of coordinates $(X_1, Y_1, Z_1) = (\lambda X, \lambda Y, \lambda Z)$ for any $\lambda \in K^\times$ isn't *really* a change of coordinates: it's just the identity operation of $\mathbb{P}^2_K$. More generally, two matrices A and λA for $A \in \mathrm{GL}_3(K)$ and $\lambda \in K^\times$ represent the same change of coordinates. This tells us that the group of automorphisms is actually the quotient of $\mathrm{GL}_3(K)$ by its normal subgroup $K^\times$, which is called the *projective general linear group*

$$\mathrm{PGL}_3(K) := \mathrm{GL}_3(K)/K^\times.$$

Therefore, $\mathrm{PGL}_3(K)$ acts on $\mathbb{P}^2_K$ by linear changes of coordinates. (Similarly, $\mathrm{PGL}_{n+1}(K)$ acts on $\mathbb{P}^n_K$ for any $n \geq 1$. The action of $\mathrm{PGL}_2(K)$ on $\mathbb{P}^1_K$ is by *Möbius transformations*.)

Unimportant Remark 1.4.17. It can be shown (using heavy machinery that I will not get into here) that *all* (algebraic) automorphisms of $\mathbb{P}^2_K$ (over K) come from $\mathrm{PGL}_3(K)$: these are all the possible automorphisms you can have. Note that this is not true of $\mathbb{A}^2_K$: the map $(x, y) \mapsto (x, y - x^3)$ is a non-linear change of coordinates (i.e., automorphism) of $\mathbb{A}^2_K$.

Exercise 1.4.18. When does a change of coordinates preserve that line at infinity $Z = 0$? You will have to figure out what this means. For more along these lines, see 2.2.2.

The flexibility of these changes of coordinates, therefore, comes down to linear algebra. For instance, we can take any point to any other point by a linear change of coordinates. The same holds for any pair of points. The same is *not* true of any triple of points, since linear changes of coordinates preserve incidence relations: so you can't take a collinear triple of points to a non-collinear triple of points. But this is the only constraint: any collinear triple of points maps to any other, and any non-collinear triple maps to any other, by a linear change of coordinates. Finally, because projective changes of coordinates preserve incidence relations, this tells us that we can use them to map any line to any other. Similarly, any pair of lines goes to any other. Finally, not all triples of lines go to each other for the same reason as above: there is a difference between lines that "form a triangle" and coincident lines. Proving all of this amounts to a little linear algebra. You are invited to explore things of this sort in 2.1.13.

We have seen one example of change of coordinates in 1.2.B, and will see many examples of linear transformations throughout the course, so I don't feel terribly guilty giving too many examples now. Changes of coordinates are extremely useful, but lest you think those are all possible transformations, let me give you a warning:

Remark 1.4.19. Not every "change of coordinates" we will be interested in arises via a linear change of coordinates (on a projective closure). There are definitely other (birational) changes of coordinates we will be interested in. For a concrete example, see 2.1.12.

1.5 26/06/22 - Review of Projective Geometry III: Smoothness

The next thing about curves we will need is the notion of smoothness. Before we give a formal definition, let's consider some examples.

Example 1.5.1. Which of the following curves is "smooth"? (a) $3x+5y=1$ (b) $x^2+y^2=1$ (c) $x^2-y^2=1$ (d) $x^2-y^2=0$ (e) $y^2=x^3+1$ (f) $y^2=x^3+x$ (g) $y^2=x^3+x^2$ (h) $y^2=x^3$.

Hopefully, the following definition feels reasonably motivated:

Definition 1.5.2. Let K be a field, and $f(x,y) \in K[x,y]$ be nonzero. Let $C = \mathbb{V}(f) \subset \mathbb{A}^2_K$ be the affine curve defined by f.

(a) Given a field extension L/K and a point $P=(x_0,y_0) \in C(L)$, we say that P is a *singular point* of C iff it is simultaneously a solution of

$$f = \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$$

in $\mathbb{A}^2(L)$, and a *smooth point* otherwise. When P is a smooth point, we define the *tangent line* $\mathrm{T}_P C$ to C at P to be the line

$$\mathrm{T}_P C := \mathbb{V}\left(\left.\frac{\partial f}{\partial x}\right|_P (x-x_0) + \left.\frac{\partial f}{\partial y}\right|_P (y-y_0)\right) \subset \mathbb{A}^2_L.$$

(b) The curve C is said to be *smooth*[a] iff for every field L/K, every point of $C(L)$ is smooth.

[a] Also known as *geometrically regular*.

Make sure this agrees with your intuitive definition of the tangent line to a curve at a point!

Exercise 1.5.3. Let $\ell \subset \mathbb{A}^2_K$ be an affine line. Show that for any $P \in \ell(K)$, we have that $\mathrm{T}_P\ell = \ell$.

Exercise 1.5.4. Figure out the "singular locus" of each of the curves from 1.5.1. Does the characteristic of the base field matter? What happens for the curve $x^2+y^2=0$ over $K=\mathbb{R}$?

Unimportant Remark 1.5.5. Why do we need to insist on checking all extensions L/K? We don't need to; a version of Hilbert's Nullstellensatz tells us that it suffices to check one: any algebraically closed field Ω (e.g., $\Omega = \overline{K}$) containing K (e.g., if $K=\mathbb{Q}$, it suffices to look at $\Omega=\mathbb{C}$, i.e., $\mathbb{C}$-valued points). In other words, if every point of $C(\Omega)$ is a smooth point, then C is smooth. (This needs an argument, which I will not give here. Ask me if you're interested.) Here's an example to explain why we can't just check K-points: take $K=\mathbb{F}_3(t)$ and $f(x,y) := y^2-x^3+t \in K[x,y]$. The partial derivatives here are 0 and $2y$ respectively. We see that the "singular locus" is cut out by the equations $y = x^3-t=0$, which has no K-point, but is nonetheless nonempty: it contains the L-point $(t^{1/3},0)$ where $L := \mathbb{F}_3(t^{1/3})$. Such examples are pathologies we wish to avoid by taking our definition above, which is consistent with that of modern algebraic geometry. This example will return later when we discuss singularities of plane cubics.

When dealing with projective curves, we could either work with affine patches, or give a global definition. If we take the former approach, we would then have to justify that the choice of affine patch doesn't matter. This can be done, but the global definition is probably easier:

Definition 1.5.6 (Smoothness). Let K be a field, $F \in K[X,Y,Z]$ a nonzero homogeneous polynomial, and $D = \mathbb{V}(F) \subset \mathbb{P}^2_K$ the plane curve with defining equation F.

(a) Given a field extension L/K and a point $P \in D(L)$, we say that P is a *singular point* of D iff it is simultaneously a solution of

$$F = \frac{\partial F}{\partial X} = \frac{\partial F}{\partial Y} = \frac{\partial F}{\partial Z} = 0,$$

in $\mathbb{P}^2(L)$, and a *smooth point* otherwise. When P is a smooth point, we define the *(projective) tangent line* $\mathbb{T}_P D$ *to* D *at* P to be the line

$$\mathbb{T}_P D := \mathbb{V}\left(\left.\frac{\partial F}{\partial X}\right|_P X + \left.\frac{\partial F}{\partial Y}\right|_P Y + \left.\frac{\partial F}{\partial Z}\right|_P Z \right) \subset \mathbb{P}^2_L.$$

Here the notation means: pick a representative $(X, Y, Z) \in L^3 \smallsetminus \{(0,0,0)\}$ for $P = [X : Y : Z]$, and evaluate the partials simultaneously on that representative, noting that the triple of partials only changes up to simultaneous scaling by a different choice of representative, from which it follows that the corresponding line is well-defined.

(b) The curve D is said to be *smooth*[a] iff for every field L/K, every point of $D(L)$ is smooth.

[a] Also known as *geometrically regular*.

At this point, we need to insert an exercise to make sure that these two definitions really are the same. In particular, the projective definition behaves well under taking affine patches. Because this is important enough, I have isolated this and added this to Exercise Sheet 2 (2.2.6(a)). The advantage of the projective definition is that it is highly symmetrical, and in particular stable under projective changes of coordinates (2.2.6(b)).

Unimportant Remark 1.5.7. If $F \in K[X, Y, Z]$ is homogeneous of degree $d \in \mathbb{Z}_{\geq 0}$, then *Euler's formula* tells us that

$$X\frac{\partial F}{\partial X} + Y\frac{\partial F}{\partial Y} + Z\frac{\partial F}{\partial Z} = d \cdot F.$$

(Prove this! See 2.1.11.) In particular, if $\operatorname{ch} K \nmid d$, then the locus given by the vanishing of partials is automatically contained in the vanishing locus of F. The example of 1.5.5 shows that this is not true if $\operatorname{ch} K \mid d$ in general.

Example 1.5.8. If we take $K = \mathbb{Q}$ and $F = Y^2Z - X^3$, then for any field extension $L/\mathbb{Q}$, there is a unique singular point of $D(L)$, namely $[0 : 0 : 1]$. The tangent line to D at the $\mathbb{Q}$-point $P = [1 : 1 : 1]$ is $\mathbb{V}(-3X + 2Y + Z) \subset \mathbb{P}^2_{\mathbb{Q}}$. In affine coordinates in the usual patch, this is saying that the curve $y^2 = x^3$ is singular precisely at $(0, 0)$ and has tangent line $-3x + 2y + 1 = 0$ at $(1, 1)$.

Next time, we will see how to apply these ideas in practice, and work through a few examples.

1.6 26/06/23 - Review of Projective Geometry IV: Wrapping up Smoothness, and Bézout's Theorem

Let me say a few more things about smoothness first, and let's go over a few examples. Then we'll talk about Bézout's Theorem. After this, starting next time, we'll start talking about cubic curves and group laws on them.

1.6.A Wrapping up Smoothness

Let's do a few examples.

Example 1.6.1. Let's study the example of the curve $C = \mathbb{V}(Y^2Z - X^3 - Z^3) \subset \mathbb{P}^2_K$ we saw in 1.3.12. For simplicity, we will assume that $\operatorname{ch} K \neq 2, 3$; we will figure out what happens in these characteristics later.

First, let's study what happens in the two affine patches.

(a) In the patch $Z \neq 0$ with coordinates $(x, y) = (X/Z, Y/Z)$, we have the curve $\mathbb{V}(y^2 - x^3 - 1)$. This equation has partials $-3x^2, 2y$. If $\operatorname{ch} K \neq 2, 3$, then any common solution has $x = y = 0$, but the point $(0, 0)$ doesn't lie on the curve. This proves that this patch is smooth. The tangent line at say $R = (2, 3)$ looks like $-2x + y + 1 = 0$.
(b) In the patch $Y = 0$ with coordinates $(t, s) = (X/Y, Z/Y)$, we have the curve $\mathbb{V}(s - t^3 - s^3)$. This equation has partials $-3t^2, 1 - 3s^2$. Again, there are no common solutions, and this shows that this patch is smooth as well. The tangent line at $R = (2/3, 1/3)$ then is $-2t + 1 + s = 0$.

Therefore, we have shown that the curve C is smooth, at least when $\operatorname{ch} K \neq 2, 3$.

We could do this "globally" instead of looking at affine patches. To study its singular points, for $F = Y^2Z - X^3 - Z^3$, we have partial derivatives

$$-3X^2, 2YZ, Y^2 + 3Z^2.$$

Again, since we are assuming $\operatorname{ch} K \neq 2, 3$, this forces $X = 0$, and either $Y = 0$ or $Z = 0$, and either forces the other by $Y^2 + 3Z^2 = 0$. Therefore, there are no solutions, i.e., the curve is smooth. Finally, the tangent line at $R = [2 : 3 : 1]$ is $\mathbb{T}_R(C) = \mathbb{V}(-2X + Y + Z)$.

Do you see how the "projective" and "affine computations" ((a) and (b)) relate to each other? You should now be in a good position to attempt Exercise 2.2.6.

Let's see what happens with the same curve in characteristic 3; I'll leave characteristic 2 to you below.

Example 1.6.2. Let K be a field with $\operatorname{ch} K = 3$, and $C = \mathbb{V}(Y^2Z - X^3 - Z^3) \subset \mathbb{P}^2_K$. As before, in the affine patches, we get:

(a) In the patch $Z \neq 0$, we get the singularity $(x, y) = (-1, 0)$.
(b) In the patch $Y \neq 0$, we see that there are no singularities, i.e., the patch $Y \neq 0$ is smooth.

Globally, we see that a singularity would have $Y = 0$, which forces it to be the point $[-1 : 0 : 1]$, which is indeed a singularity. Note that this is not in the affine patch $Y \neq 0$. In particular, this projective curve has a unique singular point, namely $[-1 : 0 : 1] \in C(K)$.

Exercise 1.6.3. Figure out what happens when $\operatorname{ch} K = 2$. How many singular points do we have?

As one more example, let's classify all smooth conics in the plane.

Proposition 1.6.4. Let K be a field, and $C \subset \mathbb{P}^2_K$ be a smooth conic with a K-rational point $P \in C(K)$. Then there is a projective change of coordinates taking C to $\mathbb{V}(Y^2 - XZ) \subset \mathbb{P}^2_K$. In particular, C is the (isomorphic) image of a quadratic map $\mathbb{P}^1_K \to \mathbb{P}^2_K$, i.e., can be parametrized via map of the form $[S : T] \mapsto [A(S, T) : B(S, T) : C(S, T)]$ for some homogeneous quadratic polynomials $A, B, C \in K[S, T]$ with no common roots (over an algebraic closure).

Remark 1.6.5. Note that the result of this example is true in any characteristic (including $\operatorname{ch} K = 2!$), and not true if we assume only that $C \subset \mathbb{P}^2_K$ is a smooth conic, since C might not have any K-points at all! A simple example is given by $\mathbb{V}(X^2 + Y^2 + Z^2) \subset \mathbb{P}^2_{\mathbb{R}}$, which is isomorphic to $\mathbb{V}(Y^2 - XZ)$ over $\mathbb{C}$ but not over $\mathbb{R}$.

Proof. The second statement follows from the first, since the conic $\mathbb{V}(Y^2 - XZ)$ can be parametrized as $[S : T] \mapsto [S^2 : ST : T^2]$. Therefore, we just have to figure out the right change of coordinates.

Here's the proof idea: take the point P and the tangent line L to P. Take some *other* line M through P. This M will intersect C in another K-point $Q \in C(K)$ by a baby version of Bézout's Theorem. Now take the tangent line N to C through Q. This is different than L (it doesn't contain P), and so intersects L in a third K-point R. Now take the projective change of coordinates sending (P, Q, R) to $([0 : 0 : 1], [1 : 0 : 0], [0 : 1 : 0])$, and note that this works, up to a minor rescaling of one coordinate.

Let's work this out explicitly in practice. We can certainly take a change of coordinates sending P to $[0 : 0 : 1]$ and the tangent line $\mathbb{T}_P C$ (which exists since we assumed C is smooth!) to $L = \mathbb{V}(X)$. In these coordinates, C looks like

$$aX^2 + bXY + cY^2 + dXZ + eYZ + fZ^2 = 0.$$

Since $P = [0 : 0 : 1]$ lies on C, we see that $f = 0$. Since $\mathbb{T}_P(C) = \mathbb{V}(X)$, we see that $d \neq 0$ and $e = 0$. Therefore, the equation really has the form

$$aX^2 + bXY + cY^2 + dXZ = 0. \tag{1.6.6}$$

Now take the other line M to be $M = \mathbb{V}(Y)$. To figure out its intersection with C, we set $Y = 0$ in 1.6.6 to get $X(aX + dZ) = 0$, so the other point of intersection Q is $Q = [d : 0 : -a]$. This fulfils our promise of finding a different point Q. Now you can check that $P \notin \mathbb{T}_Q C$ (do this!), and this allows us to do what we promised above: take R to be the intersection of L and $\mathbb{T}_Q(C)$, and now note that P, Q, R are non-collinear (check!), which allows us to send them to $[0 : 0 : 1], [1 : 0 : 0], [0 : 1 : 0]$. When we do this, we still get an equation of the form (1.6.6), but now since Q lies on it, we have $a = 0$. Finally, the fact that $\mathbb{T}_Q(C) = \mathbb{V}(Z)$ implies that $b = 0$ as well. This leaves us with an equation of the form $cY^2 + dXZ = 0$. Since C is smooth, neither c nor d can be zero, and hence up to rescaling Z we get an equation of the desired form. ∎

Exercise 1.6.7. Fill in the gaps in the above proof: show that $P \notin \mathbb{T}_Q C$, and that P, Q, and R are not collinear.

Note that 1.6.4 gives us another way to parametrize smooth conics with a rational point in the plane: figure out the change of coordinates explicitly, and then see how the parametrization $[S^2 : ST : T^2]$ changes under it.

Exercise 1.6.8. Apply the above technique to give a second solution to Exercise 2.1.6(a).

That's all we will say about smoothness in general for now, but it's an idea we'll keep revisiting.

1.6.B Bézout's Theorem

Let's first ask a very natural question: given a field and two curves $C, D \subset \mathbb{A}^2_K$, how many points can $C(K) \cap D(K)$ have?

Are there always finitely many points? No. A silly example is when $C = D$. For a less silly example, consider $C = \mathbb{V}(x(x^2 + y^2 - 1))$ and $D = \mathbb{V}(x(y - x^2))$. The problem, of course, is that C and D contain a common "component" of the y-axis $\mathbb{V}(x)$. We'll define components formally soon, but let's pretend we understand what they are. It is then at least believable that there can only be finitely many points of intersection.

How many? Let's consider a few examples. If we intersect $C = \mathbb{V}(x^2 + y^2 - 1)$ with the line $D = \mathbb{V}(x - 2)$. Well, there are no solutions over $\mathbb{Q}$ or $\mathbb{R}$, but we know this is a fault of these fields (c.f. 1.4.3). This suggests we work over an algebraically closed field such as $K = \mathbb{C}$, where we would expect 2 solutions. Over such a field, this example certainly works, and so does an example of the form $C = \mathbb{V}(y - f(x))$ for some polynomial $f(x) \in K[x]$ and the x-axis $D = \mathbb{V}(y)$, where we expect $\deg f$ intersection points. Similarly, we would expect a cubic and a conic to intersect in $2 \cdot 3 = 6$ points. More generally, we would expect curves of degrees d and e to intersect in de points. However, even over such a field, we could take $D = \mathbb{V}(x - 1)$, which gives us just one point $(1, 0)$. A similar example comes from taking $C = \mathbb{V}(y - f(x))$ for some polynomial $f(x) \in K[x]$ and $D = \mathbb{V}(y)$, the x-axis, when $f(x)$ has a repeated root. This suggests we should count things with multiplicites.

Even when you try to do that, sometimes you run into problems. The simplest example of this is when you take $C = \mathbb{V}(x)$ and $D = \mathbb{V}(x - 1)$. In this case, there are no points because the lines are parallel. A more subtle example is when we take $C = \mathbb{V}(y - x^2)$ and $D = \mathbb{V}(sy - x)$ for various s. Again for $s \neq 0$, there are two points of intersection $(0, 0)$ and $(1/s, 1/s^2)$; what happens as $s \to 0$ is that the second point "escapes to infinity", as in 1.3.2 . This suggests that we should look at the projective plane as before. It is a miracle that nothing else goes wrong:

Theorem 1.6.9 (Bézout)**.** Let K be a field, $C, D \subset \mathbb{P}^2_K$ be curves with no common component. Then for any algebraically closed field $\Omega \supset K$, the intersection $C(\Omega) \cap D(\Omega)$ consists of precisely $(\deg C) \cdot (\deg D)$ points, when counted with multiplicites.

This is named after the same guy, the 18$^{\text{th}}$-century French mathematician Étienne Bézout, after whom the identity involving greatest common divisors is named. The history of the proof of this result is quite intriguing, and I will refer you to the corresponding Wikipedia page to read more if you're interested. The result itself (1.6.9) is somewhat nontrivial, and belongs properly to a course on algebraic geometry. Part of the difficulty lies in coming up with a good definition of the intersection multiplicities at points, which is a subtle notion in general. We won't prove it; if you are interested in seeing a full proof, I can refer you to [5, 1.14.1] or [6, §5.3].

Since we won't prove this result, we won't use it either. The only case we will really need is when one of C and D, say D, is a line; that is the case we will prove next time. In the course of proving this result, we will see how to make each of the terms in the statement of the theorem precise.

1.7 26/06/24 - Review of Projective Geometry V: Proof of Bézout's Theorem For Intersection with a Line; The Geometry of Cubic Curves I: The Group Law

Today, we'll wrap up our discussion of projective geometry, and start talking about the geometry of cubic curves in particular.

1.7.A Proof of Bézout's Theorem for Intersection with a Line

As promised, let us now prove Bézout's Theorem for intersection with a line.

> **Theorem 1.7.1** (Special Case of 1.6.9)**.** Let K be a field, $C \subset \mathbb{P}^2_K$, and $L \subset \mathbb{P}^2_K$ be a line such that C and L don't have a common component, i.e., that $L \not\subset C$. If Ω is an algebraically closed field containing K, then C and L intersect in exactly $\deg C$ points over Ω when counted with multiplicities.

The definitions of the intersection multiplicities will turn up naturally in the proof, which you will see is very easy and natural.

Proof. Without loss of generality, we may assume that L is the X-axis $L = \mathbb{V}(Y)$. In this case, intersecting with L amounts to setting $Y = 0$ in a defining equation for C. Let $F \in K[X, Y, Z]$ be a definiting equation for C; the intersection points then correspond to homogeneous solutions $[X : 0 : Z]$ to $F(X, 0, Z) = 0$.

Since $L \not\subset C$, the polynomial $F(X, 0, Z) \in K[X, Z]$ is not identically zero. Therefore, it is a homogeneous polynomial of degree $d := \deg F = \deg C \in \mathbb{Z}_{\geq 1}$. Since Ω is algebraically closed, $F(X, 0, Z)$ splits completely into linear factors in $\Omega[X, Z]$ as

$$F(X, 0, Z) = \prod_{i=1}^{k} (\mu_i X - \lambda_i Z)^{m_i}$$

corresponding to the say $k \in \mathbb{Z}_{\geq 1}$ points of intersection $P_i = [\lambda_i : 0 : \mu_i] \in C(\Omega)$, where P_i has multiplicity $m_{P_i}(C, L) := m_i \in \mathbb{Z}_{\geq 1}$. Therefore, the number of intersection points over Ω counted with multiplicity is precisely

$$\sum_{i=1}^{k} m_i = d.$$

■

Example 1.7.2. If $K = \mathbb{Q}$ and $F(X, 0, Z) = X^6 Z - 2X^5 Z^2 - X^4 Z^3 + 4X^3 Z^4 - 2X^2 Z^5$, then this factors as $F(X, 0, Z) = X^2 Z (X - Z)^2 (X - \sqrt{2} Z)(X + \sqrt{2} Z)$ over an algebraically closed field, say $\Omega = \mathbb{C}$, and the corresponding points $[0 : 0 : 1], [1 : 0 : 0], [1 : 0 : 1], [\sqrt{2} : 0 : 1], [-\sqrt{2} : 0 : 1]$ have multiplicities $2, 1, 2, 1, 1$ respectively, adding up to $7 = \deg F(X, 0, Z)$.

That was straightforward! Actually, the above proof shows something a little stronger:

> **Corollary 1.7.3** (of the Proof of 1.7.1.)**.** In the set-up of 1.7.1, suppose that $\deg C - 1$ of the points in $C \cap L$ have coordinates in K. Then so does the last one.

Proof. This is just the projective version of the fact that a polynomial $h(x) \in K[x]$ of degree d that has $d - 1$ roots in K (counted with multiplicity) must have its last root in K as well. ■

One last thing I want to point out about the above discussion is that the nonvanishing of $F(X, 0, Z)$ is equivalent to the statement that the monomial Y does not divide F. This would be automatic if F is *irreducible* as a polynomial, of degree $\deg F \geq 2$. A plane curve with irreducible defining polynomial is said to be an *integral* curve. (This applies to both affine and projective plane curves.)

Example 1.7.4. For any field K, the curves $\mathbb{V}(X^2 - Y^2) \subset \mathbb{P}^2_K$ and $\mathbb{V}(X^2) \subset \mathbb{P}^2_K$ are not integral.

Remark 1.7.5. This notion of irreducibility of polynomials (and hence integrality of curves) depends crucially on the base field. For instance, the polynomial $X^2 + Y^2$ is irreducible in $\mathbb{R}[X, Y, Z]$ (check!), but the same polynomial is not irreducible in $\mathbb{C}[X, Y, Z]$, since it factors as $(X + iY)(X - iY)$. A polynomial $F \in K[X, Y, Z]$ is said to be *geometrically irreducible* if it remains irreducible even after replacing K by any field extension L/K. (Again, by arguments similar to those in 1.5.5, it suffices to check one algebraically closed field $L = \Omega$ containing K.) Therefore, the polynomial $X^2+Y^2 \in \mathbb{R}[X, Y, Z]$ is irreducible but not geometrically irreducible. A curve cut out by a geometrically irreducible defining polynomial is said to be *geometrically integral*. Therefore, the curve $C = \mathbb{V}(X^2 + Y^2) \subset \mathbb{P}^2_{\mathbb{R}}$ is integral but not geometrically integral.

As a fun exercise, you could try

Exercise 1.7.6. Show that the "circle" $\mathbb{V}(X^2 + Y^2 - Z^2) \subset \mathbb{P}^2_{\mathbb{R}}$ is geometrically integral.

We will show later that for any field K, the curve $\mathbb{V}(Y^2Z - X^3 - Z^3) \subset \mathbb{P}^2_K$ is geometrically integral. Let's now actually begin talking about the main subject matter for the summer: cubic curves.

1.7.B The Group Law on Points of a Cubic Curve

Let's start by talking about cubic curves. We'll focus on the smooth case for now, and leave the singular case for later.

So suppose that K is field, $C \subset \mathbb{P}^2_K$ is a smooth cubic curve. Suppose that we are given two points $P, Q \in C(K)$. We can then draw the line $L = L_{PQ}$ through them. By 1.6.9, this will intersect C in a third point, say R. (This R could be P, e.g., if L_{PQ} is tangent to C at P. It could be both P and Q if $P = Q$ is a flex!) In fact, since P and Q have coordinates in K, 1.7.3 tells us that $R \in C(K)$ as well. This gives us a way of producing a new K-point from two K-points on C. Following [2], let us temporarily denote the R obtained by such a procedure from P and Q by $P * Q$. What should we do when $P = Q$? Of course, we should take L to be the tangent line $L = \mathbb{T}_P C$, and take $R = P * P$ to be the third point of intersection.

What properties does this $*$ operation have? We would like it to be a group law, but a little experimentation shows that this doesn't work. For instance, could it have an identity element? Suppose it did, and call it $\mathscr{O} \in C(K)$. Then for $\mathscr{O} = \mathscr{O} * \mathscr{O}$ to be true, we would need $\mathscr{O}$ to be a flex point. (That's not horrible yet.) But now for every $P \in C(K)$, we must have $P = \mathscr{O} * P$, which is to say that the line joining $\mathscr{O}$ and P is tangent to C at P. This would mean that *every line through* $\mathscr{O}$ is tangent to C. It is geometrically very believable that it would be very hard to find such a point! We can show with a little effort that such a point does not exist for any smooth cubic curve C over any field K, but let's not bother with this.[3] We have concluded that we need to modify this procedure slightly. Perhaps it is believable from the above discussion, that the procedure is missing a "reflection in $\mathscr{O}$" step. Therefore, let us consider a slightly different operation given as follows.

Fix a smooth cubic $C \subset \mathbb{P}^2_K$ and a point $\mathscr{O} \in C(K)$. (The point $\mathscr{O}$ doesn't have to be a flex point to begin with, but we'll see that taking it to be a flex point does simplify some operations below. We will also see that the choice of $\mathscr{O}$ doesn't matter–at least when considering only the abstract isomorphism type of the group. See below.) Given two points $P, Q \in C(K)$, we can take the line $L = L_{PQ}$ joining them, and then take the third point of intersection $R \in C(K)$ of L with C (where again R is a K-point thanks to 1.7.3). Next, take the line $L_{R\mathscr{O}}$ joining R and $\mathscr{O}$, and take *its* third intersection point with C. Call this intersection point $P \oplus Q$. As above, if $P = Q$, then we start by taking $L = \mathbb{T}_P C$ instead, with the rest being the same. Symbolically, we have in all cases that $P \oplus Q := \mathscr{O} * (P * Q)$.

The fundamental result in the theory is then

[3] Here's a quick proof for the cognoscenti: If such a point exists, then $\pi_P : C \to \mathbb{P}^1_K$ is a purely inseparable morphism of degree 2. This forces $\operatorname{ch} K = 2$ and $g(C) = g(\mathbb{P}^1_K) = 0$ ([7, Proposition IV.2.5]), but this is not possible because $g(C) = 1$. Such "strange" points which lie on every tangent to a given curve can only exist for (lines in any characteristic and) conics in characteristic two; see [7, Theorem IV.3.9].

Theorem 1.7.7 (Fundamental Theorem of Elliptic Curve Theory)**.** For any field K, smooth cubic $C \subset \mathbb{P}^2_K$, and $\mathscr{O} \in C(K)$. The operation $P, Q \mapsto P \oplus Q$ described above makes $C(K)$ an abelian group with identity element $\mathscr{O}$. Further, for any field extension L/K, the natural inclusion $C(K) \subset C(L)$ makes $C(K)$ a subgroup of $C(L)$.

Before, we prove this, let's remark that this is really cool. We started with the set of points on a curve, and gave it an algebraic structure similar to that found on $\mathbb{Z}$ or $\mathbb{Z}/3$. Finally, we can use this additional structure to say something about the set of points on this curve. It is probably an understatement to say that this fundamental theorem is precisely the result that makes the theory of elliptic curves so rich and vast. No such phenomenon exists for (smooth plane) curves of higher degree than 3.

Unimportant Remark 1.7.8. In fact, much more is true than 1.7.7. The group operations on C really come from *algebraic operations*, and this makes C into a *group variety* (or even *abelian variety*). We're not going to use this result (or even this terminology). If you're really interested, you can find a proof of this fact in various places such as [7, IV.4.8], [8, Theorem 3.6], or [9, 19.10.3-4].

Let's try to prove 1.7.7.

Proof Sketch of 1.7.7. . Let's go through the abelian group axioms one at a time. Commutativity is clear, as is the fact that $\mathscr{O}$ is the identity. To create inverses, one could try $\ominus P := \mathscr{O} * P$, but it is easy to see this doesn't work in general. In fact, a little experimentation shows that the right thing to do is to let $\mathscr{O}' := \mathscr{O} * \mathscr{O}$, and then define $\ominus P := \mathscr{O}' * P$. (This is where we see it would be nice to take $\mathscr{O}$ to be an inflection point, because then $\mathscr{O}' = \mathscr{O}$, and the negation is simply $\ominus P = \mathscr{O} * P$. However, you should remember that this really needs $\mathscr{O}$ to be an inflection point.)

The only thing that remains to be checked is associativity. This is the hardest. Let's see what that involves. A simple geometric argument tells us that associativity is the same as the statement that if $P, Q, R \in C(K)$, then $(P \oplus Q) * R = P * (Q \oplus R)$ (draw a picture!). This step genuinely requires argument. There are three proofs of this result I know, each somewhat nontrivial. One pedestrian approach is to write everything out in coordinates and check that the algebra works. However, this is a very daunting task. The classical geometers had a better proof of this result using something known as Chasles's Theorem. This really amounts to either using some dimensional estimates on the set of cubic curves or some local properties of plane curve intersections (an exposition of this proof I gave two years ago at Ross can be found at [5, 1.15.13-14]; you can also read about in [1, §7]). Finally, the third and modern approach uses divisors on curves and the Riemann-Roch Theorem; such a proof can be found in modern books such as [8, III.3.4(e)] or [9, 19.9.3, 19.9.10]. ∎

By popular demand, I will give a brief outline of the third proof (using the Riemann-Roch Theorem) next time.

1.8 26/06/25 - The Geometry of Cubic Curves II: Proof Sketch of Associativity

Today, our goal will be to finish the proof sketch of 1.7.7. To do this, I will outline the proof of the associativity of the group law on the cubic using the circle of ideas involving divisors, genus, and the Riemann-Roch Theorem (although I will not use this language).

But before that, I want to take a second to restate the Mordell-Weil theorem in this language as

> **Theorem 1.8.1** (Mordell-Weil Theorem)**.** Let K be a number field (e.g., $K = \mathbb{Q}$). Then for any smooth cubic curve $C \subset \mathbb{P}^2_K$ with an $\mathscr{O} \in C(K)$, the abelian group $C(K)$ described in 1.7.7 is finitely generated.

Make sure you understand that this is saying the same thing as: there are finitely many "seed" points on $C(K)$ such that every other point can be obtained from these by iteration of the "chord and tangent processes". One of our goals for the summer will be to prove (a simple case of) 1.8.1.

Unimportant Remark 1.8.2. Note that the theorem 1.8.1 is specific to number fields (or more generally global fields). It is not true if we take something like $K = \mathbb{C}$ (an uncountable group has a hard time being finitely generated!), and it is obviously true over a finite field $K = \mathbb{F}_q$, since then $C(\mathbb{F}_q)$ is just finite! This theorem is less obvious but true over fields of the form $K = \mathbb{F}_q(t)$, which are also global fields.

Given this discussion, let's move on to

1.8.A Proof Sketch of Associativity

To prove associativity of the group law, we need to study what it really means to say something like $S = P \oplus Q$. Specifically, we will give an equivalent definition of the group law $\oplus$ on $C(K)$ (1.8.8)–modulo two blackboxes (1.8.11 and 1.8.14), which you can fill in later–and see that this definition is visibly associative. (The following argument is an expanded version of one of the two found in [1, §7].)

Throughout this section, we will work with the following example to see ideas in practice: let K be a field of characteristic other than 2 or 3, and let $C := \mathbb{V}(Y^2Z - X(X+Z)(X+4Z))$ with $\mathscr{O} := [0 : 1 : 0]$.

Exercise 1.8.3. Check that when $\operatorname{ch} K \neq 2, 3$, the cubic curve $C = \mathbb{V}(Y^2Z - X(X+Z)(X+4Z)) \subset \mathbb{P}^2_K$ is smooth.

We'll take the points $P = [-4 : 0 : 1]$ and $Q = [-2 : 2 : 1]$, so that $U := P * Q = [2 : 6 : 1]$ and $S := P \oplus Q = [2 : -6 : 1]$. The affine patch $Z \neq 0$ on this curve is pictured in 1.8.4.

Recall that if A is a homogeneous polynomial in $K[X, Y, Z]$, then it doesn't quite make sense to "evaluate" A at points of $\mathbb{P}^2(K)$. However, if A and B are both homogeneous polynomials of the same degree, then it at least makes sense to evaluate A/B at points of $\mathbb{P}^2(K)$ where B doesn't vanish. We've done this before already when we've taken $A = X, Y$ and $B = Z$; this is what gives us the coordinates $x = X/Z$ and $y = Y/Z$.

For another example, in the curve of 1.8.4, we could take A to be a linear form defining the line L_{PQ}, say $A := Y - X - 4Z$, and similarly B to be a linear form defining the vertical line $L_{U\mathscr{O}}$, say $B := X - 2Z$. Then the quotient would be

$$f := \frac{A}{B} = \frac{Y - X - 4Z}{X - 2Z} = \frac{y - x - 4}{x - 2} = \frac{1 - t - 4s}{t - 2s}, \tag{1.8.5}$$

where as before $(x, y) = (X/Z, Y/Z)$, and $(t, s) = (X/Y, Z/Y)$. It now does make sense to evaluate this fraction 1.8.5 away from the locus where $B = 0$. In particular, we can evaluate this function on our curve C, away from the three intersection points $\mathscr{O}, S, U$. For instance, what is its value at $(x, y) = (0, -1)$? It is $5/2$.

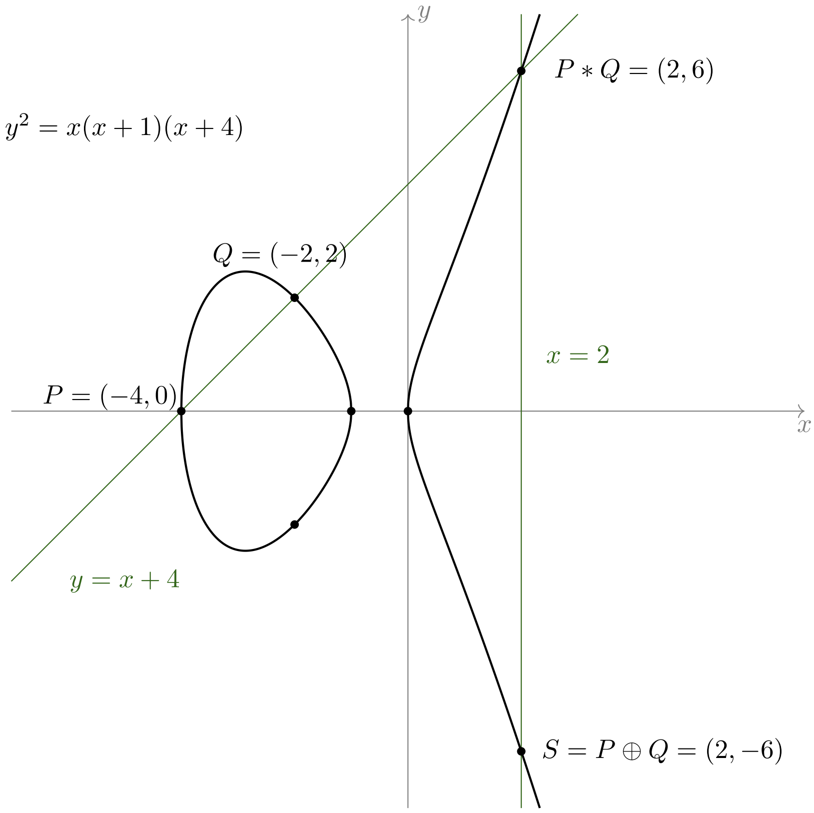


Figure 1.8.4: The $\{Z \neq 0\}$ points of the projective curve $\mathbb{V}(Y^2Z - X(X+Z)(X+4Z)) \subset \mathbb{P}^2_K$.

Where does this function vanish on our curve? Of course at the points P, Q, and U–the locus of intersection of $\mathbb{V}(A)$ and C. Where is it not defined, or where does it blow-up, or where is it infinite? Of course, at the points $\mathscr{O}, S, U$–the locus of intersection of $\mathbb{V}(B)$ and C. But wait: U belongs to both of these. What is happening there?

Exercise 1.8.6. (This exercise is for those who know a little calculus.) Show that if $K = \mathbb{R}$, then

$$\lim_{x \to 2} \frac{\sqrt{x(x+1)(x+4)} - x - 4}{x - 2}$$

exists and is nonzero by calculating it.

In fact, for any field K, the "zero" coming from A and the "pole" coming from B exactly "cancel out" at U, leaving us with no zeroes or poles at this point. Therefore, we have produced an *algebraic function* $f = A/B$ on C with the property that it has zeroes at P and Q, and poles at $\mathscr{O}$ and S.

Exercise 1.8.7. If you know what these words mean, show that the morphism $C \smallsetminus \{U\} \to \mathbb{P}^1_K$ defined by $[A : B]$ extends to give a morphism $C \to \mathbb{P}^1_K$ which maps U to a point different than 0 and ∞. (This is an instance of the "curve-to-projective" extension theorem.) If you don't know what these words mean, you may either try to interpret them or ignore this exercise.

We could take this to be a *definition* of $S = P \oplus Q$:

Definition 1.8.8. In the above set-up, given $P, Q, S \in C(K)$, we define $P \oplus Q = S$ iff there is a nonzero algebraic function f on C with zeroes at P and Q and poles at $\mathscr{O}$ and S (and no zeroes and poles elsewhere).

Of course, for this to be a definition of $P \oplus Q$, we need to do three things:

(a) we need to define what it means to be an algebraic function on the curve C, and what zeroes and poles of an algebraic function mean,

(b) we need to show that given any P and Q, such an S exists, and
(c) we need to show that given any P and Q, such an S is unique.

Let's deal with these issues one at a time.

Given an integral curve $C \subset \mathbb{P}^2_K$, we define the *field of algebraic functions* on C, denoted $K(C)$, to be the field consisting of functions of the form 1.8.5. To say more, recall what it means to say for C to be integral: it means that any defining equation $F \in K[X, Y, Z]$ for C is irreducible. Since $K[X, Y, Z]$ is a UFD, this is the same as saying that F is prime, or that $K[X, Y, Z]/(F)$ is an integral domain. Therefore, we may consider its fraction field, denoted

$$K(\mathrm{Cone}(C)) := \operatorname{Frac} K[X, Y, Z]/(F).$$

Now this field is too big and not what we're looking for: it contains things like X, which don't make sense as functions on C. (The notation is suggesting what this is: it is the field of algebraic functions on the cone over C, which is $\mathrm{Cone}(C) := \mathbb{V}(F) \subset \mathbb{A}^3_K$.) What we need is a smaller subfield $K(C) \subset K(\mathrm{Cone}(C))$, which consists of all fractions of the form $[A]/[B]$, where A and B can be chosen to be homogeneous of the same degree.

Definition 1.8.9. Let K be a field and $C \subset \mathbb{P}^2_K$ be an integral projective curve with defining equation $F \in K[X, Y, Z]$. Then we define the *algebraic function field* $K(C)$ of C to be

$$K(C) := \left\{ \frac{[A]}{[B]} : \begin{array}{l} [A], [B] \in K[X, Y, Z]/(F), \text{ such that } [B] \neq 0 \text{ and} \\ A \text{ and } B \text{ can be chosen to be homogeneous of the same degree} \end{array} \right\} \subset K(\mathrm{Cone}(C)).$$

The objects of $K(C)$ can then be thought of as genuine K-valued functions on $C(K)$, except possibly for finitely many points where they have the value "∞".

Example 1.8.10. Let $K = \mathbb{C}$ and $C = \mathbb{V}(Y^2Z - X(X + Z)(X + 4Z)) \subset \mathbb{P}^2_{\mathbb{C}}$. Then $\mathbb{C}(C)$ contains things like $x := [X]/[Z]$, $y := [Y]/[Z]$, $t := [X]/[Y]$, and $s := [Z]/[Y]$. The class $[X]/[1]$ belongs to $\mathbb{C}(\mathrm{Cone}(C))$ but not to $\mathbb{C}(C)$. The fraction

$$\frac{[X^3 - Z^3]}{[Y^2Z - X^3 - Z^3]} = \frac{x^3 - 1}{y^2 - x^3 - 1} = \frac{t^3 - s^3}{s - t^3 - s^3}$$

defines an element of $\mathbb{C}(C)$, but the fraction

$$\frac{[X^3 - Z^3]}{[Y^2Z - X(X + Z)(X + 4Z)]}$$

does not, since the denominator is zero. As elements of $\mathbb{C}(C)$, we have that

$$y^2 = x(x + 1)(x + 4) \text{ and } s = t(t + s)(t + 4s).$$

In fact,

$$\mathbb{C}(C) \cong \operatorname{Frac} \mathbb{C}[x, y]/(y^2 - x(x + 1)(x + 4)) \cong \operatorname{Frac} \mathbb{C}[t, s]/(s - t(t + s)(t + 4s)).$$

More generally, the function fields of a curve is the same as the "function field" of any of its nonempty affine patches (2.3.7).

Now we're going to need one important fact:

Fact 1.8.11. Let $C \subset \mathbb{P}^2_K$ be a smooth integral[a] curve with algebraic function field $K(C)$.

(a) For each $P \in C(K)$, we have a (discrete) valuation

$$v_P : K(C)^* \to \mathbb{Z}$$

which measures the order of vanishing of an algebraic function at P.

(b) Given an $f \in K(C)^*$ and a $P \in C(K)$, we say that P is a *zero* (resp. *pole*) of f of order $v_P(f)$ (resp. $-v_P(f)$) if $v_P(f) > 0$ (resp. $v_P(f) < 0$). A zero (resp. pole) of order one is said to be a simple zero (resp. simple pole), and similarly for double/triple zeroes/poles, etc.

(c) If we have a function f of the form $f = [A]/[B]$ where $A, B \in K[X, Y, Z]$ are homogeneous of the same degree and $[B] \neq 0$, then for any $P \in C(K)$, if $A(P) = 0$ but $B(P) \neq 0$, then P is a zero of f. Similarly, if $A(P) \neq 0$ but $B(P) = 0$, then P is a pole of f. (More care is needed if $A(P) = B(P) = 0$; c.f. 1.8.6).

(d) Moreover, if A and B are linear forms, $A(P) = 0$ but $B(P) \neq 0$, and the line $\mathbb{V}(A)$ is *not* tangent to C at P, then $v_P(f) = 1$, i.e., f has a simple zero at P. Similarly, if $A(P) \neq 0$ and $B(P) = 0$, and the line $\mathbb{V}(B)$ is not tangent to C at P, then $v_P(f) = -1$, i.e., f has a simple pole at P. More generally, if $A(P) = 0$ but $B(P) \neq 0$, and $L = \mathbb{V}(A)$, then $v_P(f) = m_P(C, L)$ is the intersection multiplicity of C and L at P as in the proof of 1.7.1.

[a] See also 2.3.8.

For a proof of the first part of 1.8.11, see [8, §II.1] and the references cited there, where v_P is denoted by ord_P. For a proof of the second part (unfortunately not using exactly the same language), see [6, Theorem 1, §3.2].

Remark 1.8.12. The operation v_P on the field $K(C)$ behaves very similarly to the p-adic valuation v_p on the field $\mathbb{Q}$ of rational numbers. This suggests that maybe we can interpret the p-adic valuation in this language. For instance, where does the "function" 12/5 have zeroes and poles? Of course, it has a "double zero" at 2, a "simple zero" at 3, and a "simple pole" at 5. On what geometric object are rational numbers rational functions?

Unimportant Remark 1.8.13. The same as in 1.8.11 can be done for other closed points $P \in C$, possibly with bigger residue fields. (E.g., this might correspond to something like the $(t^2 + 1)$-adic valuation on the field $\mathbb{F}_3(t)$.) If you have no idea what this means, you can safely ignore this bit.

At this point, we can at least make sense of the individual components of 1.8.8: given a field K, a smooth cubic $C \subset \mathbb{P}^2_K$ with $\mathscr{O} \in C(K)$, and given points $P, Q, S \in C(K)$, we say that $P \oplus Q = S$ iff there is an $f \in K(C)^\times$ such that $v_P(f) = v_Q(f) = 1$ and $v_{\mathscr{O}}(f) = v_S(f) = -1$ (when $P, Q, \mathscr{O}, S$ are all distinct; otherwise, say, if $P = Q$, then we require that $v_P(f) = 2$, etc.) and such that f has no zeroes or poles elsewhere (interpreted correctly, see 1.8.15). To show that this is a definition of $P \oplus Q$, we still need to show that such an S exists and is unique. The existence part, however, is OK: we have seen above that the point $S = P \oplus Q$ defined by our "chord and tangent process" works. (The multiplicities are automatically taken care of by various parts of 1.8.11; details omitted for now.) The uniqueness, however, needs another important fact, namely

Fact 1.8.14. Let K be a field, $C \subset \mathbb{P}^2_K$ a smooth integral curve with algebraic function field $K(C)$. If $S_1, S_2 \in C(K)$ are two distinct points, then there does not exist an $f \in K(C)^\times$ with a simple zero at S_1, a simple pole at S_2, and no zeroes or poles elsewhere, i.e., with $v_{S_1}(f) = -v_{S_2}(f) = 1$, and $v_S(f) = 0$ for all $S \neq S_1, S_2$.

Unimportant Remark 1.8.15. Here again, "elsewhere" needs to be interpreted correctly as: at no other closed points (c.f. 1.8.13). This is not a problem if we assume K is algebraically closed, which we can certainly do for the purposes of proving 1.7.7: simply pass to an algebraic closure $\overline{K}$ and prove the result there, and then note that $C(K) \subset C(\overline{K})$ forms a subgroup. The reason I wanted to not avoid this approach from the outset is that I wanted to emphasize that the function field $K(C)$ can be defined even when K is not algebraically closed; some authors (such as Silverman in [8]) decide to work almost exclusively with $\overline{K}(C)$ instead, where $\overline{K}$ denotes an algebraic closure of K.

This fact (1.8.14) is something special to plane cubic curves (or actually smooth plane curves of degree ≥ 3):

Exercise 1.8.16. Let K be a field, $C \subset \mathbb{P}^2_K$ be a line or a smooth conic. Show that for any two distinct points $S_1, S_2 \in C(K)$, there is an $f \in K(C)^\times$ with $v_{S_1}(f) = -v_{S_2}(f) = 1$, and $v_S(f) = 0$ for all $S \neq S_1, S_2$. (Hint: Use a nice coordinate system. See also 1.6.4.)

Unimportant Remark 1.8.17. For the cognoscenti, the rough idea behind the proof of 1.8.14 is as follows. If f is such a function, then f induces a morphism $C \to \mathbb{P}^1_K$ (c.f. 1.8.7). If f has the mentioned properties, then f has degree 1, and is hence necessarily an isomorphism. But C and $\mathbb{P}^1_K$ are *not* isomorphic: they have different *genera*, namely $g(C) = 1$ but $g(\mathbb{P}^1_K) = 0$. The notion of genus is easiest to understand when $K = \mathbb{C}$, where it just means "the number of holes". However, it also makes sense in positive characteristic, with one definition being: it is the integer that makes the Riemann-Roch Theorem true. For the notion of genus, see for instance [6, §8.3]. For the Riemann-Roch Theorem, see [6, §8.6]. To see how 1.8.14 follows from the Riemann-Roch Theorem, see [8, Lemma III.3.3].

At this point, uniqueness is disposed of by

Lemma 1.8.18. Let K be a field, $C \subset \mathbb{P}^2_K$ a smooth cubic curve with $\mathscr{O} \in C(K)$. Suppose we are given points $P, Q, S_1, S_2 \in C(K)$, and assume for simplicity that $P \neq Q$ and $S_1, S_2 \neq \mathscr{O}$. Suppose for $i = 1, 2$ that there are $f_i \in K(C)^\times$ such that $v_P(f_i) = v_Q(f_i) = 1$ and $v_{\mathscr{O}}(f_i) = v_{S_i}(f_i) = -1$, with both f_1 and f_2 having no zeroes or poles elswhere. Then $S_1 = S_2$.

Proof. If $S_1 \neq S_2$, then $f := f_2/f_1$ has the property that $v_{S_1}(f) = 1$ and $v_{S_2}(f) = -1$, and that $v_S(f) \neq 0$ elsewhere (1.8.15); this contradicts 1.8.14. ∎

The cases where $P = Q$ or one of the S_i coincide with $\mathscr{O}$ is similar and left to the reader:

Exercise 1.8.19. Show that the result of 1.8.18 is also true if $P = Q$ or if one of the S_i is $\mathscr{O}$. For instance, show that if $P = Q$ but $S_1, S_2 \neq \mathscr{O}$, and for $i = 1, 2$ there are $f_i \in K(C)^\times$ such that $v_P(f_i) = 2$ and $v_{\mathscr{O}}(f_i) = v_{S_i}(f_i) = -1$, with both f_1 and f_2 having no zeroes or poles elsewhere, then $S_1 = S_2$. State and prove a similar result when (a) $P \neq Q$, but one of the S_i coincides with $\mathscr{O}$, and (b) $P = Q$ and one of the S_i coincides with $\mathscr{O}$. (Hint: The same proof works.)

Unimportant Remark 1.8.20. The above exercise 1.8.19 shows that it would be very convenient to have the language of adding and subtracting zeroes and poles as abstract objects; this would allow us to give a uniform proof of 1.8.18 and the result in 1.8.19, which clearly contain the same ideas. This is indeed possible, and is achieved by introducing the notion of a *divisor* on a curve, which is basically an integer-linear combination of points on it. If you're interested in this notion, come and talk to me, or see [6, §8.1] or [8, §II.3].

Finally, at this stage, 1.8.8 makes sense: we've defined what the terms mean, and shown that such an S exists and is unique. The associativity statement now follows from

Lemma 1.8.21. Let K be a field, $C \subset \mathbb{P}^2_K$ be an (integral) smooth cubic curve, and let $\mathscr{O} \in C(K)$. Suppose we are given points $P, Q, R, T \in C(K)$, and let $S = P \oplus Q$. Suppose for simplicity that $\mathscr{O}, P, Q, R, S, T$ are all distinct. Then we have that

$$T = (P \oplus Q) \oplus R$$

if and only if there is an $h \in K(C)^\times$ which has simple zeroes at P, Q, R, a double pole at $\mathscr{O}$, and a double pole at T, and no other zeroes or poles, i.e.,

$$v_P(h) = v_Q(h) = v_R(h) = 1 \text{ and } v_{\mathscr{O}}(h) = -2 \text{ and } v_T(h) = -1,$$

and such that $v_U(h) = 0$ for all $U \neq \mathscr{O}, P, Q, R, T$.

Proof. We need to show two implications. In both cases, from the equality $S = P \oplus Q$ and 1.8.8, there is an $f \in K(C)^\times$ such that $v_P(f) = v_Q(f) = -v_{\mathscr{O}}(f) = -v_S(f) = 1$, and which has no other poles or zeroes.

(a) Suppose first that $T = (P \oplus Q) \oplus R$, and we have to produce an h with the required properties. From $T = S \oplus R$, there is a $g \in K(C)^\times$ such that $v_S(g) = v_R(g) = -v_{\mathscr{O}}(g) = -v_T(g) = 1$, and such that g has no other poles or zeroes. Then taking $h = f \cdot g$ works (check!).

(b) Conversely, suppose that there is an h with the required properties, and we have to show that $T = S \oplus R$. Well, by 1.8.8, it suffices to produce a $g \in K(C)^\times$ such that $v_S(g) = v_R(g) = -v_{\mathscr{O}}(g) = -v_T(g) = 1$, and such that g has no other poles or zeroes. But the choice $g := h/f$ works (check!).

∎

Again, edge cases are left as an

Exercise 1.8.22.

(a) Show that an appropriate version of the result of 1.8.21 also holds when some of the $\mathscr{O}, P, Q, R, S, T$ coincide.
(b) Show that associativity of $\oplus$ follows from 1.8.21 and the edge cases covered in (a).

This completes our proof sketch of the associativity of the group law on a smooth cubic with a rational point.

1.9 26/06/29 - Weierstrass Equations I

Today, we will roughly follow §1.3-1.4 of [2].

Last time, we showed that if K is a field and $C \subset \mathbb{P}^2_K$ a smooth cubic curve, then for any $\mathscr{O} \in C(K)$, the operation $\oplus$ given by $C(K) \ni P, Q \mapsto P \oplus Q := \mathscr{O} * (P * Q)$ makes $C(K)$ into an abelian group. To finish up that discussion, I only want to remark one more thing: a priori, this group structure depends on the choice of $\mathscr{O}$, so it's not clear what happens if you take two different points $\mathscr{O}_1, \mathscr{O}_2 \in C(K)$. Are the resulting group structures on $C(K)$ the same? They can't *literally* be the same if $\mathscr{O}_1 \neq \mathscr{O}_2$, since they have different identity elements. Is there a different way they are the same? This I leave you to explore in Exercise 2.3.9.

At this point, we are finally in a position to be able to define the eponymous *elliptic curves*.

1.9.A Elliptic Curves and Weierstrass Equations

Let me start by giving two different definitions of elliptic curves.

Definition 1.9.1. Let K be a field. An *elliptic curve* over K is a pair $(E, \mathscr{O})$, where E is a smooth projective geometrically integral curve of genus one, and $\mathscr{O} \in E(K)$.

Definition 1.9.2. Let K be a field. An *elliptic curve* over K is a curve $E \subset \mathbb{P}^2_K$ with equation of the form

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3, \tag{1.9.3}$$

where $a_1, a_2, a_3, a_4, a_6 \in K$ are such that $\Delta \neq 0$, where Δ is given by the formula

$$\begin{aligned}\Delta = &-a_1^6a_6 + a_1^5a_3a_4 - a_1^4a_2a_3^2 - 12a_1^4a_2a_6 + a_1^4a_4^2 + 8a_1^3a_2a_3a_4 + a_1^3a_3^3 \\ &+ 36a_1^3a_3a_6 - 8a_1^2a_2^2a_3^2 - 48a_1^2a_2^2a_6 + 8a_1^2a_2a_4^2 - 30a_1^2a_3^2a_4 + 72a_1^2a_4a_6 + 16a_1a_2^2a_3a_4 \\ &+ 36a_1a_2a_3^3 + 144a_1a_2a_3a_6 - 96a_1a_3a_4^2 - 16a_2^3a_3^2 - 64a_2^3a_6 + 16a_2^2a_4^2 + 72a_2a_3^2a_4 \\ &+ 288a_2a_4a_6 - 27a_3^4 - 216a_3^2a_6 - 64a_4^3 - 432a_6^2,\end{aligned}$$

equipped with the point $\mathscr{O} = [0:1:0] \in E(K)$.

Note that this definition (1.9.2) is correct as stated in any characteristic for K, including $\operatorname{ch} K = 2, 3$. However, this leaves many questions unanswered. For instance, where does this monstrous formula for Δ come from, and how does one come up with it? More on this soon. The form of the curve in 1.9.3 is called the (projective) Weierstrass form, named after the 19$^{\text{th}}$ century German mathematician Karl Weierstrass (also spelled "Weierstraß", notably in German), who laid the foundation for the modern theory of elliptic curves.

Given the tools we have so far, the first definition (1.9.1) doesn't even make sense, since we haven't defined the genus of a curve (but see 1.8.17). I am also lying a little bit when I say that we defined everything else in the first definition, since I have only given you the definition of smooth projective *plane* curves; but there are curves more general than these, and we will meet some of these below (1.9.5(c)). Finally, I need to explain where the convention for naming the coefficients of the defining equation of an elliptic curve comes from; this I will explain below in 1.9.15(b). But before all of this, let me mention the

Fact 1.9.4. These two definitions of elliptic curves (1.9.1 and 1.9.2) are equivalent.

We will not need this fact, and so I will not prove it here. We will sketch below (1.9.9) the proof that every object of the kind mentioned in 1.9.2 satisfies the conditions of 1.9.1 (except for the genus one bit), but going the other direction really needs the Riemann-Roch Theorem; the interested reader can find the proof in [8, Proposition III.3.1]. We will adopt 1.9.2 as our official definition. This might raise the question of why bother with the first definition (1.9.1) at all; let me try to explain this now.

The basic idea is that smooth projective geometrically integral curves do show up in lots of other contexts. Let me give you a few examples (without proof).

Example 1.9.5.

(a) If $C \subset \mathbb{P}^2_K$ is a smooth curve and there is an $\mathscr{O} \in C(K)$, then $(C, \mathscr{O})$ is an elliptic curve according to 1.9.1. Indeed, smoothness and projectivity comes from the definition, geometric integrality come from a good salvage to 2.3.8(b), and the fact that C has genus one comes from, say, the degree-genus formula for smooth plane curves (see, e.g., [9, 18.6.7.1]). The transformation from C into Weierstrass form (1.9.3) is easy to achieve when $\mathscr{O} \in C(K)$ is an inflection point (1.9.6), but it harder to achieve when $\mathscr{O}$ is not an inflection point (you can find one way to do this, due to Nagell, in [1, §8]). This allows us to put lots of curves in Weierstrass form, and then use special techniques to study curves of this sort. This is exactly what happened when we were studying the (generalized) Fermat problem for $n = 3$ in §1.2.B; we found a change of coordinates that put $x^3 + y^3 = \alpha$ (or more precisely its projective closure) into Weierstrass form to rephrase the problem. (We are yet to finish this proof.) We will also use 1.7.7 combined with our implication that 1.9.2 implies 1.9.1 to define the grofup law on an elliptic curve in Weierstrass form.

(b) Let K be a field. If a curve C is a smooth projective curve with a degree two cover of $\mathbb{P}^1_K$ branched at four points, then C has genus one, and is hence an elliptic curve in the sense of 1.9.1 if we can equip it with a K-rational point.[4] This is what enabled us to transform a curve of the form $v^2 = g(u)$ for $g(u) \in K[u]$ a quartic into Weierstrass form as in 2.1.12, giving us another way of approaching the Fermat problem for $n = 4$.

(c) Suppose we are interested in four-term sequences of integers $p < q < r < s$ such that p^2, q^2, r^2, s^2 are in arithmetic progression. (We already have enough tools to study such sequences of length three; see 2.3.4.) I hope that you agree that this is a reasonably natural question. To do this, we might look at the homogeneous equations

$$p^2 + r^2 = 2q^2 \text{ and } q^2 + s^2 = 2r^2.$$

These are homogeneous equations, and hence each cut out surfaces, say S_1 and S_2 in $\mathbb{P}^3_K$ when we equip it with homogeneous coordiantes $[p : q : r : s]$. It them turns out that the intersection $C := S_1 \cap S_2 \subset \mathbb{P}^3_{\mathbb{Q}}$ (at least in sufficiently large characteristic; I think ≥ 5 works) is a smooth projective geometrically integral curve of genus one ([9, 18.6.R]). Since it clearly has a rational point, say $[1 : -1 : -1 : 1] \in C(\mathbb{Q})$, this tells us that the curve C is an elliptic curve in the sense of 1.9.1, and so the abstract theory guarantees that we can put it in Weierstrass form 1.9.2. If we can make this transformation explicit, then this give us a way to understand four-term sequences of squares in arithmetic progression. I invite you to do this in an elementary fashion in 2.3.5.

One general cultural remark (really due to Dan Shapiro): if you're studying a new field of mathematics, then you should think about questions or theorems, the statements of which do not involve any concepts from your theory, but the "most natural" answers or proofs of which use the theory in an essential way. Further, if you are having a really hard time coming up with examples of such things, then you should carefully reevaluate why you might be interested in studying that particular theory. I hope that the Fermat examples and the example of 1.9.5(c) can provide a few such natural questions which might motivate you to study the theory of elliptic curves.

In lieu of proving 1.9.4 in full generality, let me at least tell you how to transform a smooth cubic $C \subset \mathbb{P}^2_K$ over some field K with an inflection point $\mathscr{O} \in C(K)$ into Weierstrass form by a change of coordinates.

Lemma 1.9.6. Let K be a field, $C \subset \mathbb{P}^2_K$ be a smooth cubic, and let $\mathscr{O} \in C(K)$ be an *inflection point*. Then there is a projective change of coordinates that brings C into the projective Weierstrass form (1.9.2), with $\mathscr{O}$ mapping to $[0 : 1 : 0]$.

Recall that $\mathscr{O} \in C(K)$ is said to be an *inflection point* or a *flex point* iff $\mathbb{T}_{\mathscr{O}}C$ intersects C with multiplicity 3 at $\mathscr{O}$, or equivalently iff $\mathscr{O} * \mathscr{O} = \mathscr{O}$. The following proof is very similar to that of 1.6.4, which might be helpful to review now.

[4] To be more precise, I should say: the morphism $C \to \mathbb{P}^1_K$ should be finite separable of degree two, where the ramification locus $R \subset \mathbb{P}^1_K$ is a closed subscheme of length 4 with at least one K-rational point in $R(K)$. If you have no idea what this means, you can safely ignore this remark.

Proof. Take a projective change of coordinates which sends $\mathscr{O}$ to $[0:1:0]$, and sends $\mathbb{T}_{\mathscr{O}}C$ to the line $L_\infty = \mathbb{V}(Z)$ at infinity. In this coordinates system, pick a defining equation F for C, and express it in the form

$$F = A_0Y^3 + A_1Y^2 + A_2Y + A_3,$$

where the $A_i \in K[X,Z]$ is homogeneous of degree i for $i = 0,\dots,3$. The fact that $\mathscr{O} = [0:1:0] \in C(K)$ implies that $A_0 = 0$. The fact that $\mathbb{T}_{\mathscr{O}}C = \mathbb{V}(Z)$ implies (check!) that A_1 is a nonzero scalar multiple of Z. Therefore, by rescaling F appropriately, we may assume that $A_1 = Z$. Finally, the fact that $\mathscr{O} = [0:1:0]$ is an inflection point on C implies that the tangent line $L_\infty = \mathbb{V}(Z)$ intersects C with multiplicity three at $\mathscr{O}$, which forces that $Z \mid A_2$ (check!). This means that F has the form

$$Y^2Z + (a_1X + a_3Z)YZ - (a_0X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3)$$

for some $a_0, a_1, a_2, a_3, a_4, a_6 \in K$. Can a_0 be zero? If this is the case, then $Z \mid F$, and so C is not integral; but this can't happen if C is smooth, basically by Bézout's Theorem (see 1.7.1 and 2.3.8(b)). Therefore, $a_0 \neq 0$, and we may multiply F throughout by a_0^2 and let $(X', Y') := (a_0X, a_0Y)$ to get an equation in Weierstrass form (1.9.2), up to minor relabeling of the coefficients. It remains to justify that the smoothness of C implies that $\Delta \neq 0$; this we do in a more general context below (1.9.9). ■

Exercise 1.9.7. Fill in the gaps in the above proof: check that $A_0 = 0$ follows from $\mathscr{O} \in C(K)$, check that $A_1 = Z$ (up to scaling) follows from $\mathbb{T}_{\mathscr{O}}C = \mathbb{V}(Z)$, and check that $Z \mid A_2$ follows from the fact that $\mathscr{O} \in C(K)$ is an inflection point. It may be helpful to use local coordinates $(t,s) = (X/Y, Z/Y)$ around the point $(0,0)$.

Exercise 1.9.8. Let K be a field of characteristic other than 3, and let $\alpha \in K^\times$. Show that the curve $E_\alpha \mathbb{V}(X^3 + Y^3 - \alpha Z^3) \subset \mathbb{P}^2_K$ of §1.2.B and 2.2.4 has an inflection point at $[1:-1:0] \in E_\alpha(K)$, and that the linear change of coordinates 1.2.8 is obtained exactly by the procedure described in the proof of 1.9.6, up to minor modifications (which you should figure out).

Let us now sketch the proof of the result I promised which implies that a curve in Weierstrass form (1.9.2) has the properties of 1.9.1, except for the part involving the genus, which we will not need (and a proof of which can be found at [9, 18.6.7.1]).

Proposition 1.9.9. Let K be a field (of any characteristic!) and $E \subset \mathbb{P}^2_K$ be a curve in Weierstrass form, as described in 1.9.2. Then E is smooth iff $\Delta \neq 0$. In particular, if this is the case, then E is a smooth projective geometrically integral curve of genus one, with $\mathscr{O} = [0:1:0] \in E(K)$, i.e., an elliptic curve in the sense of 1.9.1.

A few remarks are in order. Firstly, as I mentioned above, I will not prove the part involving the genus. Secondly, as before, I will invoke 2.3.8(b) to reduce to showing only that E is smooth iff $\Delta \neq 0$. I don't feel too bad about this, because we might later show that any curve in Weierstrass form (smooth or not!) is automatically geometrically integral. Thirdly, we will also show later a simpler version of this statement in characteristic other than 2 or 3. Finally, I will only give the key ideas behind the proof of 1.9.9, and leave some details to the reader. A full proof, using slightly different terminology, can be found at [8, Proposition III.1.4(a)].

Proof Sketch of 1.9.9. Let's first begin by study the points at infinity. Setting $Z = 0$ in 1.9.3 gives us $X^3 = 0$, which shows that $\mathscr{O} = [0:1:0] \in E(K)$ is the unique point at infinity on a curve in Weierstrass form. Either by using the coordinates $(t,s) = (X/Y, Z/Y)$, or by taking $\partial F/\partial Z$, it is easy to see (check!) that $\mathscr{O}$ is a smooth point with tangent line $L_\infty = \mathbb{V}(Z)$. (In fact, the fact that setting $Z = 0$ gives us $X^3 = 0$ tells us that $\mathscr{O}$ is an inflection point on $E(K)$, but we won't need this.)

Having checked the unique point at infinity for smoothness (which is true irrespective of whether $\Delta \neq 0$), we can now restrict to studying the affine equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \tag{1.9.10}$$

which corresponds to the vanishing of $f := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 \in K[x,y]$. To check smoothness, we evaluate the partial derivatives

$$\frac{\partial f}{\partial x} = a_1y - 3x^2 - 2a_2x - a_4 \text{ and } \frac{\partial f}{\partial y} = 2y + a_1x + a_3. \tag{1.9.11}$$

For simplicity, first assume that $\operatorname{ch} K \neq 2, 3$ and that $a_1 \neq 0$. Then if $\partial f/\partial y = 0$, the we can express y in terms of x as

$$y = -\left(\frac{a_1 x + a_3}{2}\right). \tag{1.9.12}$$

Plugging this into the expression for $\partial f/\partial x$ in 1.9.11 yields a quadratic equation for x, which we can solve for using the quadratic formula. Having solved for x, we can then use 1.9.12 to express y in terms x and plug both into 1.9.10 to get a huge equation involving just one huge square root. We can then move this square root to one side, take squares, and rearrange along with some very minor modifications to arrive precisely at Δ. This tells us that there is a singular point iff $\Delta = 0$. (This procedure might explain the huge size of Δ.)

This proof doesn't quite cover all cases. We still need to worry about small characteristics and whether $a_1 = 0$. For the purposes of illustration, let me do one more case: that of $\operatorname{ch} K = 2$ and $a_1 = 0$. Note that in characteristic two, the expression for Δ simplifies tremendously to

$$\Delta = a_1^6 a_6 + a_1^5 a_3 a_4 + a_1^4 a_2 a_3^2 + a_1^4 a_4^2 + a_1^3 a_3^3 + a_3^4. \tag{1.9.13}$$

In particular, if further $a_1 = 0$, then $\Delta = a_3^4$ simply. If there is a singular point, then $\partial f/\partial y = 0$ implies $a_3 = 0$ and hence $\Delta = 0$. Conversely, if $\Delta = 0$, then $\partial f/\partial y = 0$ identically. The expression for $\partial f/\partial x$ simplifies to $x^2 + a_4$, so that we may solve to get $x = a_4^{1/2}$. Finally, plugging $x = a_4^{1/2}$ into 1.9.10 yields the quadratic equation (check!)

$$y^2 + (a_1 a_4^{1/2} + a_3) y + a_2 a_4 + a_6 = 0,$$

which can certainly be solved for y (over an algebraically closed field $\Omega \supset K$!), yielding a singular point $(x, y) \in E(\Omega)$, and showing that E is not smooth.

The other cases are similar and left to the reader as an extended exercise. ■

Exercise 1.9.14. Do a few more cases to convince yourself that the result is true as claimed in all cases. Then, if you feel the need to, look up the general proof in [8, Proposition III.1.4(a)].

Remark 1.9.15.

(a) The affine equation 1.9.10 is what is often called the *Weierstrass form* of the elliptic curve. When we say "an elliptic curve with Weierstrass equation 1.9.10", what is really meant is the projective closure 1.9.3, which is an elliptic curve iff $\Delta \neq 0$. In particular, even though we'll work with the affine equation 1.9.10, it should be kept in mind that the curve $E \subset \mathbb{P}^2_K$ we are considering is a projective curve with the point $\mathscr{O} = [0 : 1 : 0] \in E(K)$ at infinity.

(b) The affine equation 1.9.10 explains the notation for the coefficients a_i: if we assign the weight 2 to x and 3 to y, then 1.9.10 becomes a homogeneous equation iff we assing weight i to a_i for $i = 1, 2, 3, 4, 6$. This also explains why there is no a_5. There are many reasons why we might want to do this; for now, it suffices to note that this will make every constant we define come with a natural weight. For instance, you can check that although Δ (given in 1.9.2) is not homogeneous in $a_1, \dots, a_6$ if we assign the weight 1 to each a_i, it *is* homogeneous of degree 12 if we assign weight i to a_i. This might be helpful to keep in mind when we define the constants c_4 and j later.

Unimportant Remark 1.9.16. The above discussion may not fully convincing about the real origins of Δ, even though it is easier to check that this statement (1.9.9) is correct. Indeed, the formula for Δ in 1.9.2 really comes from *elimination theory*, which answers the following question: what is the "best" way to eliminate the variables x and y from the system of equations (1.9.10) and (1.9.11)? This amounts to finding generators for the ideal J given as the intersection

$$J := I \cap \mathbb{Z}[a_1, a_2, a_3, a_4, a_6] \subset \mathbb{Z}[a_1, a_2, a_3, a_4, a_6], \tag{1.9.17}$$

where I is the ideal

$$I = \left(f, \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) \in \mathbb{Z}[a_1, \dots, a_6, x, y].$$

The ideal J of 1.9.17 is known as the *elimination ideal*, and it can be shown, say using the theory of Gröbner bases, that J is principal, generated precisely by Δ. (If you don't trust me–and indeed, you shouldn't be willing to take things on faith–then you can easily check this fact in software such as, say, Macaulay2. There is a good geometric reason why the ideal J should be principal; I'm not going to include this here, but if you are interested, you should ask me.) This is where Δ really comes from–it is the generator of the elimination ideal. If you want to read more about elimination theory, you can find it in [10] when working over fields and [11, Chapter 4] over more general rings such as $\mathbb{Z}$ (which is basically what Macaulay2 is actually doing).

In closing today, I wish to remark that when the characteristic of K is not small, then we may come up with changes of coordinates to simply 1.9.10 even further. For instance, if $\operatorname{ch} K \neq 2$, then we may complete the square on the left hand side of 1.9.10 by taking $x_1 = x$ and

$$y_1 = y + \frac{a_1 x + a_3}{2}$$

to get a shorter Weierstrass equation of the form

$$y_1^2 = x_1^3 + a x_1^2 + b x_1 + c. \tag{1.9.18}$$

Note that you *cannot* do something along these lines $\operatorname{ch} K = 2$, since the curve 1.9.18 is necessarily singular in characteristic two–check!. Finally, if $\operatorname{ch} K \neq 2, 3$, then we may simplify 1.9.18 even further by taking $(x_2, y_2) = (x_1 + (a/3), y_1)$ to depress the cubic, yielding an even shorter Weierstrass equation of the form

$$y_2^2 = x_2^3 + b' x_2 + c'. \tag{1.9.19}$$

Again, you may not be able to do this if $\operatorname{ch} K = 3$. We will study these shorter Weierstrass forms in more detail next time.

Remark 1.9.20. One can check that Δ remains invariant under the operations of the form that allowed to us bring curves into short Weierstrass form. This is almost miraculous. For an example of what I mean by this: if we use that substitution $(x_1, y_1) = (x, y + (a_1 x + a_3)/2)$ when $\operatorname{ch} K \neq 2$, then you can check that 1.9.10 transforms into 1.9.18, where

$$(a, b, c) = \left(a_2 + \frac{1}{4}a_1^2, a_4 + \frac{1}{2}a_1 a_3, a_6 + \frac{1}{4}a_3^2\right).$$

(Check this!) What I mean by the invariance of Δ is then that

$$\Delta(a_1, a_2, a_3, a_4, a_6) = \Delta(0, a, 0, b, c) := \Delta\left(0, a_2 + \frac{1}{4}a_1^2, 0, a_4 + \frac{1}{2}a_1 a_3, a_6 + \frac{1}{4}a_3^2\right),$$

both sides being equal to

$$\Delta = -16(4a^3 c - a^2 b^2 - 18abc + 4b^3 + 27c^2). \tag{1.9.21}$$

A similar thing happens with the substitution that depressed the cubic to be of the form 1.9.19. (Check this!) This is something special to the mysterious polynomial Δ, and obviously not true of all polynomials in the a_i. This mysterious invariance property of Δ is incredibly helpful when working with shorter Weierstrass forms 1.9.18 or 1.9.19, because then we can just use the simplified expression 1.9.21.

Unimportant Remark 1.9.22.

(a) For analogs of these shorter Weierstrass forms when $\operatorname{ch} K$ is 2 or 3, you may consult [8, Appendix A] or [12, Chapter 3].

(b) One might ask why I have bothered at all to define the longer Weierstrass form 1.9.10 at all, when we could work with the shorter forms 1.9.18 or 1.9.19, as is done, for instance, in [2]. There are a few reasons for this. One of them is that the longest Weierstrass form 1.9.10 works in *any* characteristic; in particular, it works in characteristics 2 and 3, in which the shorter ones might not. Secondly, software such as Sage is used to taking input into the form of a five-tuple $(a_1, a_2, a_3, a_4, a_6)$ produce elliptic curves, and I wanted you to understand what Sage is doing. Finally, even when working exclusively with elliptic curves over $\mathbb{Q}$, we sometimes need the long Weierstrass form 1.9.10, for instance, when finding convenient forms in which to express 2-isogenies (more on this soon!), or trying to find minimal models to study primes of bad reduction, or when giving short forms in which to express the equation of the curve when the coefficients become huge, as is needed for the curves involved in state-of-the-art records on elliptic curve ranks over $\mathbb{Q}$ (more on this later!). Even though it might be easiest to start with the shortest Weierstrass form, I think that it is not much more difficult pedagogically to start with the longer Weierstrass form 1.9.10, which allows an easier transition into topics mentioned in this remark.

1.10 26/06/30 - Weierstrass Equations II and Sage

In the first part of class today, I want to talk a little more about Weierstrass equations and changes of coordinates, and then focus on using Sage for the rest of the lecture.

1.10.A One More Thing About the Elliptic Discriminant

For simplicity now, let us focus on the case of a field K of characteristic not two. We saw last time that any elliptic curve E over K can be put into the shorter Weierstrass form

$$y^2 = x^3 + ax^2 + bx + c$$

for some a, b, c. For this to define a nonsingular curve, we need the discriminant Δ, given by the formula

$$\Delta = -16(4a^3c - a^2b^2 - 18abc + 4b^3 + 27c^2),$$

to be nonzero. In this setting of $\operatorname{ch} K \neq 2$, let us review where this formula comes from. Let $f(x) := x^3 + ax^2 + bx + c \in K[x]$, so E is defined by $y^2 = f(x)$. The singular points of E are cut out by the vanishing locus of $y^2 - f(x)$, and its partial derivatives $2y$ and $-f'(x)$. Since we are assuming $\operatorname{ch} K \neq 2$, if $2y = 0$, then $y = 0$. Therefore, having verified above that the point at infinity is not a singularity for curves in Weierstrass form, we see that E has a singular point iff the polynomials $f(x)$ and $f'(x)$ share a common root. By what you know from the Ross sets, this happens iff a polynomial $\operatorname{disc}(f)$ in the coefficients a, b, c of f vanishes, and indeed we have that

$$\Delta = 16 \cdot \operatorname{disc}(f).$$

This is essentially where Δ comes from, modulo some more care to accommodate all characteristics at once. The 16 is a reminder that the above discussion only works when $\operatorname{ch} K \neq 2$!

1.10.B Admissible Changes of Coordinates

Example 1.10.1. For simplicity, let's work over $K = \mathbb{Q}$.

(a) Consider the two elliptic curves

$$E : y^2 = x(x+1)(x+4) \text{ and } E' : (y')^2 = (x'-1)((x')^2 - 4).$$

Get Desmos to draw plots of both of these curves. What do you notice? Indeed, these curves are related by the change of coordinates $(x, y) = (x' - 2, y')$.

(b) Consider the two elliptic curves

$$E : y^2 + y = x^3 - x \text{ and } E' : (y')^2 = (x')^3 - 16x' + 16.$$

Repeating the above procedure, we see that E and E' are related by the change of coordinates

$$(x, y) = \left(\frac{1}{4}x', \frac{1}{8}y' - \frac{1}{2}\right).$$

At this point, I want to go back to working over general K (of any characteristic). We need a general procedure to understand when two elliptic curves are related by a simple change of coordinates. Suppose that K is a field, and that $E, E' \subset \mathbb{P}^2_K$ are elliptic curves in Weierstrass form with coefficients a_i and a'_i respectively. Suppose that $\varphi : \mathbb{P}^2_K \to \mathbb{P}^2_K$ is a linear change of coordinates (where we think of the first $\mathbb{P}^2_K$ has having coordinates X', Y', Z', and the second one as having coordinates X, Y, Z), taking E' to E and $\mathscr{O}' \in E(K)$ to $\mathscr{O} \in E'(K)$. Then by 2.2.6(b), φ takes $\mathbb{T}_{\mathscr{O}'}E' = \mathbb{V}(Z')$ to $\mathbb{T}_{\mathscr{O}}(E) = \mathbb{V}(Z)$, and fixes the point $[0 : 1 : 0]$. It follows from 2.2.2 that φ corresponds to a matrix in $\mathrm{PGL}_3(K)$ of the form

$$\begin{bmatrix} p & 0 & r \\ n & q & t \\ 0 & 0 & 1 \end{bmatrix}$$

for some $n, p, q, r, t \in K$ with $pq \neq 0$. In other words, the change of coordinates is given by $(X, Y, Z) = (pX' + rZ', nX' + qY' + tZ', Z')$, or in affine coordinates $(x, y) = (px' + r, nx' + qy' + t)$. Using this coordinate change to transform the equation of E into that of E', we get immediately by comparing the leading terms in a Weierstrass equation that $q^2 = p^3$. In particular, $(p, q) = (u^2, u^3)$ for some $u \in K^\times$, namely $u := q/p$. Finally, if we let $s := nu^{-2}$, then we end up with the formula

$$(x, y) = (u^2x' + r, u^3y' + u^2sx' + t) \tag{1.10.2}$$

for the change of coordinates. A change of coordinates as in 1.10.2 is said to be an *admissible change of coordinates*. Such a change of coordinates related the coefficients a_i and a_i' in a very particular way; for instance, $ua_1' = a_1 + 2s$.

Exercise 1.10.3. Check this: check that under 1.10.2, the coefficients a_1 and a_1' are related by $ua_1' = a_1 + 2s$. In particular, if $\operatorname{ch} K = 2$, then whether or not $a_1 = 0$ for a particular elliptic curve E/K is a (boolean) invariant of the curve E, invariant under admissible changes of coordinates (and hence by 1.10.6 under isomorphisms). We will revisit this idea later.

Similarly, you can work out for yourself how the rest of the coefficients a_i and a_i' are related (in terms of (u, r, s, t)) (or you can find it at [8, Table 3.1]). I want to highlight two things: that if $r = s = t = 0$, the transformation is simply $u^ia_i' = a_i$ for all $i = 1, 2, 3, 4, 6$ (c.f. 1.9.15(b)), and that the discriminants Δ and Δ' of E and E' are miraculously related by the very simple formula

$$u^{12}\Delta' = \Delta. \tag{1.10.4}$$

(You need not take my word for it, and can verify this for yourselves.) In particular, if $u = 1$, then $\Delta' = \Delta$. In particular, all the changes we did to transform curves into shorter Weierstrass forms when $\operatorname{ch} K \neq 2, 3$, which involved things like completing squares and depressing cubics, did not change Δ (c.f. 1.9.20). We summarize this as:

Proposition 1.10.5. Let K be a field. Two elliptic curves E and E' over K in Weierstrass form are the same iff they are related by an admissible change of coordinates of the form 1.10.2.

Unimportant Remark 1.10.6. Technically speaking, we haven't fully proved 1.10.5; the giveaway is the usage of the imprecise terminology of being "the same". It can be shown that any isomorphism $E \to E'$ of (abstract) elliptic curves in the sense of 1.9.1 taking $\mathscr{O}$ to $\mathscr{O}'$ is induced by an admissible change of coordinates when E and E' are put in Weierstrass form. This again uses the Riemann-Roch Theorem. We will not need this fact, and we won't prove it. The interested reader can find a proof at [8, Proposition III.3.1(b)].

1.10.C Working with Sage

The rest of the lecture was spent discussing how to work with Sage. We learned how to work with various fields in Sage, how to construct elliptic curves over various fields and study their properties, and how to work with explicit admissible changes of coordinates. This is a pain to typeset (well) in LaTeX, so I am not going to do this here, but a version of this discussion can be found in [13, Appendix A], to which I refer the interested reader.

1.11 26/07/01 - Weierstrass Equations III and the j-Invariant

Today, I want to do a few things. Firstly, I want to start by saying something I haven't yet explicitly, but which has been implicit in what we are doing.

Let K be a field and $E \subset \mathbb{P}^2_K$ be an elliptic curve in Weierstrass form (1.9.2). Since $\Delta \neq 0$, the result of 1.9.9 tells us that E is a smooth cubic curve. Equipping it with the point $\mathscr{O} = [0:1:0] \in E(K)$ at infinity, we can then apply 1.7.7 to get that the $\oplus$ operation on $E(K)$ defined there makes $E(K)$ into an abelian group with identity element $\mathscr{O}$. Let's summarize this in

Proposition 1.11.1. Let K be a field, $E \subset \mathbb{P}^2_K$ be an elliptic curve with $\mathscr{O} = [0:1:0] \in E(K)$. Then the operation $\oplus : E(K) \times E(K) \to E(K)$ given by $(P,Q) \mapsto P \oplus Q := \mathscr{O} * (P * Q)$ makes $E(K)$ into an abelian group with identity $\mathscr{O}$. If L/K is any field extension, then the inclusion $E(K) \subset E(L)$ makes $E(K)$ a subgroup of $E(L)$.

Henceforth, we will drop the notation $\oplus$ altogether, and simply refer to the group law using the usual $+$. In particular, we will write $P+Q$ for what we called $P \oplus Q$, and nP for $P + \cdots + P$, where the sum has n terms, for any $n \in \mathbb{Z}_{\geq 0}$. Finally, since an admissible change of coordinates is a linear transformation, it preserves the group law $+$, showing

Proposition 1.11.2. Let K be a field, and E and E' elliptic curves in Weierstrass form over K. An admissible change of coordinates transforming E to E' yields a group isomorphism $E(K) \to E'(K)$, or indeed $E(L) \to E(L')$ for any field extension L/K.

We will use the notation $E \cong_K E'$ to express that two curves are related by an admissible change of coordinates over K. (Note that this notion depends on K; see 1.11.9.) The above proposition is just saying that if $E \cong_K E'$, then $E(K) \cong E'(K)$ as abelian groups (sometimes denoted $E(K) \cong_{\mathsf{Ab}} E'(K)$.)

Let's see what the group operation $+$ means concretely for two points $P, Q \in E(K)$ in the affine patch $Z \neq 0$, so points on the affine curve given by $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. We draw the line L_{PQ} joining P and Q and take its third point of intersection with E, which is $P * Q$. Either $P * Q = \mathscr{O}$ (this happens iff L_{PQ} is vertical), or $P * Q$ lies in this affine patch. If it's the former, then because $\mathscr{O}$ is an inflection point, we get that $P + Q = \mathscr{O}$, i.e., $P = -Q$. If it's the latter, then we draw the line joining $P * Q$ and $\mathscr{O}$, which in this case is just the vertical line through $P * Q$, and take its other intersection point with E; this is $P + Q$. Finally, if $P = Q$, then L_{PQ} should be taken to mean the tangent line $\mathrm{T}_P E$ as usual. The figure 1.8.4 illustrates this procedure for the elliptic curve $E : y^2 = x(x+1)(x+4)$ over $K = \mathbb{Q}$ with the points $P = (-4, 0)$ and $Q = (-2, 2)$, where $P * Q = (2, 6)$ and so $P + Q = (2, -6)$.

1.11.A Automorphism Groups

We saw last time that an admissible change of coordinates over a field K is given by specifying a quadruple $(u; r, s, t)$, where $u \in K^\times$ and $r, s, t \in K$, with the transformation corresponding to the matrix

$$\begin{bmatrix} u^2 & 0 & r \\ u^2 s & u^3 & t \\ 0 & 0 & 1 \end{bmatrix}. \tag{1.11.3}$$

The set of all such transformations forms a subgroup of $\mathrm{PGL}_3(K)$, which as far as I know doesn't have a standard name, so I'm just going to call it $G(K)$.

Exercise 1.11.4. As a set, $G(K) = \{(u; r, s, t) : u \in K^\times, r, s, t \in K\}$. What is the group operation on $G(K)$ corresponding to matrix multiplication when $(u; r, s, t)$ corresponds to the matrix 1.11.3? Fill in the blanks in $(u; r, s, t) \circ (u'; r', s', t') = (uu'; \dots)$.

An admissible change of coordinates takes on elliptic curve E' into another E, but nothing is stopping us from taking $E = E'$, in which case an admissible change of coordinates corresponds to a symmetry of the elliptic curve E. Such an admissible change of coordinates is called an *automorphism* of

E, and the subgroup of $G(K)$ consisting of all such $(u;r,s,t)$ fixing E is called the *automorphism group of E over K*, denoted $\mathrm{Aut}_K(E)$ (or more precisely $\mathrm{Aut}_K(E,\mathscr{O})$, but let's not worry too much about that). This depends on K in general.

Example 1.11.5. Let $K = \mathbb{Q}$ and $E : y^2 = x^3 + 1$. Consider the following admissible changes of coordinates:

(a) the transformation $(x,y) = (x', -y')$ corresponding to $(u;r,s,t) = (-1;0,0,0)$, and
(b) the transformation $(x,y) = (\omega x', y')$ corresponding to $(u;r,s,t) = (\omega^2;0,0,0)$.

Here ω denotes a primitive cube root of unity. The first transformation is defined over $\mathbb{Q}$ and expresses a symmetry $E(\mathbb{Q}) \to E(\mathbb{Q})$, i.e., belongs to $\mathrm{Aut}_{\mathbb{Q}}(E)$. The second transformation is *not* defined over $\mathbb{Q}$, but is defined over $\Omega = \overline{\mathbb{Q}}$ or $\Omega = \mathbb{C}$ (or even the number field $L = \mathbb{Q}[\omega] = \mathbb{Q}[\sqrt{-3}]$), and expresses a symmetry $E(\Omega) \to E(\Omega)$, i.e., belongs to $\mathrm{Aut}_{\Omega}(E)$.

In general we have a diagram of subgroups of the form

$$\begin{tikzcd} \mathrm{Aut}_K(E) \arrow[r, hook] \arrow[d, hook] & G(K) \arrow[r, hook] \arrow[d, hook] & \mathrm{PGL}_3(K) \arrow[d, hook] \\ \mathrm{Aut}_{\overline{K}}(E) \arrow[r, hook] & G(\overline{K}) \arrow[r, hook] & \mathrm{PGL}_3(\overline{K}). \end{tikzcd}$$

The group $\mathrm{Aut}_{\overline{K}}(E)$ is sometimes called the *geometric* automorphism group of the elliptic curve E, but we won't use this terminology much. You are invited to explore some more automorphism groups (and geometric automorphism groups) of elliptic curves over some finite fields in 2.3.3.

1.11.B Admissible Changes For Curves in Short Weierstrass Form

Suppose now that the field K has $\mathrm{ch}\, K \neq 2$, and the elliptic curves E and E' are in the short Weierstrass form

$$E : y^2 = x^3 + ax^2 + bx + c \text{ and } E' : (y')^2 = (x')^3 + a'(x')^2 + b'x' + c'.$$

Suppose we have an admissible change of coordinates $(x,y) = (u^2x' + r, u^3y' + u^2sx' + t)$ transforming E into E'. Then a simple inspection shows that we must in fact have $s = t = 0$. In other words, any admissible change of coordinates taking a curve in short Weierstrass form to another of the same form must look like $(x,y) = (u^2x' + r, u^3y')$. That's a much easier form to work with.

Similarly, suppose now that $\mathrm{ch}\, K \neq 2,3$, and that $a = a' = 0$ as well. Then a very similar argument to the one above tells us that in fact $r = 0$ as well, i.e., the transformation looks like $(x,y) = (u^2x', u^3y')$. Therefore, if for $b,c \in K$ such that $4b^3 + 27c^2 \neq 0$, we let $E_{b,c}$ denote the elliptic curve

$$E_{b,c} : y^2 = x^3 + bx + c,$$

then for any allowable b,c,b',c', we have that $E_{,b,c} \cong_K E_{b',c'}$ iff there is a $u \in K^\times$ such that $(b,c) = (u^4b', u^6c')$.

Example 1.11.6. Let $K = \mathbb{C}$ (or indeed any field of characterstic not 2 or 3). The elliptic curves $E : y^2 = x^3 + x$ and $E' : y^2 = x^3 + 1$ over K are not isomorphic (over K, or indeed any field containing K).

This leaves the question of whether two curves can be non-isomorphic over K but isomorphic over some bigger field containing K. We'll answer this question in the next section (1.11.9).

Remark 1.11.7. When $\mathrm{ch}\, K = 3$, a careful version of the above discussion implies that whether or not $a = 0$ in the short Weierstrass form is a (boolean) invariant of the curve, i.e., does not depend on the choice of coordinates. We will revisit this idea soon.

1.11.C The j-Invariant and the Isomorphism Classification of Elliptic Curves

The previous discussion suggests that the quantity b^3/c^2 is an invariant of the isomorphism type of $E_{b,c}$. Actually, this is slightly imprecise since c could be zero without b being zero, causing this "invariant" to be undefined. Is there a linear combination of b^3 and c^2 we know is never going to vanish, and which we could use as the denominator instead? Of course there is, and it is $\Delta = -16(4b^3 + 27c^2)$. This leads us to

Definition 1.11.8 (The j-Invariant)**.** Let K be a field of characteristic other than 2 or 3. For any $b, c \in K$ with $\Delta = -16(4b^3 + 27c^2) \neq 0$, we let the j-invariant of the curve $E_{b,c}$ be

$$j(E_{b,c}) := \frac{(-48b)^3}{\Delta}.$$

At this point, we have shown that if $b, c, b', c' \in K$ are such that $\Delta\Delta' \neq 0$, and if $E_{b,c} \cong_K E_{b',c'}$, then $j(E_{b,c}) = j(E_{b',c'})$. Does the converse hold? Well, suppose that $j(E_{b,c}) = j(E_{b',c'})$; we're trying to show that there is a $u \in K^\times$ such that $(b, c) = (u^4 b', u^6 c')$. Let's split into a few cases.

(a) First suppose that $bc \neq 0$. Then $j(E_{b,c}) = j(E_{b',c'})$ forces $b'c' \neq 0$ as well (check!). From the equality $b^3/c^2 = (b')^3/(c')^2$, it follows that if let $v := (b'c)/(bc') \in K^\times$, then $(b, c) = (v^2 b', v^3 c')$. Now if there is a square root u of v in K, then this u works and we are done. Therefore, we are certainly done when we can take square roots of elements in K, which happens for instance, if K is algebraically closed. In general, however, we may not be able to do this, and this would give us an example of two curves that are isomorphic over $\overline{K}$ but not K. Such curves are called *twists* of each other. (In this special case of $bc \neq 0$, they are called *quadratic twists.*) For an explicit example, see 1.11.9(a).

(b) Next, suppose that $b \neq 0$ but $c = 0$, which happens iff $j(E_{b,c}) = 1728$ (check!). Then $j(E_{b',c'}) = j(E_{b,c}) = 1728$ forces $b' \neq 0$ and $c = 0$. Now $E_{b,0} \cong_K E_{b',0}$ iff there is a $u \in K^\times$ such that $b = u^4 b'$. Again, if a fourth root of b/b' exists in K (e.g., K algebraically closed), we are done; else we get curves which are *quartic twists* of each other (in general)–see 1.11.9(b).

(c) Finally, suppose that $b = 0$ but $c \neq 0$, which happens iff $j(E_{b,c}) = 0$ (clear!). Then $j(E_{b',c'}) = j(E_{b,c})$ forces $b' = 0$ and $c' \neq 0$. Now $E_{0,c} \cong_K E_{0,c'}$ iff there is a $u \in K^\times$ such that $c = u^6 c'$. Again, if a sixth root of c/c' exists in K (e.g., K algebraically closed), we are done; else, we get curves which are *sextic twists* of each other (in general)–see 1.11.9(c).

Example 1.11.9 (Twists)**.** Let $K = \mathbb{Q}$ for simplicity.

(a) (Quadratic Twists) The elliptic curves $E : y^2 = x^3 + x + 1$ and $E' : (y')^2 = (x')^3 + 4x' + 8$ are quadratic twists of each other: they are isomorphic over $\overline{K} = \overline{\mathbb{Q}}$ (or indeed over the smaller subfield $L = \mathbb{Q}[\sqrt{2}]$) by the change of coordinates $(x, y) = (2^{-1}x', 2^{-3/2}y')$, but they are *not* isomorphic over $\mathbb{Q}$, since there does not exist a $u \in \mathbb{Q}^\times$ such that $(1, 1) = (4u^4, 8u^6)$. Note that both curves E and E' have the same j-invariant, namely $j(E) = j(E') = 2^8 3^3 31^{-1}$.

(b) (Quartic Twists) The elliptic curves $E_i : y^2 = x^3 + 2^i x$ for $i = 0, 1, 2, 3$ are pairwise isomorphic over $\overline{K} = \overline{\mathbb{Q}}$ (or indeed over $L = \mathbb{Q}[2^{1/4}]$), but they are pairwise non-isomorphic over $K = \mathbb{Q}$. Note that all the curves E_i have the same j-invariant $j(E_i) = 1728$. For each $i = 0, 1, 2, 3$ (considered cyclically), the curves E_i and E_{i+1} are quartic twists of each other. (Note that the curve $E_4 : y^2 = x^3 + 16x$ *is* isomorphic to $E_0 : y^2 = x^3 + x$ over $K = \mathbb{Q}$! The curves $E_0 : y^2 = x^3 + x$ and $E_2 : y^2 = x^3 + 4x$ are technically *quadratic* twists of each other, since they are isomorphic over $L = \mathbb{Q}[\sqrt{2}]$ (check!); in general, the degree of a twist is the smallest degree of the field extension of K over which the resulting curves are truly isomorphic.)

(c) (Sextic Twists) The elliptic curves $E_i : y^2 = x^3 + 2^i$ for $i = 0, 1, \ldots, 5$ are pairwise isomorphic over $\overline{K} = \overline{\mathbb{Q}}$ (or indeed over $L = \mathbb{Q}[2^{1/6}]$), but they are pairwise non-isomorphic over $K = \mathbb{Q}$. Note that all the curves E_i have the same j-invariant $j(E_i) = 0$. For each $i = 0, 1, \ldots, 5$ (considered cyclically), the curves E_i and E_{i+1} are sextic twists of each other. Again, E_0 and E_6 are isomorphic over $\mathbb{Q}$, the curves E_0 and E_3 are technically quadratic twists of each other, and E_0, E_2, E_4 and E_1, E_3, E_5 are triples of *cubic twists.*

We have proven

Theorem 1.11.10. Let K be a field with $\operatorname{ch} K \neq 2, 3$. Let $b, c, b', c' \in K$ be such that $\Delta\Delta' \neq 0$, and let $E_{b,c}$ and $E_{b',c'}$ denote the corresponding elliptic curves. If $E_{b,c} \cong_K E_{b',c'}$, then $j(E_{b,c}) = j(E_{b',c'})$. The converse holds if K is algebraically closed, but not in general (1.11.9).

This is really cool: one number determines whether or not two curves are isomorphic! Suppose I hand you two elliptic curves E and E' over $\mathbb{C}$, and ask you to tell me whether they are isomorphic over $\mathbb{C}$, all you have to do is compute their j-invariants and check if they agree. If they do, these are isomorphic; if they don't, they aren't. This basically gives us a complete understanding of isomorphism classes of elliptic curves over algebraically closed fields of characteristic other than 2 or 3. (It still leaves the question of exactly which j-invariants are possible; this I leave for an exercise on the Exercise Sheet for next week.)

While we are here, we might as well prove

Theorem 1.11.11. Let K be a field with $\operatorname{ch} K \neq 2, 3$, and let $\overline{K}$ denote an algebraic closure of K. Let E be an elliptic curve over K. The size of the geometric automorphism group of E is given by

$$\#\operatorname{Aut}_{\overline{K}}(E) = \begin{cases} 2 & j \neq 0, 1728, \\ 4 & j = 1728, \\ 6 & j = 0. \end{cases}$$

Proof. Since any curve E can be put[5] into the shortest Weierstrass form $y^2 = x^3 + bx + c$, and this transformation only changes the (geometric) automorphism group up to isomorphism, we might as well assume that E is in this form. A geometric automorphism then corresponds to a $u \in \overline{K}^{\times}$ such that $(b, c) = (u^4 b, u^6 c)$. Let's split into the same cases as before:

(a) If $bc \neq 0$, which happens iff $j \neq 0, 1728$, then such a u must satisfy $u^4 = u^6 = 1$ and hence $u^2 = 1$, which implies $u \in \{\pm 1\}$. This gives us precisely two automorphisms $(x, y) \mapsto (x, \pm y)$.

(b) If $b \neq 0$ but $c = 0$, which happens iff $j = 1728$, then such a u must satisfy $u^4 = 1$, which gives us four automorphisms corresponding to $u = \pm 1, \pm i$.

(c) Finally, if $b = 0$ and $c \neq 0$, which happens iff $j = 0$, then such a u must satisfy $u^6 = 1$, which gives us six automorphisms corresponding to $u = \pm\omega^k$ for $k = 0, 1, 2$, where ω is a primitive cube root of unity.

■

Note that the statement of 1.11.11 really needs geometric automorphism groups to be true.

Example 1.11.12. Let $K = \mathbb{Q}$. The curve $E : y^2 = x^3 + x$ has $j(E) = 1728$ and geometric automorphism group $\operatorname{Aut}_{\overline{K}}(E)$ of size 4. In fact, these geometric automorphisms are given explicitly by $(x, y) \mapsto (x, \pm y)$ and $(x, y) \mapsto (-x, \pm iy)$. Of these, only the first two are defined over $\mathbb{Q}$, and hence $\operatorname{Aut}_{\mathbb{Q}}(E) \cong \mathbb{Z}/2$ of size 2.

In fact, a slight strengthening of the above argument (left to the readers) shows easily that the geometric automorphism groups when $j = 1728$ and $j = 0$ are necessarily isomorphic to $\mathbb{Z}/4$ and $\mathbb{Z}/6$.

Exercise 1.11.13. Check this!

1.11.D The Case of Characteristics 2 and 3

So far, we have stuck with the case of characteristics other than 2 or 3. The story in these characteristics is very similar. In general, we have

Definition 1.11.14. Let K be any field (of any characteristic!), and let $a_1, \ldots, a_6 \in K$ be as usual, such that $\Delta \neq 0$, so that the usual equation (1.9.10) defines an elliptic curve E over K. In this case, we define the quantity c_4 associated to the elliptic curve E by

$$c_4 := a_1^4 + 8a_1^2 a_2 - 24a_1 a_3 + 16a_2^2 - 48a_4,$$

[5] Technically, we haven't even defined the j-invariant for curves not in the shortest Weierstrass form. We will rectify this below (1.11.14); see also 1.11.16. For now, you can take this to be a statement about curves in the shortest Weierstrass form.

and the j-invariant of E to be

$$j(E) := \frac{c_4^3}{\Delta}.$$

Note how this specializes to the definition 1.11.8 when $a_1 = a_2 = a_3 = 0$. Note also that c_4 is homogeneous of degree 4 in the a_i with the usual weights, and actually similarly to Δ (1.10.4), transforms very cleanly under admissible changes of coordinates: namely via

$$u^4 c_4' = c_4. \tag{1.11.15}$$

(Again, don't take this on faith–you can check it for yourself!) These two equations (1.11.15 and 1.10.4) combined imply that $j(E)$ is actually invariant under admissible changes of coordinates, proving one half of

Theorem 1.11.16. Let K be any field (of any characteristic). Let $a_i, a_i' \in K$ for $i = 1, 2, 3, 4, 6$ be such that $\Delta\Delta' \neq 0$, and let E and E' denote the corresponding elliptic curves. If $E \cong_K E'$, then $j(E) = j(E')$. The converse holds if K is algebraically closed, but not in general.

To prove this theorem, it only remains to show that if $j(E) = j(E')$, then there is an admissible change of coordinates over $\overline{K}$ taking E to E'. The proof of this statement is very similar to what we did above in $\operatorname{ch} K \neq 2, 3$, and again when the characteristic is 2 or 3, you have to treat the cases $j = 0 = 1728$ and $j \neq 0$ separately. In characteristic 2, you also see *octic twists* showing up. I won't bother to type out the details; the interested readers can either supply the proof themselves, or can they can find it in [8, Proposition III.1.4(b)]. See also [14] for more on twisting in characteristics 2 and 3. Finally, analogously to 1.11.11, in $\operatorname{ch} K = 2, 3$ a curve with $j \neq 0$ still has geometric automorphism group $\mathbb{Z}/2$; when $\operatorname{ch} K = 3$ and $j = 0 = 1728$, you get the geometric automorphism group Dic_3 of size 12, and when $\operatorname{ch} K = 2$ and $j = 0 = 1728$, you get the geometric automorphism group $\mathrm{SL}_2(\mathbb{F}_3) \cong Q_8 \rtimes C_3$ of size 24; see [8, Theorem III.10.1 and Exercise A.1]. All I want to emphasize is that there is nothing particularly deep going on here, and the dedicated reader(s) could work these ideas out for themselves.

1.12 26/07/02 - The Name "Elliptic Curves" and Torsion Points on Elliptic Curves

I want to start today by addressing the pressing question of

1.12.A Why are elliptic curves called "elliptic" when they are not ellipses?

The story starts from the attempts by geometers going back to the 18^{th} century to find a formula for the perimeter of an ellipse. For this, suppose that we are given $a, b \in \mathbb{R}$ with $a > b > 0$, and let $C_{a,b} : (x/a)^2 + (y/b)^2 = 1$ be the ellipse with semi-major axis a and semi-minor axis b. We can parametrize the ellipse via $(x(t), y(t)) = (a \cos t, b \sin t)$, and this allows us to express the perimeter of $C_{a,b}$ as the integral

$$P = 4 \int_0^{\pi/2} \sqrt{x'(t)^2 + y'(t)^2}\, \mathrm{d}t = 4 \int_0^{\pi/2} \sqrt{a^2 \sin^2 t + b^2 \sin^2 t}\, \mathrm{d}t.$$

Pulling out the a and expressing in terms of the eccentricity $e := \sqrt{1-(b/a)^2} \in [0, 1)$ (and a minor resubstitution) gives us the integral

$$P = 4a \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t}\, \mathrm{d}t.$$

As a sanity check, if $e = 0$, then this is just $P = 2\pi a$. In general, the trigonometric substitution $u = \sin t$ gives us the expression

$$P = 4a \int_0^1 \sqrt{\frac{1-e^2u^2}{1-u^2}} \mathrm{d}u = 4a \int_0^1 \frac{1-e^2u^2}{\sqrt{(1-u^2)(1-e^2u^2)}} \mathrm{d}u.$$

This is the integral people could not figure out how to evaluate (where by evaluation we mean finding a formula in terms of elementary functions or constants). It turns out that this is for good reason–it's essentially because there is no expression for the antiderivative in terms of elementary functions. Proving this would take us very far afield, but the basic reason behind that is that if you look at the curve

$$C : v^2 = (1-u^2)(1-e^2u^2),$$

then P amounts to evaluating the integral

$$\int_\gamma \frac{1-e^2u^2}{v} \mathrm{d}u$$

for a suitable path γ on C. And the point is, that this curve C (after a suitable compatification), is a smooth projective geometrically integral genus one curve with a rational point, i.e., an elliptic curve in the sense of 1.9.2 (see 1.9.5(b)). Indeed, you don't need any fancy machinery to check that the change of coordinates

$$(x, y) = \left(\frac{\beta}{u-1}, \frac{\beta^2 v}{(u-1)^2} \right)$$

for $\beta = 1/(2(e^2 - 1))$ transforms C into a curve of the form

$$E_0 : y^2 = f(x)$$

for a suitable monic cubic $f(x) \in \mathbb{R}[x]$ with distinct roots (2.1.12), i.e., an elliptic curve. In fact, we can go one step further and put E_0 into shortest Weierstrass form to get a model of the form

$$E : y^2 = (x - 6(1+e^2))(x + 3(1 + 6e + e^2))(x + 3(1 - 6e + e^2)).$$

As you can check, this last curve E has discriminant $\Delta = 2^8 3^{12} e^2 (1-e^2)^4$, and hence *for* $e \in (0, 1)$ defines an elliptic curve. The reason there isn't an expression of P in terms of elementary functions is ultimately because of the fact that E has genus one (1.8.17). This is ultimately the reason people got interested in curves of the form $y^2 = f(x)$ for $f(x)$ a cubic, and this is why such curves are called *elliptic* curves–the name comes from *elliptic integrals*, which are so because such integrals arise naturally when you try to solve for the perimeter for an ellipse.

1.12.B Torsion in Abelian Groups

Let's quickly review the terminology surrounding torsion.

Definition 1.12.1. Let A be an abelian group, with group operation written additively.

(a) An element $a \in A$ is said to be a *torsion element* if there is an $n \in \mathbb{Z}_{\geq 1}$ such that $n \cdot a = 0$ in A; the least such n is called the *order* of a, and this case a is said to be an n-torsion element.
(b) For each $n \in \mathbb{Z}_{\geq 1}$, the subset $A[n] := \{a \in A : na = 0\}$ of elements of order dividing n is the kernel of the multiplication-by-n homomorphism $A \to A$, and is hence a subgroup of A, and is called the *n-torsion subgroup* of A. Similarly, the subset $\operatorname{Tors}(A) = \bigcup_{n \in \mathbb{Z}_{\geq 1}} A[n] \subset A$ of all torsion elements is a subgroup of A, and is called the *torsion subgroup* of A.

We can easily do other things of this sort. For instance, for any prime p, we can define the subgroup $A[p^\infty] := \bigcup_{m \geq 1} A[p^m] \subset A$ consisting of all elements annihilated by some power of p; this is known as the *p-power-torsion* or *p-primary* subgroup of A.

Example 1.12.2.

(a) We have $\operatorname{Tors} \mathbb{Z} = \{0\}$.
(b) If A is a finite group, then $\operatorname{Tors} A = A$.
(c) If $A = \mathbb{Q}/\mathbb{Z}$, then also $\operatorname{Tors} A = A$.
(d) If $A = \mathbb{Z} \oplus \mathbb{Z}/3$, then $\operatorname{Tors} A = 0 \oplus \mathbb{Z}/3 \cong \mathbb{Z}/3$. In fact, we have $A[m] = \{0\}$ if $3 \nmid m$, and $A[m] = \operatorname{Tors} A \cong \mathbb{Z}/3$ if $3 \mid m$.

The goal for us now is to study torsion points on elliptic curves. Why might we want to do that? If we look back at Bachet's example from 1.1.6, we see that Bachet's procedure is given exactly by sending $P \mapsto -2P$ in the group law on the elliptic curve $y^2 = x^3 + c$. Therefore, the sequence $\{P_0, P_1, P_2, P_3, \dots\}$ produced from a seed point $P = P_0$ is precisely $\{P, -2P, 4P, -8P, \dots.\}$. When does this sequence contain only finitely many distinct terms? You can check that this happens iff $P \in E(K)$ is a torsion point.

Exercise 1.12.3. Let A be an abelian group, and let $a \in A$. Show that the sequence of elements $\{a, -2a, 4a, -8a, \dots\}$, i.e., given by $a_k = (-2)^k a$ for $k \in \mathbb{Z}_{\geq 0}$, contains only finitely many distinct elements iff a is a torsion point, i.e., if there is an $n \in \mathbb{Z}_{\geq 1}$ such that $na = 0$.

Therefore, studying when Bachet-like procedures give rise to infinitely many points is exactly equivalent to studying torsion points on elliptic curves. If we have a good understanding of which points are torsion, and which are not, then we can use this as a tool towards studying rational points on curves.

1.12.C Two Torsion on Elliptic Curves and Supersingularity

Let K be a field, and let $\overline{K}$ be an alegbraic closure of K. Let $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ be an elliptic curve over K. Since $\mathscr{O} \in E(K)$ is an inflection point, we know from §1.7.B that to find the negative of a $P \in E(K) \smallsetminus \{\mathscr{O}\}$ say $P = (x_0, y_0)$, we just do $-P = \mathscr{O} * P$. For this, we have to draw the line $L_{\mathscr{O}P}$ joining P and $\mathscr{O}$, and take its third intersection point with E. This line is just the vertical line through P which has the equation $x = x_0$. Substituting this into the equation for E and using that $P \in E(K)$ gives us the quadratic equation for y of the form

$$y^2 + a_1x_0y + a_3y = y_0^2 + a_1x_0y_0 + a_3y_0,$$

which factors as

$$(y - y_0)(y + y_0 + a_1x_0 + a_3) = 0.$$

One root is the expected $y = y_0$ corresponding to P, and the other root is therefore

$$-P = (x_0, -y_0 - a_1x_0 - a_3). \tag{1.12.4}$$

Note that this formula is valid in all characteristics, including characteristic 2. However, if $\operatorname{ch} K \neq 2$ and we use the short Weierstrass form with $a_1 = a_3 = 0$, then the formula becomes the much easier $-P = (x_0, -y_0)$: you negate the point by negating the y-coordinate.

Now suppose we are trying to study the 2-torsion points on E, i.e., the subgroup $E[2](K) = E(K)[2]$. Note that a point $P \in E(K)$ satisfies $2P = \mathscr{O}$ iff $P = -P$. Now if $P \neq \mathscr{O}$, then $P = (x_0, y_0)$ is an affine point, and all we have to do is equate the formulae for P and $-P$ from 1.12.4, which amounts to the single equation

$$2y_0 + a_1x_0 + a_3 = 0. \tag{1.12.5}$$

As you can see from this, things proceed differently from here depending on whether $\operatorname{ch} K = 2$.

Case 1. Suppose first that $\operatorname{ch} K \neq 2$. Then if $P = (x_0, y_0) \in E[2](K)$, we get an expression for y_0 in terms of x_0 as

$$y_0 = -\frac{a_1x_0 + a_3}{2}.$$

We can plug this expression for y back into the Weierstrass equation to get a cubic equation for x, the roots of which correspond precisely to the 2-torsion points on E. The fact that $\Delta \neq 0$ actually implies that the resulting cubic polynomial is separable, i.e., has no repeated roots over an algebraic closure. This is not hard to check, but let's not bother with it in the general case, and just focus on the case of short Weierstrass equations in which $a_1 = a_3 = 0$. In this case, if $P = (x_0, y_0) \in E[2](K)$, then in fact $y_0 = 0$ and so x_0 is a root of $f(x) = x^3 + a_2x^4 + a_4x + a_6 = x^3 + ax^2 + bx + c$; we have seen in §1.10.A that nonvanishing of Δ is equivalent to the fact that $f(x)$ has three distinct roots over an algebraic closure. However, if we are only looking for roots in K, then exactly one of the following three things happens:

Case 1.a. The polynomial $f(x)$ is irreducible, and so there are no 2-torsion K-points other than $\mathscr{O}$, i.e., $E[2](K) = \{\mathscr{O}\}$. This happens, for instance, if $K = \mathbb{Q}$ and $f(x) = x^3 + 3x + 1$ (c.f. 2.2.5).

Case 1.b. The polynomial $f(x)$ factors into a linear and an irreducible quadratic factor. In this case, there is precisely one nontrivial 2-torsion K-point $T = (x_0, 0)$, where $x_0 \in K$ satisfies $f(x_0) = 0$. In this case, we have $E[2](K) = \{\mathscr{O}, T\}$, with group structure $\mathbb{Z}/2$. This happens, for instance, if $K = \mathbb{Q}$ and $f(x) = x^3 + 1 = (x + 1)(x^2 - x + 1)$.

Case 1.c. The polynomial $f(x)$ splits completely over K. In this case, there are three nontrivial 2-torsin K-points, say T_1, T_2, T_3, where for $i = 1, 2, 3$ we have $T_i = (x_i, 0)$, where the x_i are the roots of $f(x)$. In this case, $E[2](K) = \{\mathscr{O}, T_1, T_2, T_3\}$ is a group of size 4 in which every nontrivial element has order 2, and this forces $E[2](K)$ to be isomorphic to $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. This happens, for instance, if $K = \mathbb{Q}$ and $f(x) = x(x + 1)(x + 4)$.

Of course, only Case 1.c occurs if K is algebraically closed (of characteristic not two). Therefore, in all subcases of Case 1, we have that $E[2](\overline{K}) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2$.

Case 2. Finally, suppose that $\operatorname{ch} K = 2$. Here the defining equation for 2-torsion just because $a_1x_0 + a_3 = 0$, and we see that things behave very differently depending on whether or not $a_1 = 0$.

Case 2.a. Suppose that $a_1 \neq 0$. Then the only allowable x-coordinate for a 2-torsion point on E is $x_0 = -a_3/a_1 = a_3/a_1$. Plugging this back into the equation for the elliptic curve tells us that the y-coordinate must satisfy

$$y_0^2 = \frac{a_3^3 + a_1a_2a_3^2 + a_1^2a_3a_4 + a_1^3a_6}{a_1^3}. \tag{1.12.6}$$

If we're looking for roots over an algebraically closed field (or even perfect field!), then then this equation *has a unique* solution; this is because if one solution is $y_0 = \alpha$, then the equation 1.12.6 factors as $(y_0 - \alpha)^2 = 0$, since we're in characteristic two. However, if we are only looking for roots in K, then again exactly one of the following two things happens:

Case 2.a.i The unique root y_0 of 1.12.6 lies in K. In this case, there is precisely one nontrivial 2-torsion K-point $T = (-a_1^{-1}a_3, y_0)$, so $E[2](K) = \{\mathscr{O}, T\}$ with group structure $\mathbb{Z}/2$. This happens, for instance if $K = \mathbb{F}_2$ and $E : y^2 + xy = x^3 + 1$, in which $E[2](\mathbb{F}_2) = \{\mathscr{O}, (0, 1)\}$.

Case 2.a.ii The unique root y_0 of 1.12.6 in $\overline{K}$ *does not* lie in K. In this case, there are no 2-torsion K-points other than $\mathscr{O}$, i.e., $E[2](K) = \{\mathscr{O}\}$. In this case, the "remaining" 2-torsion point is not visible over K, but only over a purely inseparable extension of K. This happens, for instance, if $K = \mathbb{F}_2(t)$ and $E : y^2 + xy = x^3 + t$; the point $T = (0, t^{1/2}) \in E[2](\overline{K}) \smallsetminus \{\mathscr{O}\}$ is the other two-torsion point visible over $L = \mathbb{F}_2(t^{1/2})$, and so $E[2](L) = [2](\overline{K}) = \{\mathscr{O}, T\}$, which is isomorphic to $\mathbb{Z}/2$.

Again, if K is a perfect field (e.g., if K is finite, or if K is algebraically closed of characteristic two), then only Case 2.a.i occurs.

Case 2.b. Suppose finally that $a_1 = 0$. In this case, from the smoothness $\Delta \neq 0$ and the expression for Δ in characteristic two (1.9.13), we see immediately that $a_3 \neq 0$. In this case, there is *no solution* to $a_3 = 0$, even if K is algebraically closed. Therefore, in this case, there are no 2-torsion K-points other than $\mathscr{O}$, and $E[2](K) = E[2](\overline{K}) = \{\mathscr{O}\}$. This happens, for instance, when $K = \mathbb{F}_2$ and $E : y^2 + y = x^3$.

These six cases cover all the possibilities for what 2-torsion K-points on an elliptic curve E over a field K can look like. We summarize our discussion in

Theorem 1.12.7 (2-Torsion on Elliptic Curves)**.** Let K be a field, $\overline{K}$ an algebraic closure of K, and E/K an elliptic curve.

(a) If $\operatorname{ch} K \neq 2$, then $E[2](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/2$. The subgroup $E[2](K) \subset E[2](\overline{K})$ is either trivial, $\mathbb{Z}/2$, or $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, and all possibilities occur in general (e.g., over $K = \mathbb{Q}$).
(b) If $\operatorname{ch} K = 2$, then $E[2](\overline{K})$ is isomorphic to either $\mathbb{Z}/2$, which happens iff $a_1 \neq 0$, or trivial, which happens iff $a_1 = 0$. In the former case, $E[2](K) \subset E[2](\overline{K})$ is either trivial or $\mathbb{Z}/2$, and only the latter can happen if K is perfect (e.g., finite or algebraically closed).

In characteristic 2, the cases $a_1 \neq 0$ and $a_1 = 0$ are thus genuinely different (this was foreshadowed in 1.10.3). These have different names:

Definition 1.12.8 (Supersingularity in Characteristic Two)**.** Let K be a field of characteristic two. An elliptic curve E/K is said to be *supersingular* if the following equivalent conditions hold:

(a) In some Weierstrass equation for E, we have $a_1 = 0$.
(b) In *every* Weierstrass equation for E, we have $a_1 = 0$ (c.f. 1.10.3).
(c) For every field L/K, we have that $E[2](L) = \{\mathscr{O}\}$.
(d) If $\overline{K}$ is an algebraic closure of K, then $E[2](\overline{K}) = \{\mathscr{O}\}$.

The elliptic curve E is said to be *ordinary* otherwise.

Therefore, 1.12.7 says that if $\operatorname{ch} K = 2$ and $\overline{K}/K$ as above, then an elliptic curve E/K is ordinary iff $E[2](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/2$ and supersingular iff $E[2](\overline{K}) \cong_{\mathsf{Ab}} \{0\}$.

Example 1.12.9. Let $K = \mathbb{F}_2$.

(a) If $E : y^2 + xy = x^3 + 1$, then $E(\mathbb{F}_2) = \{\mathscr{O}, (0,1), (1,0), (1,1)\}$. This is a group of size 4 with only one 2-torsion point, and hence $E(\mathbb{F}_2) \cong_{\mathsf{Ab}} \mathbb{Z}/4$, with generators $(1,0)$ and $(1,1)$. The curve E is ordinary and $j(E) = 1$.
(b) If $E : y^2 + y = x^3$, then $E(\mathbb{F}_2) = \{\mathscr{O}, (0,0), (0,1)\}$. This is a group of size 3, and hence $E(\mathbb{F}_2) \cong_{\mathsf{Ab}} \mathbb{Z}/3$. The curve E is supersingular and $j(E) = 0$.

Unimportant Remark 1.12.10. Note that when $\operatorname{ch} K = 2$, then the formula for c_4 in 1.11.14 simplifies drastically to $c_4 = a_1^4$, and similarly the j-invariant is simply $j(E) = a_1^{12}/\Delta$. This tells us that in characteristic two, supersingularity is equivalent to the vanishing of the j-invariant. This is true also in characteristics 3 and 5, but the reason I want to de-emphasize this is because this should be thought of as an accident: this equivalence is not true for general $p > 0$. We will define supersingularity for a field of any positive characteristic p below, and 1.12.28 gives an example of supersingular elliptic curve $E/\mathbb{F}_7$ with $j(E) = -1$.

There are many other equivalent definitions of supersingularity, and we will meet some of them on the next Exercise Sheet; see also [7, §IV.4] and [8, Theorem V.3.1].

Unimportant Remark 1.12.11. Note that this is really unfortunate terminology, since an elliptic curve is always smooth and hence "nonsingular", i.e., doesn't have any singular points. (This is partly why I like to avoid the term "nonsingular" altogether.) The clashing nomenclature comes from the unrelated facts that singular points on curves behave strangely and different than other (i.e., smooth) points and that supersingular elliptic curves are very strange (super = very, singular = strange).

So far, we have avoided developing explicit formulae for the group law on an elliptic curve. However, we do really need at least a general form of these formulae when discussing three-torsion, so we might as well bite the bullet now.

1.12.D Explicit Formulae for the Group Law

Let K be any field, and let $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ be an elliptic curve over K. For $i = 1, 2, 3$, let $P_i = (x_i, y_i) \in E(K) \smallsetminus \{\mathscr{O}\}$ be points. Suppose that $P_1 + P_2 = P_3$; in particular, we are assuming that $P_1 + P_2 \neq \mathscr{O}$. In this case, we can easily find an expression relating the coordinates of the P_i.

Indeed, let $L_{P_1P_2}$ denote the line joining P_1 and P_2. (The case $P_1 = P_2$ is allowed, in which case $L_{P_1P_2} = \mathrm{T}_{P_1}E$ is the tangent line to E at P_1 as usual.) The hypothesis $P_1 + P_2 \neq \mathscr{O}$ tells us that $L_{P_1P_2}$ has an equation of the form

$$L_{P_1P_2} : y = \lambda x + \nu,$$

where

$$\lambda := \begin{cases} \dfrac{y_2 - y_1}{x_2 - x_1}, & \text{if } x_1 \neq x_2, \\ \dfrac{3x^2 + 2a_2x + a_4 - a_1y}{2y + a_1x + a_3}, & \text{if } x_1 = x_2, \end{cases}$$

and where

$$\nu := y_1 - \lambda x_1 = y_2 - \lambda x_2.$$

To intersect $L_{P_1P_2}$ with E, we plug in the equation for y into the equation for E to get a cubic equation in x of the form

$$x^3 - (\lambda^2 + a_1\lambda - a_2)x^2 + \cdots = 0.$$

The three roots of this equation are $x = x_1, x_2, x_3$, and so these are related by

$$x_1 + x_2 + x_3 = \lambda^2 + a_1\lambda - a_2.$$

In particular, if we know x_1 and x_2, we can recover x_3 as $x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2$. The y-coordinate of the third intersection point of $L_{P_1P_2}$ with E is the $\lambda x_3 + \nu$; to get y_3, we then simply apply the negation formula 1.12.4 to get $y_3 = -(\lambda x_3 + \nu) - a_1x_3 - a_3$. In this way, we have found an explicit formula for adding two points on an elliptic curve: the formula is

$$(x_3, y_3) = (\lambda^2 + a_1\lambda - a_2 - x_1 - x_2, -(\lambda x_3 + \nu) - a_1x_3 - a_3),$$

where λ, ν are as above.

In particular, we can take $P_1 = P_2$ to get (after a little algebraic simplification),

Proposition 1.12.12 (The Duplication Formula)**.** Let K be a field, $E : y^2 + a_1xy + a_3x = x^3 + a_2x^2 + a_4x + a_6$ be an elliptic curve over K, and let $P = (x, y) \in E(K)$. If $2P \neq \mathscr{O}$, then the x-coordinate of $2P$ can be expressed in terms of that of P as

$$x(2P) = \frac{x^4 - b_4x^2 - 2b_6x - b_8}{4x^3 + b_2x^2 + 2b_4x + b_6},$$

where the polynomials b_i are defined by

$$\begin{aligned} b_2 &= a_1^2 + 4a_2, \\ b_4 &= 2a_4 + a_1a_3, \\ b_6 &= a_3^2 + 4a_6, \text{ and} \\ b_8 &= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2. \end{aligned}$$

Let me emphasize that there is nothing mysterious going on here; this is just algebra. For instance, it is clear from the formula for λ when $x_1 = x_2$ that the denominator of x_3 will look like $(2y + a_1x + a_3)^2$. We can expand this out to get

$$4(y^2 + a_1xy + a_3y) + a_1^2x^2 + 2a_1a_3 + a_3^2.$$

If P lies on the elliptic curve, then we can replace the quantity $y^2 + a_1xy + a_3y$ by $x^3 + a_2x^2 + a_4x + a_6$, and the polynomials b_k essentially come from giving names to the resulting quantities. Similarly, it is clear from the general form of the equation that the numerator will be a quartic polynomial in x and y (of "total weight 8"), and to make it into a polynomial in x only, you again replace every occurence of $y^2 + a_1xy + a_3y$ by $x^3 + a_2x^2 + a_4x + a_6$.

Remark 1.12.13.

(a) Let me remark here, for future use, the polynomial equalities

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6 \tag{1.12.14}$$

and

$$4b_8 = b_2b_6 - b_4^2, \tag{1.12.15}$$

both of which from straightforward algebraic checks.

(b) If $\operatorname{ch} K \neq 2$ and we work with the shorter Weierstrass form with $a_1 = a_3 = 0$ and use the labels $(a_2, a_4, a_6) = (a, b, c)$, so the curve E has the form $y^2 = f(x) = x^3 + ax^2 + bx + c$, then the duplication formula can be expressed more succinctly as

$$x(2P) = \frac{f'(x)^2}{4f(x)} - a - 2x = \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4f(x)}. \tag{1.12.16}$$

1.12.E Three Torsion on Elliptic Curves

Let's now study three-torsion on elliptic curves. There are two perspectives we could take here: the geometric perspective, and the algebraic perspective. For the geometric perspective, note that the group law on an elliptic curve can be summarized by the

Observation 1.12.17. Let K be a field, and $E \subset \mathbb{P}^2_K$ be an elliptic curve in Weierstrass form. Then for $P, Q, R \in E(K)$, we have that $P + Q + R = \mathscr{O}$ in the group law iff P, Q, R are collinear.

In particular, applying this to $P = Q = R$ immediately gives us that $3P = \mathscr{O}$ in the group law iff P is an inflection point of E. In other words, the 3-torsion points on E are precisely the inflection points on E. This is really cool relationship between the algebraic and geometric perspectives. If we have a good way of getting all the inflection points on E, then we have a handle on 3-torsion, and vice-versa. When $\operatorname{ch} K = 0$, there is in fact a good way of getting a handle on all inflection points; this proceeds by something known as the Hessian of the curve. I will not tell this story here, but you can find it, for instance, in [6, Problem 5.23].

The algebraic perspective is much more direct and mechanical. Suppose that $P = (x_0, y_0) \in E(K) \setminus \{\mathscr{O}\}$ and that $3P = \mathscr{O}$. Then $2P = -P \neq \mathscr{O}$, and so

$$x(2P) = x(-P) = x(P).$$

Conversely, if $x(2P) = x(P)$, then the line $x = x(P)$ intersects E in the points $\mathscr{O}, P$ and $2P$. This means that $2P = \pm P$, but $2P = P$ can't be true since $P \neq \mathscr{O}$, and so we must have $2P = -P$, i.e., $3P = \mathscr{O}$. We summarize this in

Proposition 1.12.18. Let K be a field (of any characteristic), and let $E \subset \mathbb{P}^2_K$ be an elliptic curve in Weierstrass form. Let $P \in E(K) \setminus \{\mathscr{O}\}$. The following conditions are equivalent:

(a) We have $3P = \mathscr{O}$ in the group $E(K)$.
(b) The point P is an inflection point on E.
(c) We have that $x(2P) = x(P)$.

Now, the duplication formula 1.12.12 gives us an easy way to check when $x(2P) = x(P)$. Indeed, simply cross-multiplying and subtracting, we see that $x(2P) = x(P)$ is equivalent to saying that $x = x(P)$ is a root of the polynomial

$$f_3(x) := 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8. \tag{1.12.19}$$

The polynomial $f_3(x)$, whose roots correspond to the x-coordinates of the three-torsion points on E, is called the *third division polynomial* of E. To study how many roots $f_3(x)$ has, let's first see if it has any repeated roots, which can be checked by computing its discrimimant:

$$\operatorname{disc}(f_3) = -81\Delta^2.$$

You don't have to–and shouldn't–take my word on faith; you can just compute the discriminant yourself, or get a computer algebra software to do it for you. Again, different things happen when $\operatorname{ch} K \neq 3$ and when $\operatorname{ch} K = 3$.

Case 1. Suppose first that $\operatorname{ch} K \neq 3$. Then smoothness $\Delta \neq 0$ implies that $\operatorname{disc}(f_3) \neq 0$. This means that over an algebraic closure $\overline{K}$, the polynomial f_3 factors completely as

$$f_3(x) = 3\prod_{i=1}^{4}(x - \beta_i)$$

for four distinct $\beta_i \in \overline{K}$. For each $i = 1, \ldots, 4$, we can plug $x = \beta_i$ into the equation for E to get the corresponding quadratic equation

$$y^2 + (a_1\beta_i + a_3)y - (\beta_i^3 + a_2\beta_i^2 + a_4\beta_i + a_6) = 0. \tag{1.12.20}$$

Can this quadratic equation 1.12.20 for y have a repeated root? Well, if it does, then then this repeated root y_0 is also a root of its derivative

$$2y + (a_1\beta_i + a_3) = 0.$$

But this was precisely our condition (1.12.5) for (β_i, y_0) to be a 2-torsion point! Since we're assuming it's a 3-torsion point, it *cannot* be 2-torsion (since it is not $\mathscr{O}$), and this tells us that y_0 is not a repeated root of the quadratic equation 1.12.20. Therefore, for each $i = 1, \ldots, 4$, we get two different roots $y = \delta_i^{\pm}$ of this equation, giving us the 8 nontrivial 3-torsion points $(\beta_i, \delta_i^{\pm})$ for $i = 1, \ldots, 4$. In particular, $\#E[3](\overline{K}) = 9$. Up to isomorphism, there are two abelian groups of size 9, and only one of them has the property that every nontrivial element has order 3. This is sufficient to conclude that $E[3](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/3 \oplus \mathbb{Z}/3$.

Case 2. Finally, suppose that $\operatorname{ch} K = 3$. In this case, $f_3(x)$ becomes

$$f_3(x) = b_2x^3 + b_8.$$

Therefore, we see that things behave very differently depending on whether or not $b_2 = 0$.

Case 2.a. Suppose that $b_2 \neq 0$. Then the only allowable x-coordinate for a three-torsion point on E is $x = (-b_8/b_3)^{1/3} \in \overline{K}$. (Why is this?Well, if α solves $b_2\alpha^3 + b_8 = 0$, then in the equation $b_2x^3+b_8$ actually factors completely as $b_2(x-\alpha)^3 = 0$, since we are in characteristic 3.) Again, plugging this in above gives us a quadratic equation for y, which doesn't have repeated roots for the same reason as in Case 1. We conclude that in this case, $E[3](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/3$. (Again, we don't need K to be algebraically closed; we only need it to be perfect for this to be true. In particular, something like $K = \mathbb{F}_3$ works just fine.)

Case 2.b. Suppose finally that $b_2 = 0$. In this case, the formula 1.12.14 when $\operatorname{ch} K = 3$ simplifies drastically to $\Delta = b_4^3$, so smoothness $\Delta \neq 0$ implies that $b_4 \neq 0$. But now the relation (1.12.15) along with $b_2 = 0$ forces $b_8 = -b_4^2$, from which we conclude that $b_8 \neq 0$. In particular, the equation $f_3(x) = 0$ has no roots whatsoever, and we conclude that $E[3](\overline{K}) = \{\mathscr{O}\}$.

Remark 1.12.21. Suppose that $\operatorname{ch} K \neq 2, 3$, and we are working with the short Weierstrass form with $a_1 = a_3 = 0$. Then either by using the formula 1.12.16 or by plugging in $a_1 = a_3 = 0$ into 1.12.19, we get the formula

$$f_3(x) = 3x^4 + 4ax^3 + 6bx^2 + 12cx + 4ac - b^2 = 2f(x)f'(x) - f'(x)^2,$$

where $E : y^2 = f(x)$ with $f(x) = x^3 + ax^2 + bx + c$. To check if $f_3(x)$ has any repeated roots, we need to check if it shares any roots with its derivative

$$f_3'(x) = 2f(x)f'''(x) = 12f(x).$$

But any common root of f_3 and f_3', would then be a common root of f and f', which does not exist by smoothness (1.10.A). This is again enough to conclude that the polynomial $f_3(x)$ is separable, i.e., has distinct roots over an algebraic closure $\overline{K}$. This is basically the argument given in [2, Theorem 2.1]. The only disadvantage of this approach is that it does not work when $\operatorname{ch} K = 2$, whereas our proof above does.

We have proven

Theorem 1.12.22 (3-Torsion on Elliptic Curves)**.** Let K be a field, $\overline{K}$ an algebraic closure of K, and E/K an elliptic curve.

(a) If $\operatorname{ch} K \neq 3$, then $E[3](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/3 \oplus \mathbb{Z}/3$.
(b) If $\operatorname{ch} K = 3$, then $E[3](\overline{K})$ is isomorphic to either $\mathbb{Z}/3$, which happens iff $b_2 \neq 0$, or trivial, which happens iff $b_2 = 0$.

An analysis similar to 1.12.7 (but more complicated) can be done to figure out what the subgroup $E[3](K) \subset E[3](\overline{K})$ looks like in general; this is an exercise on the Exercise Sheet for next week. Finally, as in 1.12.7, we see from the proof that if K is perfect (e.g., finite) and E has $b_2 \neq 0$, then in fact $E[2](K) \cong_{\mathsf{Ab}} \mathbb{Z}/3$; we don't need to go to $\overline{K}$. Finally, we have

Definition 1.12.23 (Supersingularity in Characteristic Three)**.** Let K be a field of characteristic three. An elliptic curve E/K is said to be *supersingular* if the following equivalent conditions hold:

(a) In some Weierstrass equation for E, we have $b_2 = 0$.
(b) In *every* Weierstrass equation for E, we have $b_2 = 0$.
(c) For every field L/K, we have that $E[3](L) = \{\mathscr{O}\}$.
(d) If $\overline{K}$ is an algebraic closure of K, then $E[3](\overline{K}) = \{\mathscr{O}\}$.

The elliptic curve E is said to be *ordinary* otherwise.

Note that when $\operatorname{ch} K = 3$ and we're in short Weierstrass form with $a_1 = a_3 = 0$, then $b_2 = a_2$. In other words, a curve of the form $y^2 = x^3 + ax^2 + bx + c$ in characteristic 3 is supersingular iff $a = 0$ (c.f. 1.11.7).

Example 1.12.24. Let $K = \mathbb{F}_3$.

(a) If $E : y^2 = x^3 + x^2 + 1$, then $\Delta(E) = j(E) = -1$. It is easy to see that

$$E(\mathbb{F}_3) = \{\mathscr{O}, (0, \pm 1), (-1, \pm 1), (1, 0)\},$$

so that $\#E(\mathbb{F}_3) = 6$. Up to isomorphism, there is a unique abelian group of order six, and so we must have $E(\mathbb{F}_3) \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/3 \cong_{\mathsf{Ab}} \mathbb{Z}/6$. The points $(0, \pm 1)$ are easily seen to have order six, and hence must be the two generators. Note that $x^3 + x^2 + 1 = (x - 1)(x^2 - x - 1) \in \mathbb{F}_3[x]$; therefore, we are in Case 1.b of §1.12.C. Correspondingly,

$$E[2](\mathbb{F}_3) = \{\mathscr{O}, (1, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2,$$

whereas

$$E[2](\overline{\mathbb{F}}_3) = \{\mathscr{O}, (1, 0), (-1 \pm i, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Finally, we have that

$$E[3](\mathbb{F}_3) = E[3](\overline{\mathbb{F}}_3) = \{\mathscr{O}, (-1, \pm 1)\}.$$

The curve E is therefore ordinary.

(b) If $E : y^2 = x^3 + x$, then $\Delta(E) = -1$ and $j(E) = 0$. It is easy to see that

$$E(\mathbb{F}_3) = \{\mathscr{O}, (0, 0), (-1, \pm 1)\},$$

so that $\#E(\mathbb{F}_3) = 4$. Since $x^3 + x = x(x^2 + 1) \in \mathbb{F}_3[x]$, we are in Case 1.b of §1.12.C. Correspondingly,

$$E[2](\mathbb{F}_3) = \{\mathscr{O}, (0, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2.$$

This is enough to conclude that $E(\mathbb{F}_3) \cong_{\mathsf{Ab}} \mathbb{Z}/4$, and also

$$E[2](\overline{\mathbb{F}}_3) = \{\mathscr{O}, (0,0), (\pm i, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Finally, we see that

$$E[3](\mathbb{F}_3) = E[3](\overline{\mathbb{F}}_3) = \{\mathscr{O}\},$$

i.e., the curve E is supersingular.

Unimportant Remark 1.12.25. Note that when $\operatorname{ch} K = 3$, the formula for c_4 in 1.11.14 simplifies drastically to $c_4 = b_2^2$ (check!), and similarly the j-invariant is simply $j(E) = b_2^6/\Delta$. This tells us that in characteristic three, supersingularity is again equivalent to the vanishing of the j-invariant. See 1.12.10.

1.12.F Supersingularity in General

At this point, based on what we have seen already, the following results shouldn't be too surprising.

Theorem 1.12.26. Let K be a field of characterstic zero, and $\overline{K}$ an algebraic closure of K. Then for any elliptic curve E/K and $n \in \mathbb{Z}_{\geq 1}$, we have that

$$E[n](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/n \oplus \mathbb{Z}/n.$$

Theorem 1.12.27. Let K be a field of characteristic $p > 0$, and $\overline{K}$ be an algebraic closure of K. Then for any given elliptic curve E/K, exactly one of the following holds:

(a) For all $n \in \mathbb{Z}_{\geq 1}$, if we write $n = p^r m$ for $r, m \in \mathbb{Z}_{\geq 0}$ with $p \nmid m$, then

$$E[n](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/m \oplus \mathbb{Z}/m \oplus \mathbb{Z}/p^r.$$

(b) For all $n \in \mathbb{Z}_{\geq 1}$, if we write $n = p^r m$ for $r, m \in \mathbb{Z}_{\geq 0}$ with $p \nmid m$, then

$$E[n](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/m \oplus \mathbb{Z}/m.$$

Curves satisfying (a) are known as *ordinary*, and curves satisfying (b) are called *supersingular*.

Note that in either case, if $\operatorname{ch} K \nmid n$, then we always have $E[n](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/n \oplus \mathbb{Z}/n$; the difference comes when trying to analyze p-power torsion. For instance, in the ordinary case for any $r \geq 0$ we have $E[p^r](\overline{K}) \cong_{\mathsf{Ab}} \mathbb{Z}/p^r$, whereas in the supersingular case, we have $E[p^r](\overline{K}) \cong_{\mathsf{Ab}} \{0\}$. We won't prove these results here, although one proof of a special case can be found in (the double-starred exercise) 2.2.10. The general algebraic proof of these results can be given by studying the *invariant differential* on elliptic curves; material along these lines can be found in [8]. However, the *real* reason for why we should expect results of this sort to be true (I think) comes from the picture over $K = \mathbb{C}$, which we will see next time. We end with an

Example 1.12.28. Let $K = \mathbb{F}_7$, and consider the elliptic curve $E : y^2 = x^3 - x$. This has $\Delta(E) = 1$ and $j(E) = -1$. An easy check shows that

$$E(\mathbb{F}_7) = \{\mathscr{O}, (0,0), (\pm 1, 0), (-2, \pm 1), (-3, \pm 2)\}.$$

In particular, $\#E(\mathbb{F}_7) = 8$. Now $x^3 - x = x(x+1)(x-1) \in \mathbb{F}_7[x]$ tells us that we are in Case 1.c of §1.12.C, and hence

$$E[2](\mathbb{F}_7) = \{\mathscr{O}, (0,0), (\pm 1, 0)\} \cong_{\mathsf{Ab}} \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

This is enough to conclude (check!) that in fact $E(\mathbb{F}_7) \cong_{\mathsf{Ab}} \mathbb{Z}/4 \oplus \mathbb{Z}/2$, with the four points $(-2, \pm 1)$ and $(-3, \pm 2)$ each having order 4. Next, we note that $f_3(x) = 3x^4 + x^2 - 1 \in \mathbb{F}_7[x]$, which factors as

$$f_3(x) = 3(x^2 + x + 3)(x^2 - x + 3) \in \mathbb{F}_7[x].$$

In particular, $E[3](\mathbb{F}_7) = \{\mathscr{O}\}$ (we knew this already), but over $\mathbb{F}_{2401} \cong \mathbb{F}_7[\alpha] := \mathbb{F}_7[x]/(x^4 - 2x^2 - 3x + 3)$, you can check that $\#E[3](\mathbb{F}_{2401}) = 9$ with $\#E[3](\mathbb{F}_{2401}) \cong_{\mathsf{Ab}} \mathbb{Z}/3 \oplus \mathbb{Z}/3$. Finally, it follows from $\#E(\mathbb{F}_7) = 8$ and Lagrange's Theorem that $E[7](\mathbb{F}_7) = \{\mathscr{O}\}$. In fact, E is supersingular with $j(E) = -1$.

Chapter 2

Exercise Sheets

2.1 Exercise Sheet 1

2.1.A Numerical and Exploration

Exercise 2.1.1. Let $b, c \in \mathbb{R}$. When does the equation $x^3 + bx + c = 0$ have exactly one (resp. two, resp. three) real root(s)? (Hint: Play around on Desmos. A little calculus might come in handy.)

Exercise 2.1.2.

(a) Let R be a UFD with fraction field K. State and prove the analog of the Rational Root Theorem (1.1.2) with $\mathbb{Z}$ replaced by R, and conclude that there is an algorithmic procedure (given the ability to find all factors of an element of R[1]) for finding roots in K of single-variable polynomials with coefficients in K.

(b) Using (a), or otherwise, determine all solutions in the field $\mathbb{Q}(i)$ to

$$2ix^3 + (2-4i)x^2 - (2-5i)x - (2-i) = 0.$$

Exercise 2.1.3.

(a) Find all integer solutions (x, y) to $3x + 5y + 4 = 0$ (and prove you have found all of them!).

(b) Do the same for $4x + 6y + 5 = 0$.

(c) For $a, b, c \in \mathbb{Z}$, give a necessary and sufficient criterion for the equation $ax + by + c = 0$ to have (i) infinitely many solutions, (ii) a finite nonzero number of solutions, and (iii) no solutions (x, y) in the integers.

(d) Generalize (c) to an arbitrary number of variables. More precisely, given any $n \in \mathbb{Z}_{\geq 1}$ and integers $a_1, \ldots, a_n, c \in \mathbb{Z}$, find necessary and sufficient conditions for the equation

$$a_1x_1 + \cdots + a_nx_n + c = 0$$

to have integer solutions $(x_1, \ldots, x_n)$ as in (i), (ii), and (iii).

Exercise 2.1.4.

(a) Find all rational solutions (x, y) to $x^2 + 3y^2 = 2$.

(b) Let p be a prime. Find all rational solutions (x, y) to the equation $x^3 + py^3 = p^2$.

Exercise 2.1.5.

(a) Finish the computation started in class (1.1.6) to verify that the described procedure ("chord and tangent process") yields Bachet's formula 1.1.8.

(b) In the case $c = 1$, why does Bachet's formula get "stuck" at the points $(0, \pm 1)$?

Exercise 2.1.6.

(a) Find all rational solutions (x, y) to $3x^2 - 2y^2 = 1$.

(b) $(**)$ Find all integer solutions (x, y) to $3x^2 - 2y^2 = 1$.

Exercise 2.1.7.

(a) Find a prime at which the curve $x^2 + y^2 = 3$ is locally obstructed.

(b) $(*)$ Find *all* primes at which the curve $x^2 + y^2 = 3$ is locally obstructed.

Exercise 2.1.8. (Adapted from [2, Exercise 2.4].)

(a) Consider the plane curve $y^2 = x^3 + 1$. For each prime p, let n_p be the number of solutions (x, y) to this equation modulo p (i.e., the number of $\mathbb{F}_p$-points on this affine curve). Using your favorite computer software, compute the values n_p for all primes $p \leq 100$. Do you see any patterns? Make and prove conjectures. (Hint: Consider the cases $p \equiv \pm 1 \pmod 3$ separately.)

(b) Repeat the same procedure for the curve $y^2 = x^3 + x$. (Hint: Consider the cases $p \equiv \pm 1 \pmod 4$ separately.)

Exercise 2.1.9 (Fisher). Let E be the elliptic curve over $\mathbb{Q}$ defined by the equation $y^2 + y = x^3 - x$.

(a) Draw a graph of the real points of E.

[1] This is the same as having the ability to find *one* factorization of any element of R, along with the knowledge of all units of R.

(b) Let $P := (0, 0) \in E(\mathbb{Q})$. Get Sage to compute nP for $n = 0, 1, \ldots, 20$. What do you notice about the denominators? Make and prove conjectures. (Hint: Can you express the operation of adding P formulaically?)

2.1.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.1.10 (Hasse-Minkowski for Quadratic Equations)**.** Let $b, c \in \mathbb{Q}$. The quadratic equation $x^2 + bx + c = 0$ has a root in $\mathbb{Q}$ iff it has a root in $\mathbb{Q}_p$ for all primes $p \leq \infty$. (For those not too familiar with $\mathbb{Q}_p$ yet, the only fact you will need is: if a $d \in \mathbb{Q}^\times$ is a square in $\mathbb{Q}_p$, then $v_p(d)$ is even. You may assume this for now.)

Exercise 2.1.11 (Homogeneous Polynomials)**.** Let K be a field, $d \in \mathbb{Z}_{\geq 0}$, and $F \in K[X, Y, Z]$. The following conditions are equivalent:

(a) Every monomial in F has total degree d.
(b) We have the equality $F(WX, WY, WZ) = W^d F(X, Y, Z)$ in the polynomial ring $K[W, X, Y, Z]$.
(c) For all $\lambda \in K$, we have $F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z)$.
(d) We have that
$$X\frac{\partial F}{\partial X} + Y\frac{\partial F}{\partial Y} + Z\frac{\partial F}{\partial Z} = d \cdot F.$$

Exercise 2.1.12. (Adapted from [2, Exercise 1.15].) Let K be a field, $g(u) \in K[u]$ be a monic quartic polynomial, and consider the curve
$$v^2 = g(u).$$

(a) Suppose we are given a root $\alpha \in K$ of g, and a $\beta \in K^\times$. The transformation
$$(x, y) := \left(\frac{\beta}{u - \alpha}, \frac{\beta^2 v}{(u - \alpha)^2}\right)$$
transforms $v^2 = g(u)$ to a curve of the form
$$y^2 = f(x)$$
for some cubic polynomial $f(x) \in K[x]$. This transformation is birational (i.e., a one-to-one correspondence between points of the two curves, except possibly for a finite number of points). Note that this birational transformation does not arise from a linear change of coordinates on (projective closures of) these curves.
(b) If $X, Y, Z \in K$ satisfy $X^4 + Y^4 = Z^4$ with $Z \neq X$, then
$$(x, y) := \left(2\left(\frac{Z + X}{Z - X}\right), \frac{4Y^2}{(Z - X)^2}\right)$$
lies on the curve
$$y^2 = x^3 + bx$$
for some $b \in K$. This b is unique. (What is it?)

(Hint: You will need some condition on g for (a) to work. What is this condition? What implications does this condition have for the roots of f? What do parts (a) and (b) have to do with each other? Part (b) is definitely not correct as stated.)

Exercise 2.1.13. Let K be a field.

(a) Any two lines in $\mathbb{P}^2(K)$ meet in exactly one point. Any two points in $\mathbb{P}^2(K)$ simultaneously lie on exactly one line.
(b) Given any two ordered quadruples (P_1, P_2, P_3, P_4) and (Q_1, Q_2, Q_3, Q_4) of points of $\mathbb{P}^2(K)$, there is a unique linear change of coordinates φ such that $\varphi(P_i) = Q_i$ for $i = 1, 2, 3, 4$. The same result holds for ordered pentuples.

Exercise 2.1.14 (Fisher)**.** Let $C \subset \mathbb{P}^2_{\mathbb{Q}}$ be a smooth cubic curve. If $K/\mathbb{Q}$ is a quadratic or cubic number field, then $C(K) \neq \emptyset$ implies that $C(\mathbb{Q}) \neq \emptyset$.

2.2 Exercise Sheet 2

Make sure to finish all the exercises from Sheet 1 that you did not get to in the first week (e.g., 2.1.9) too!

2.2.A Numerical and Exploration

Exercise 2.2.1. For any $n \in \mathbb{Z}_{\geq 0}$ and prime p, how many points does $\mathbb{P}^n(\mathbb{F}_p)$ have? Draw a picture of $\mathbb{P}^1(\mathbb{F}_3)$. Draw a picture of $\mathbb{P}^1(\mathbb{F}_p)$ for any $p \geq 2$. Draw a picture of $\mathbb{P}^2(\mathbb{F}_2)$, including all lines on it, and capturing all the incidence relations between points and lines. (Does this picture remind you of something magical?)

Exercise 2.2.2. When does a (projective linear) change of coordinates fix the line at infinity ($Z = 0$) as a set? When does it fix it pointwise? When does it fix the line $Z = 0$ as a set, and fix the point $[0 : 1 : 0]$ on it?

Exercise 2.2.3. Let $K = \mathbb{R}$, let $b, c \in \mathbb{R}$ with $\Delta = -16(4b^3 + 27c^2) \neq 0$, and let E be the elliptic curve with Weierstrass equation $y^2 = x^3 + bx + c$.

(a) When does $E(\mathbb{R})$ consist of one connected component? When does it consist of two connected components?[2] (Hint: What does this have to do with Exercise 2.1.1?)
(b) Consider the curve with $(b, c) = (-21, -20)$. How many connected components does $E(\mathbb{R})$ have? Take the point $P := (-3, 4) \in E(\mathbb{R})$. Get Sage to compute (and maybe even plot!) the points nP for $n = 0, 1, \ldots, 10$. Which component does each of these points lie on? Observe patterns, and make and prove conjectures.
(c) Repeat the same procedure as in (b) for more elliptic curves over $\mathbb{R}$ with two components.

Exercise 2.2.4. Let K be a field with $\operatorname{ch} K \neq 3$, and let $\alpha \in K$, and let

$$E_\alpha = \mathbb{V}(X^3 + Y^3 - \alpha Z^3) \subset \mathbb{P}^2_K.$$

(a) Show that if $\alpha \neq 0$, then E_α is smooth. (What happens when $\operatorname{ch} K = 3$?)

Hence suppose that $\alpha \neq 0$.

(b) Given a point $P_0 = (x_0, y_0) \in E_\alpha(K)$, find a formula for the point $P_1 = (x_1, y_1)$ obtained by the same "chord and tangent process" as in the Bachet example (1.1.6): draw the tangent line, and then take the third point of intersection with E_α. This procedure gives a way to produce an infinite sequence of points $P_n \in E_\alpha(K)$ as before. (Hint: Your final formula should be quite elegant.)
(c) A point $P_0 \in E_\alpha(K)$ is said to be *exceptional* if the set $\{P_0, P_1, \ldots\}$ is finite; in other words, when the "chord and tangent process" only gets you finitely many points. Find at least one exceptional point on E_α for any α. When $K = \mathbb{Q}$ and $\alpha = 1, 2$, find some more exceptional points.
(d) $(*)$ Suppose now that $K = \mathbb{Q}$ and that $\alpha \geq 1$ is a cube-free integer. Use your answer to (b) to show that the only exceptional points on E_α are the ones you found in (c).
(e) Using the discussion from 1.2.B or otherwise, compute the j-invariant $j(E_\alpha)$ as a function of α.
(f) Finally, suppose now that $K = \mathbb{F}_p$ for primes $p \neq 3$. Using Sage, compute $\#E_1(\mathbb{F}_p)$ for all primes p with $5 \leq p \leq 1000$. What patterns do you observe? Make and prove conjectures. (Hint: Consider the cases $p \equiv 1, 2 \pmod 3$ separately.)

2.2.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.2.5. For any $b \in \mathbb{Z}$, the polynomial $x^3 + bx + 1 \in \mathbb{Q}[x]$ is irreducible, i.e., does not factor into two polynomials of smaller degree. The same is true for $x^3 + bx - 1$. (Hint: What does this have to do with this course? A result we talked about very early on may help.)

Exercise 2.2.6. Let K be a field, $F \in K[X, Y, Z]$ a homogeneous polynomial, $D = \mathbb{V}(F) \subset \mathbb{P}^2_K$. Let L/K be a field extension and $P = (x_0, y_0) = [x_0 : y_0 : 1] \in D(L)$.

[2] What's a connected component?

(a) Let $f(x,y) := F(x,y,1) \in K[x,y]$ denote the dehomogenization of F with respect to Z, and $C = D^\circ = \mathbb{V}(f) \subset \mathbb{A}^2_K$ the corresponding affine curve. Then $(x_0, y_0) \in C(L)$ is a smooth point of the affine curve C iff $[x_0 : y_0 : 1] \in D(L)$ is a smooth point of the projective curve D. Further, the projective tangent line $\mathbb{T}_P(D)$ is the projective closure of the affine tangent line $\mathrm{T}_P(C)$, i.e., $\mathbb{T}_P(D) = \overline{\mathrm{T}_P(C)}$.

(b) Suppose that $\varphi : \mathbb{P}^2_K \to \mathbb{P}^2_K$ is a projective linear change of coordinates. Then P is a smooth point of D iff $\varphi(P)$ is a smooth point of $\varphi(D)$, in which case we have $\varphi(\mathbb{T}_P D) = \mathbb{T}_{\varphi(P)}\varphi(D)$. (What is $\varphi(D)$?)

Exercise 2.2.7.

(a) Let K be a field and $D \subset \mathbb{P}^2_K$ a curve. Suppose that $P \in D(K)$ is a singular point. Then for any line L passing through P, we have that $m_P(D, L) \geq 2$, i.e., L intersects D in multiplicity at least 2 at P.

(b) (**) (This might require a little Galois theory, so if you haven't seen that, you can skip this part.) Suppose $D \subset \mathbb{P}^2_K$ is an integral cubic curve. Suppose that L/K is a field extension, and that $P \in D(L)$ is a singular point. Then in fact $P \in D(K)$. (Hint: See 1.5.5, which might suggest you look only at perfect K. Next, consider the cubic $\mathbb{V}(X^3+XY^2+Y^3+X^2Z+XYZ+Y^2Z+Z^3) \subset \mathbb{P}^2_{\mathbb{F}_2}$ or the cubic $\mathbb{V}(X^3-3XY^2-Y^3-3X^2Z-3XYZ-3Y^2Z+3Z^3) \subset \mathbb{P}^2_{\mathbb{Q}}$. Finally, for a very strong salvage, consider the case of *geometrically* integral D (e.g., a D in Weierstrass form) over a perfect field K. Here "geometrically integral" means that D remains integral over an algebraic closure $\overline{K}$ of K. If it makes things easier for you, you may suppose that L/K is finite.)

Exercise 2.2.8. (Adapted from [2, Exercise 2.13].) Let K be a field of characteristic other than 2, let $t \in K$, and consider the cubic $E_t \subset \mathbb{P}^2_K$ defined as the projective closure of the affine curve

$$y^2 = x^3 - (2t-1)x^2 + t^2 x.$$

(a) The curve E_t is nonsingular iff $t \notin \{0, 1/4\}$, in which case E_t is an elliptic curve over K in Weierstrass form. What is $j(E_t)$ as a function of t?

(b) In the situation in (a), the point $(t,t) \in E(K)$ has order 4.

(c) If $E \subset \mathbb{P}^2_K$ is any elliptic curve over a field K of characteristic other than 2 such that there is a point $P \in E(K)$ of order 4, then there is a projective linear change of coordinates $\varphi : \mathbb{P}^2_K \to \mathbb{P}^2_K$ and a $t \in K \smallsetminus \{0, 1/4\}$ such that $\varphi(E) = E_t$ and $\varphi(P) = (t,t)$.

(d) For a given (E,P) as in (c), the t found in (c) is unique. (Possible salvage: How many values of t work?)

Exercise 2.2.9. Let $R := \mathbb{Z}[a,b,c]$ be the polynomial ring in the variables a, b, c. Take the polynomial $f = f(x) = x^3 + ax^2 + bx + c \in R[x]$, and let $f'(x) = 3x^2 + ax + b$ and $f''(x) = 6x + a$ be the first and second formal derivatives of f with respect to x. The equation $y^2 = f(x)$ defines an elliptic curve E in Weierstrass form over $K = \mathbb{Q}(a,b,c)$, or over any field K of characteristic other than 2 when given specific $a, b, c \in K$ such that $\Delta \neq 0$ in K, which happens iff f and f' have no common roots over an algebraic closure.

(a) Let $f_3(x) := 2f(x)f''(x) - f'(x)^2$. Compute $f_3(x)$ explicitly as an element of $R[x]$. What is the degree of $f_3(x)$? What is its leading coefficient?

(b) PODASIP: For any field K and $P = (x,y) \in E(K) \smallsetminus \{\mathscr{O}\}$, the point P has order 3 (i.e., $3P = \mathscr{O}$, i.e., $P \in E[3](K)$) iff $f_3(x) = 0$. (Possible hints: $3P = \mathscr{O}$ is equivalent to $2P = -P$. Alternatively, think about inflection points. Can you check that

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \frac{f_3(x)}{4yf(x)}?$$

What does the second derivative have to do with inflection points?)

(c) PODASIP: For any field K, given $a, b, c \in K$ with $\Delta \neq 0$, the polynomial $f_3(x) \in K[x]$ never has repeated roots. (Hint: What is $f_3'(x)$? How many roots can this share with $f_3(x)$?)

(d) How many roots can it have in K when (i) $\operatorname{ch} K = 3$, (ii) when $K = \mathbb{R}$, (iii), when K is algebraically closed of characteristic other than 3? What implications does this have for the group $E[3](K)$ for various K as above? Specifically, for $K = \mathbb{R}$:

(e) Suppose $K = \mathbb{R}$, and $a, b, c \in \mathbb{R}$ are such that $\Delta \neq 0$. Then $f_3(x) \in \mathbb{R}[x]$ has exactly two real roots, say $\alpha < \beta$. We have $f(\alpha) < 0$ and $f(\beta) > 0$. In particular, $E[3](\mathbb{R})$ is cyclic of order 3.

Here's a real challenge.

Exercise 2.2.10 (Division Polynomials). (**) (Adapted from [5, Exercise 2.6.10].) Let's generalize 2.2.9. For simplicity, we'll do it assuming $a = 0$. Let $R := \mathbb{Z}[b, c]$ be the polynomial ring in two variables b, c. Take the polynomial $f = f(x) = x^3 + bx + c \in R[x]$, and let $f'(x) = 3x^2 + b$ and $f''(x) = 6x$ be the first and second formal derivatives of f with respect to x.

(a) Define the sequence $(f_n)_{n\geq 0}$ of polynomials in $R[x]$ recursively by $f_0 = 0, f_1 = f_2 = 1$,

$$\begin{aligned} f_3 &:= 2f \cdot f'' - (f')^2, \\ f_4 &:= -16f^2 + 4f \cdot f' \cdot f'' - 2(f')^3, \\ f_{2n+1} &:= f_{n+2} \cdot f_n^3 - 16f^2 \cdot f_{n-1} \cdot f_{n+1}^3 \quad \text{for } n \geq 2 \text{ odd,} \\ f_{2n+1} &:= 16f^2 \cdot f_{n+2} \cdot f_n^3 - f_{n-1} \cdot f_{n+1}^3 \quad \text{for } n \geq 2 \text{ even, and} \\ f_{2n} &:= f_n(f_{n+2} \cdot f_{n-1}^2 - f_{n-2} \cdot f_{n+1}^2) \quad \text{for } n \geq 3. \end{aligned}$$

For $n \geq 1$, we have

$$f_n = \begin{cases} nx^{(n^2-1)/2} + \cdots, & \text{for } n \text{ odd, and} \\ (n/2)x^{(n^2-4)/2} + \cdots, & \text{for } n \text{ even,} \end{cases}$$

where $\cdots$ denotes terms of lower degree. Get Sage to write down the polynomials f_n for $n \leq 5$ explicitly.

(b) The equation $y^2 = f(x)$ defines an elliptic curve E in Weierstrass form (over $K = \mathbb{Q}(b, c)$ or over any field K of characteristic other than 2 when given specific $b, c \in K$ such that $4b^3+27c^2 \neq 0 \in K$). In this case,

$$\gcd(f_n, f \cdot f_{n+1} \cdot f_{n-1}) = (1)$$

when n is odd and

$$\gcd(f \cdot f_n, f_{n+1} \cdot f_{n-1}) = (1)$$

when $n \geq 2$ is even.

(c) If $P = (x, y) \in E(K)$, then the coordinates of $nP \in E(K)$ are given as

$$nP = \left(x - \frac{4 \cdot f \cdot f_{n+1} \cdot f_{n-1}}{f_n^2}, y \cdot \frac{f_{2n}}{f_n^4}\right)$$

when n is odd and

$$nP = \left(x - \frac{f_{n+1} \cdot f_{n-1}}{4f \cdot f_n^2}, y \cdot \frac{f_{2n}}{16f^2 \cdot f_n^4}\right)$$

when n is even. (Hint: WOP, of course.)

(d) Now fix an $n \geq 1$, and suppose that K is a field with $\operatorname{ch} K \nmid 2n$.

(1) For $P = (x, y) \in E(K)$, we have $nP = \mathscr{O}$ iff the x-coordinate $x(P)$ of P satisfies $f_n(x) = 0$ when n is odd or satisfies $f(x) \cdot f_n(x) = 0$ when n is even.

(2) When n is odd, the polynomial f_n is separable (i.e., has nonzero discriminant, i.e., has distinct roots over an algebraic closure of K), and when n is even, the polynomial $f \cdot f_n$ is separable.

(3) There are exactly n^2 points of order dividing n in $E(K)$, and, in fact, we have

$$E[n](K) \cong \mathbb{Z}/n \oplus \mathbb{Z}/n.$$

(Hint: If G is an abelian group of order n^2 for some $n \geq 1$ such that for each divisor $d \mid n$ we have $\#G[d] = d^2$, where $G[d] \subset G$ is the subgroup of all points of order dividing d, then $G \cong \mathbb{Z}/n \oplus \mathbb{Z}/n$.)[3]

[3]This result is also true if we only assume $\operatorname{ch} K \nmid n$, but the proof is harder. We might do it in class if time permits.

2.3 Exercise Sheet 3

2.3.A Numerical and Exploration

Exercise 2.3.1. For each prime $p \neq 2, 3$, and $b, c = 0, \ldots, p-1$ such that $4b^3 + 27c^2 \not\equiv 0 \pmod{p}$, let $E_{p,b,c}$ be the elliptic curve $y^2 = x^3 + bx + c$ over $\mathbb{F}_p$.

(a) Using, say, a simple `for` loop and the `E.abelian_group()` function in Sage, study the abelian group structure on $E_{p,b,c}$ for each $p \leq 19$ and each allowable b, c. Find patterns and make conjectures. For instance, what can you say about the ratio $\#E_{p,b,c}(\mathbb{F}_p)/p$ in general? Can you find (p, b, c) such that $E_{p,b,c}(\mathbb{F}_p)$ is cyclic of order 11? 19? When is this group cyclic? When it is not, does it ever need more than 2 generators?

(b) Modify your `for` loop in (a) to remove all results in which the group $E_{p,b,c}(\mathbb{F}_p)$ is cyclic, so that we can focus on the non-cyclic case. (In Sage, you can test whether a group is cyclic using the `is_cyclic()` function.) Allowing $p \leq 100$ now, find patterns and make conjectures. Does this group ever need more than 2 generators? Can you find a (p, b, c) such that the group $E_{p,b,c}(\mathbb{F}_p)$ is isomorphic to $\mathbb{Z}/3 \oplus \mathbb{Z}/3$? $\mathbb{Z}/m \oplus \mathbb{Z}/m$ for $m = 4, 5, \ldots, 9, 10$? $\mathbb{Z}/33 \oplus \mathbb{Z}/3$? $\mathbb{Z}/18 \oplus \mathbb{Z}/6$? When the group $E_{p,b,c}(\mathbb{F}_p)$ is $\mathbb{Z}/m \oplus \mathbb{Z}/m$, what do you notice about $p \mod m$?

(c) Can you use your calculations in (a) and (b) to give a lower bound on the number N_p of isomorphism classes of elliptic curves over $\mathbb{F}_p$ for each p with say $5 \leq p \leq 100$?

Exercise 2.3.2. Given a prime $p \neq 2, 3$, write Sage code to compute the number N_p of isomorphism classes of elliptic curves over $\mathbb{F}_p$ (up to isomorphism over $\mathbb{F}_p$). For $p \leq 200$, compute N_p/p. What patterns do you notice? (Possible hints: You might try something similar to what you did in 2.3.1. In Sage, you can test whether a curve `E1` defined over a field say `GF(p)` is isomorphic to another curve `E2` defined over the same field by using the boolean function `E1.is_isomorphic(E2)`.)

Exercise 2.3.3.

(a) Classify all elliptic curves over $\mathbb{F}_2$ (up to isomorphism over $\mathbb{F}_2$); in particular, compute N_2. (Hint: Note that each of the quantities $a_1, a_2, a_3, a_4, a_6 \in \mathbb{F}_2$ could give rise to a curve, which means a priori that there are $2^5 = 32$ possibilities, but it is easy to eliminate a lot of these possibilities by inspection. For instance, it is clear from the formula for Δ in characteristic two (1.9.13) that not both a_1 and a_3 can be zero. You may use Sage to speed up calculations if that is helpful.)

(b) For each (isomorphism class of a) curve E as in (a),
   (i) identify the group $E(\mathbb{F}_2)$,
   (ii) find out if the curve E is supersingular,
   (iii) figure out the automorphism groups $\mathrm{Aut}_{\mathbb{F}_2}(E)$ and $\mathrm{Aut}_{\overline{\mathbb{F}}_2}(E)$, and
   (iv) calculate the j-invariant $j(E)$.

(c) $(*)$ For each pair of curves that are not isomorphic over $\mathbb{F}_2$ but have the same j-invariant, figure out the smallest field extension $L/\mathbb{F}_2$ over which those two curves *are* isomorphic. (Hint: This L is one of $\mathbb{F}_4, \mathbb{F}_{16}, \mathbb{F}_{256}$.)

(d) $(*)$ Repeat the same for $\mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5$, and $\mathbb{F}_7$. (Hint: Again, you may use Sage. This problem is not as difficult as it may look, since over $\mathbb{F}_3$ you can put elliptic curves in the short Weierstrass form $y^2 = x^3 + ax^2 + bx + c$, and over $\mathbb{F}_5$ and $\mathbb{F}_7$ you can put them in the even shorter Weierstrass form $y^2 = x^3 + bx + c$, as above. Recall that $\mathbb{F}_4 \cong \mathbb{F}_2[\omega] \cong \mathbb{F}_2[x]/(x^2 + x + 1)$. The way to work with $\mathbb{F}_4$ in Sage is to use `K.<a> = GF(4)`; this returns a field K isomorphic to $\mathbb{F}_4$ with a generator named a satisfying $a^2 + a + 1 = 0$.)

If you haven't seen the fields $\mathbb{F}_{p^n}$ for $n \geq 2$ (note that $\mathbb{F}_{p^n} \ncong \mathbb{Z}/(p^n)$ as rings!), you can skip those parts of this exercise for now and return to them later when you do.

Exercise 2.3.4.

(a) Find all rational solutions to $x^2 + y^2 = 2$ by projecting from $(-1, -1)$ and using t as the slope of the line, i.e., by mapping t to the other intersection of $y + 1 = t(x + 1)$ with $x^2 + y^2 = 2$.

(b) Using your solution to (a) or otherwise, prove that if p, q, r are positive integers with $p < q < r$ such that p^2, q^2 and r^2 are in arithmetic progression, then there are coprime $m, n \in \mathbb{Z}$ of different parity with $m > n > 0$ such that (p, q, r) is one of

$$\left(\pm(2mn - (m^2 - n^2)), m^2 + n^2, 2mn + m^2 - n^2\right).$$

Figure out exactly when each sign holds. (Hint: Your answer may have something to do with $\sqrt{2} \pm 1$.)

Exercise 2.3.5. (Adapted from [15].) For simplicitly, let's work over $K = \mathbb{Q}$. Let $C \subset \mathbb{P}^2_K$ be the cubic curve with equation

$$XY^2 - X^2Y + 2X^2Z - 2XZ^2 - 3Y^2Z + 3YZ^2 = 0.$$

(a) ($*$) Suppose $p, q, r, s \in \mathbb{Z}_{>0}$ are such that p^2, q^2, r^2, s^2 are in arithmetic progression. Suppose that $p \neq s$, so the sequence is not the constant sequence. Show that

$$[p - s : q - s : r - s] \in C(K).$$

Therefore, to get a handle on all four-term sequences of squares in arithmetic progressions, it suffices to get a handle on $C(K)$. (Hint: There are many ways to do this. One is to just plug and check. Another, possibly more illuminating, way might be to express the given condition in terms of two quadratic conditions on p, q, r, s; indeed, what *does* it mean to be in arithmetic progression? Next, let $(X, Y, Z) = (p - s, q - s, r - s)$, and write these quadratic conditions in terms of X, Y, Z, and s. Finally, try to eliminate s from the equations.)[4]

(b) Is C smooth? Find a rational inflection point on C. (Hint: Get Desmos to draw a picture of the affine patch $Z \neq 0$ of C. Then try an intersection with a coordinate axis.)

(c) Using your solution to (b) or otherwise, find the linear change of coordinates that transforms C into the elliptic curve $E \subset \mathbb{P}^2_K$ in short Weierstrass form

$$y^2 = x(x + 1)(x + 4).$$

We met this curve in 1.8.4.

(d) Using the Nagell-Lutz Theorem, find all torsion points in $E(K)$. How many of them are there? Describe the structure of the torsion subgroup $\operatorname{Tors} E(K)$. (Hint: Up to isomorphism, there are three groups of size $\#\operatorname{Tors} E(K)$. Part of your task is to figure out which of these is the isomorphism type of $\operatorname{Tors} E(K)$. What distinguishes these three types?)

(e) Later in the course, we will prove that $\operatorname{rank} E = 0$. Assuming this result, prove the theorem stated by Fermat in 1640 and first proven by Euler in 1780 which states that there are no nonconstant four-term sequence of squares in arithmetic progression in the integers.

(f) How much of the above discussion carries over when we replace K by a number field or a finite field? For which characteristics does the above discussion still apply, and for which does it not? When it does apply, what do we learn about four squares in arithmetic progressions over these fields?

2.3.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.3.6. Let K be a field.

(a) Let $C \subset \mathbb{A}^2_K$ be an affine curve. If C is integral, then so its projective closure $\overline{C} \subset \mathbb{P}^2_K$.

(b) Let $D \subset \mathbb{P}^2_K$ be a projective curve. If D is integral, then so is each affine patch $D^\circ \subset \mathbb{A}^2_K$.

The same statements are true with "integral" replaced by "geometrically integral".

Exercise 2.3.7. Let K be a field.

(a) Let $C \subset \mathbb{A}^2_K$ be an integral affine curve with defining equation $f \in K[x, y]$. The element $f \in K[x, y]$ is prime.

We define the *function field* of C to be $K(C) := \operatorname{Frac} K[x, y]/(f)$. (Make sure you see why this is independent of the choice of f!)

(b) If $C \subset \mathbb{A}^2_K$ is an integral curve with projective closure $\overline{C}$, then $K(C) \cong K(\overline{C})$.

(c) If $D \subset \mathbb{P}^2_K$ is an integral curve with affine patch say $D^\circ \subset \mathbb{A}^2_K$, then $K(D) \cong K(D^\circ)$.

[4]For the cognoscenti: The curve $\mathbb{V}(p^2 + r^2 - 2q^2, q^2 + s^2 - 2r^2) \subset \mathbb{P}^3_K$ is a smooth projective geometrically integral curve of genus one, and hence an elliptic curve when equipped with the rational point $[1 : -1 : -1 : 1]$. We are trying to find a Weierstrass equation for it, which can be done by projecting from the point $[1 : 1 : 1 : 1]$ to embed the curve as a cubic $C \subset \mathbb{P}^2_K$, and then applying a linear change of coordinates.

Exercise 2.3.8. Let K be a field.

(a) Let $C_1, C_2 \subset \mathbb{A}^2_K$ be affine curves, say $C_i = \mathbb{V}(f_i)$ for nonzero $f_1, f_2 \in K[x, y]$. Let $C = \mathbb{V}(f_1 \cdot f_2)$, so that for all fields L/K we have $C(L) = C_1(L) \cup C_2(L)$. (Note that C is not the "union" of C_1 and C_2 in the naive sense; for instance we could have $f_1 = f_2 = x$, in which case C would be $C = \mathbb{V}(x^2)$.) If L/K is a field extension, and $P \in C_1(L) \cap C_2(L)$, then $P \in C(L)$ is a singular point. (This might help explain why curves like $\mathbb{V}(x^2)$ are everywhere singular.) A similar result is true for projective curves.

(b) If $C \subset \mathbb{A}^2_K$ is a smooth affine curve, then C is (geometrically) integral. The same is true for projective curves. (Hint: You may use the full strength of Bézout's Theorem (1.6.9), which we have admittedly not proven.)

Exercise 2.3.9. Let K be a field, $C \subset \mathbb{P}^2_K$ a smooth cubic curve, and $\mathscr{O}_1, \mathscr{O}_2 \in C(K)$. For $i = 1, 2$, let $\oplus_i$ denote the operation $P \oplus_i Q := \mathscr{O}_i * (P * Q)$ for $P, Q \in C(K)$, so that $(C(K), \oplus_i)$ is an abelian group by 1.7.7. The map

$$(C(K), \oplus_1) \to (C(K), \oplus_2), \quad P \mapsto \mathscr{O}_1 * (\mathscr{O}_2 * P)$$

is a group isomorphism. In particular, the group law on $C(K)$ does not depend on the choice of $\mathscr{O}$, up to isomorphism.

Exercise 2.3.10. (**)

(a) Let $b \in \mathbb{Z}$ be fourth-power-free, and let $E_b/\mathbb{Q}$ be the elliptic curve with Weierstrass equation

$$y^2 = x^3 + bx.$$

Then $\operatorname{Tors} E_b(\mathbb{Q})$ is either $\mathbb{Z}/2$ or $\mathbb{Z}/4$. When does each case hold?

(b) Let $c \in \mathbb{Z}$ be sixth-power-free, and let $E_c/\mathbb{Q}$ be the elliptic curve with Weierstrass equation

$$y^2 = x^3 + c.$$

Then $\operatorname{Tors} E_c(\mathbb{Q})$ is either $\mathbb{Z}/2$ or $\mathbb{Z}/3$. When does each case hold? Compare your result to your answer to 2.2.4(e).

(Hint: First, try lots of examples, using Sage. Then, if needed, refer to [1, §12] for a detailed hint.)

2.4 Exercise Sheet 4

2.4.A Numerical and Exploration

Exercise 2.4.1 (Fisher)**.** Let p be an odd prime, and let E be an elliptic curve over $\mathbb{F}_p$.

(a) Show that there exists and elliptic curve E' over $\mathbb{F}_p$ such that $\#E(\mathbb{F}_p)+\#E'(\mathbb{F}_p) = 2(p+1)$. (Hint: Modulo the right definitions of these objects, you could probably have solved this problem before you knew anything about elliptic curves.)

(b) Given the curves E and E' as in (a), what do you notice about the sizes of $E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$ and of $E(\mathbb{F}_{p^2})$? Make and prove conjectures. (Hint: It might be helpful to consider the "conjugation" map $\mathbb{F}_{p^2} \to \mathbb{F}_{p^2}$ given by $a \mapsto a^2$. If you don't know much about the fields $\mathbb{F}_{p^2}$ in general, try to attempt this problem for small special cases such as $p = 3$ where $\mathbb{F}_9 \cong \mathbb{F}_3[i] := \mathbb{F}_3[x]/(x^2+1)$ or $p = 5$ where $\mathbb{F}_{25} \cong \mathbb{F}_5[\omega] := \mathbb{F}_5[x]/(x^2+x+1)$.)

2.4.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.4.2. Let K be a field, and for $\lambda \in K \setminus \{0,1\}$, let

$$j(\lambda) := 2^8 \frac{(\lambda(1-\lambda)+1)^3}{\lambda^2(1-\lambda)^2}.$$

(a) For $\lambda, \lambda' \in K \setminus \{0,1\}$, we have that $j(\lambda) = j(\lambda')$ iff $\lambda' \in S(\lambda)$, where

$$S(\lambda) := \left\{\lambda, \frac{1}{\lambda}, 1-\lambda, \frac{1}{1-\lambda}, 1-\frac{1}{\lambda}, 1-\frac{1}{1-\lambda}\right\}.$$

(b) For any $\lambda \in K \setminus \{0,1\}$, the set $S(\lambda)$ of (a) has cardinality exactly six.

Exercise 2.4.3. Let K be a field.

(a) For any elliptic curve E over K, there is a(n) (admissible) change of coordinates which brings E into the form $E_\lambda : y^2 = x(x-1)(x-\lambda)$ for some $\lambda \in K \setminus \{0,1\}$. The j-invariant of E_λ is precisely the $j(\lambda)$ of 2.4.2, i.e., $j(E_\lambda) = j(\lambda)$.

(b) For $\lambda, \lambda' \in K \setminus \{0,1\}$, there is an admissible change of coordinates taking E_λ to $E_{\lambda'}$ iff $\lambda' \in S(\lambda)$ iff $j(\lambda) = j(\lambda')$. In particular, the j-invariant classifies elliptic curves up to admissible changes of coordinates (and hence by 1.10.6 up to isomorphism). (Hint: The "change in the x-coordinate alone" would have to send the unordered triple $\{0,1,\lambda\}$ to $\{0,1,\lambda'\}$.)

Exercise 2.4.4. Let K be a field (of any characteristic!). Given any $j \in K$, there is a unique elliptic curve E over K (up to admissible changes of coordinates over K) such that $j(E) = j$.

Exercise 2.4.5. Let K be a field. Given finitely many points $P_i \in \mathbb{P}^2(K)$, there is a line $L \subset \mathbb{P}^2_K$ that doesn't pass through any of the P_i. In particular, up to a linear change of coordinates, we can assume that each P_i lies in $\mathbb{P}^2_K \setminus \mathbb{V}(Z) \cong \mathbb{A}^2_K$.

Exercise 2.4.6. Let K be a field and $E \subset \mathbb{P}^2_K$ an elliptic curve in Weierstrass form.

(a) If $P, Q \in E(K)$ are inflection points, then so is $P * Q$.

(b) If $S \subset E(K)$ is the set of inflection points, then no four points in S are collinear, but any line passing through two points in S passes through a third point in S.

Exercise 2.4.7.

(a) Suppose $S \subset \mathbb{A}^2(\mathbb{R})$ is a set of nine points with the property that a line passing through any two of them passes through a third. Then S is contained in a line. (Hint: This is just plane geometry; it has nothing to do with elliptic curves.)

(b) If $E \subset \mathbb{P}^2_{\mathbb{R}}$ is an elliptic curve, then exactly three of its nine complex inflection points are real, i.e., even though $\#E[3](\mathbb{C}) = 9$, the subgroup $E[3](\mathbb{R}) \subset E[3](\mathbb{C})$ has size 3. In particular, $E[3](\mathbb{R}) \cong_{\mathsf{Ab}} \mathbb{Z}/3$. (See also 2.2.9.)

Exercise 2.4.8 (Fisher). Fix an integer $b \in \mathbb{Z}$.

(a) For each prime p such that $p \nmid 2b$, the equation $y^2 = x^3 + bx$ is smooth, and hence defines an elliptic curve E_b, over $\mathbb{F}_p$.

(b) In the notation of (a), suppose $p \nmid 2b$, and let $a_p \in \mathbb{Z}$ be defined by $\#E_b(\mathbb{F}_p) = p + 1 - a_p$. Then

$$a_p \equiv b^{(p-1)/4} \binom{(p-1)/2}{(p-1)/4} \pmod{p}.$$

(c) (∗∗) Under the notation and hypotheses of (b), if $p \geq 7$ and $p \equiv 3 \pmod 4$, then $a_p = 0$. How does this relate to 2.1.8? (Hint: One way to do this would be to combine your solution in (b) with [8, Theorem V.1.1], which we have also admittedly not proven. Is there an easier way?)

[Three Torsion over K and Galois action]

[Sylvester-Gallai]

[Supersingularity]

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